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Life Cycle Inventories of Renewable Materials

Data v2.0 (2007)



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Part I

Life Cycle Inventories of Tropical Wood

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1 Introduction

This report contains life cycle inventory data for:

- Azobe from managed forests in Central Africa
- Meranti from managed forests in Malaysia
- Paran  pine from plantations/managed natural forests in Brazil
- Eucalyptus ssp. from plantations in Thailand

The datasets are taken from various sources of literature, adapting them as far as possible to the quality guidelines of the ecoinvent centre. These datasets represent (close to) sustainable forest management practices, which – unfortunately – cannot be considered common practice in the tropics.

This implies that any specific environmental impact from land use change following timber harvesting – which finally might lead also to deforestation – needs to be considered on a case-specific basis.

It should be noted that natural site conditions and forest management practices in tropical forests, socio-economic circumstances as well as species-specific material properties of tropical wood can vary considerably. These factors can have decisive influence on data inventories of tropical wood. As LCI data availability on tropical forestry processes and tropical wood is very limited, the data inventories should be considered as case-specific data; the adequacy of these datasets for specific projects should be assessed carefully.

It should also be noted that tropical forests provide environmental services much beyond what can be depicted in life cycle assessment – not to talk about the socio-economic relevance and their role for poverty alleviation. Such environmental services include above all: biodiversity conservation, hydrological services, scenic beauty, etc. Using wood from tropical forests in a sustainable way gives this valuable resource a market value, which might be decisive for the longterm conservation of tropical forests. Experiences of the last decades have shown that unreflected bans on tropical wood have not been able to stop tropical deforestation. On the contrary, evidence exists that price cuts for tropical timber have even increased deforestation rates for the conversion of forests to agricultural or pastoral lands.

This implies for the interpretation of life cycle inventory results from tropical wood that the direct impacts from forest management as inventoried in these datasets are only one part of the story; further forest degradation or even deforestation as a consequence of follow-up activities on cleared lands or indirect market effects ('opportunity effects') due to a non-use of tropical timber might turn out to be much more environmentally relevant. These mechanisms need to be considered very carefully when assessing the environmental relevance of the use of tropical timber.

Most of the relevant information on the processes is compiled in the figures depicting the input and output flows to the processes together with information about their use in the "real world" and about their representation in ecoinvent. This figure shows also the sources and the uncertainty information to the individual values.

2 Tropical forest resources

2.1 Forest types in the tropics

The tropics are the geographic region of the Earth centred on the equator and limited in latitude by the Tropic of Cancer in the northern hemisphere, at approximately 23°30' (23.5°) N latitude, and the Tropic of Capricorn in the southern hemisphere at 23°30' (23.5°) S latitude. This area includes all the areas of the Earth where the sun reaches a point directly overhead at least once during the solar year¹.

Tropical ecosystems may consist of moist broadleaf forests, dry broadleaf (deciduous) forests, coniferous forests, grasslands, savannas, alpine tundra and snow-capped peaks, shrublands and deserts.

There are often significant areas of biodiversity and species endemism present particularly in rain forests and dry deciduous forests. Some examples of important biodiversity and/or high endemism ecosystems are: Costa Rican rain forests, Madagascar dry deciduous forests, Waterberg Biosphere of South Africa and eastern Madagascar rain forests.

Often the soils of tropical forests are low in nutrient content making them quite vulnerable to slash-and-burn techniques, which are sometimes an element of shifting cultivation agricultural systems.

Tropical forest ecosystems may consist of:

- moist broadleaf forests,
- dry broadleaf/deciduous forests,
- coniferous forests.

2.1.1 Tropical moist broadleaf forests

Tropical moist broadleaf forests, also known as tropical moist forests, are a tropical forest biome.

Figure 2.1 depicts areas known to experience 'rainy tropical climates' (better known as the Wet Tropics) and 'wet and dry tropical climates' (the Wet-Dry Tropics).

The biome includes several types of forests:

- *Lowland equatorial evergreen rain forests*, commonly known as tropical rain forests, are forests which receive high rainfall (more than 2000 mm annually) throughout the year. These forests occur in a belt around the equator, with the largest areas in the Amazon basin of South America, the Congo basin of central Africa, Indonesia, and New Guinea.
- *Moist deciduous and semi-evergreen forests*, receive high overall rainfall with a warm summer wet season and a cooler winter dry season. Some trees in these forests drop some or all of their leaves during the winter dry season. These forests are found in parts of South America, in Central America and around the Caribbean, in coastal West Africa, parts of the Indian subcontinent, and across much of Indochina.
- *Mountain rain forests*, some of which are known as cloud forests, are found in cooler-climate mountainous areas.
- *Flooded forests*, including freshwater swamp forests and peat swamp forests.

¹ In the temperate zones, north of the Tropic of Cancer and south of the Tropic of Capricorn, the sun never reaches an angle of 90° or directly overhead

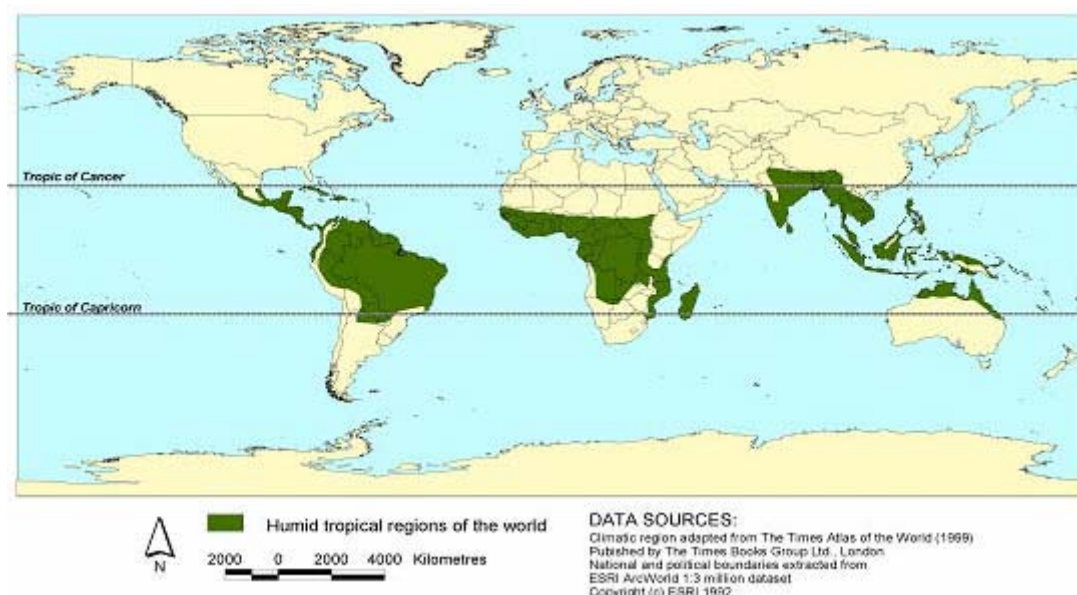


Figure 2.1: The tropics: Wet/wet and dry tropical climates (Source: www.nctwr.org.au/world-wetlands.html)

Tropical and subtropical moist broadleaf forests are common in several terrestrial eco-zones, including parts of the *Afrotropic* (equatorial Africa), *Indomalaya* (parts of the Indian subcontinent and Southeast Asia), the *Neotropic* (northern South America and Central America), *Australasia* (eastern Indonesia, New Guinea, northern and eastern Australia), and *Oceania* (the tropical islands of the Pacific Ocean). About half of the world's tropical rain forests are in the South American countries of Brazil and Peru. Rain forests now cover less than 6% of Earth's land surface. Scientists estimate that more than half of all the world's plant and animal species live in tropical rain forests.

The most severe threat to the tropical (and subtropical) moist broadleaf forests is human intervention (see chap. 2.5).

2.1.2 Tropical dry broadleaf/deciduous forests

The tropical and subtropical dry broadleaf forest biome, also known as tropical dry forest, is located at tropical and subtropical latitudes north and south of the equatorial rain forest belt and of the subtropical deserts, generally in two bands, one between 10° and 20°N latitude and the other between 10° and 20°S latitude.

The most diverse dry forests in the world occur in southern Mexico and in the Bolivian lowlands. The dry forests of the Pacific Coast of north-western South America support a wealth of unique species due to their isolation. The subtropical forests of Maputoland-Pondoland in south-eastern Africa are diverse and support many endemic species. The dry forests of central India and Indochina are notable for their diverse large vertebrate faunas. Madagascar dry deciduous forests and New Caledonia dry forests are also highly distinctive (pronounced endemism and a large number of relict taxa) for a wide range of taxa and at higher taxonomic levels.

Though these forests occur in climates that are warm year-round, and may receive several hundred centimetres of rain per year, they deal with long dry seasons which last several months and vary with geographic location. These seasonal droughts have great impact on all living things in the forest. Deciduous trees predominate in most of these forests, and during the drought a leafless period occurs, which varies with species type. Because trees lose moisture through their leaves, the shedding of leaves allows trees such as teak and mountain ebony to conserve water during dry periods. The newly bare trees open up the canopy layer, enabling sunlight to reach ground level and facilitate the growth of

thick underbrush. Trees on moister sites and those with access to ground water tend to be evergreen. Infertile sites also tend to support evergreen trees. Three tropical dry broadleaf forest eco-regions, the East Deccan dry evergreen forests, the Sri Lanka dry-zone dry evergreen forests, and the South-eastern Indochina dry evergreen forests, are characterized by evergreen trees.

Though less biologically diverse than rain forests, tropical dry forests are home to a wide variety of wildlife including monkeys, deer, large cats, parrots, various rodents, and ground dwelling birds. Mammalian biomass tends to be higher in dry forests than in rain forests, especially in Asian and African dry forests. Many of these species display extraordinary adaptations to the difficult climate.

Dry forests are highly sensitive to excessive burning and deforestation; overgrazing and exotic species can also quickly alter natural communities; restoration is possible but challenging, particularly if degradation has been intense and persistent. Degrading dry broadleaf forests often leaves thorny shrublands, thickets, or dry grasslands in their place.

2.1.3 Tropical (and subtropical) coniferous forests

Tropical (and subtropical coniferous) forests are located in regions of semi-humid climate at tropical (and subtropical latitudes). Most tropical and subtropical coniferous forest eco-regions are found in the Nearctic and Neotropic ecozones, from Mexico to Nicaragua and on the Greater Antilles, Bahamas, and Bermuda. Other tropical (and subtropical) coniferous forests eco-regions occur in Asia.

2.2 Global forests resources

This chapter is largely based on FAO (2007).

The world has just under 4 billion hectares of forest, covering about 30 percent of the world's land area. Forests are unevenly distributed around the world: of 229 countries or other reporting areas 43 have forest area exceeding 50 percent of their total land area, while 64 have forest area of less than 10 percent.

Five countries (the Russian Federation, Brazil, Canada, the United States of America and China) together account for more than half the total forest area (Figure 2.2).

Deforestation continues at an alarming rate of about 13 million hectares a year. At the same time, forest planting and natural expansion of forests have significantly reduced the net loss of forest area.

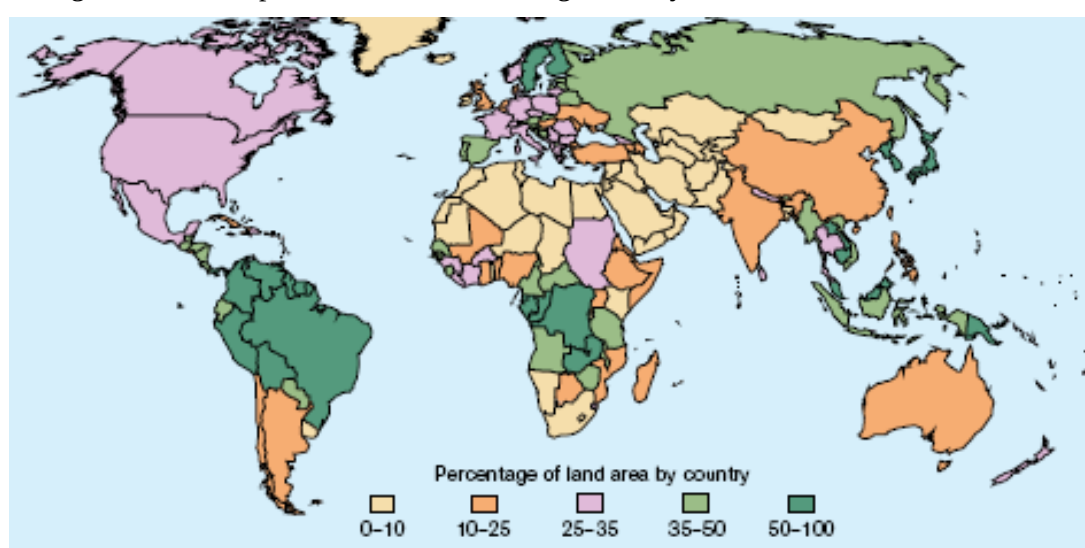


Figure 2.2: Percentage of forest land area by country in 2005 (FAO 2007, Figure 65)

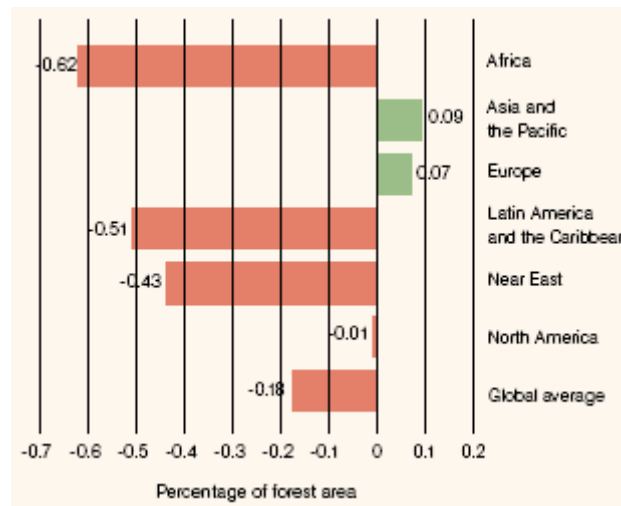


Figure 2.3: Percentage of forest land area by region in 2005 (FAO 2007, Figure 65)

Over the 15 years from 1990 to 2005, the world lost 3 percent of its total forest area, an average decrease of some 0.2 percent per year (Figure 2.3).

From 2000 to 2005, the net rate of loss declined slightly. However, the net forest loss remains 7.3 million hectares per year or 20 000 ha per day.

Globally, more than 11 percent of total forest area has been designated primarily for conservation of biological diversity. Globally, 36 percent of forests are categorized as primary forests (forests of native species in which there are no clearly visible indications of human activity and ecological processes are not significantly disturbed). The leader is Latin America and the Caribbean (75 percent), followed by North America (45 percent).

At the global level, an estimated 6 million hectares of primary forest are lost or modified each year. Nine of the ten countries that account for more than 80 percent of the world's primary forest area lost at least 1 percent of this area from 2000 to 2005, led by Indonesia (13 percent loss in just five years), Mexico (6 percent), Papua New Guinea (5 percent) and Brazil (4 percent).

As would be expected, the countries with the largest forest area represent the bulk of the world's wood volume (based on total growing stock) (Figure 2.4).

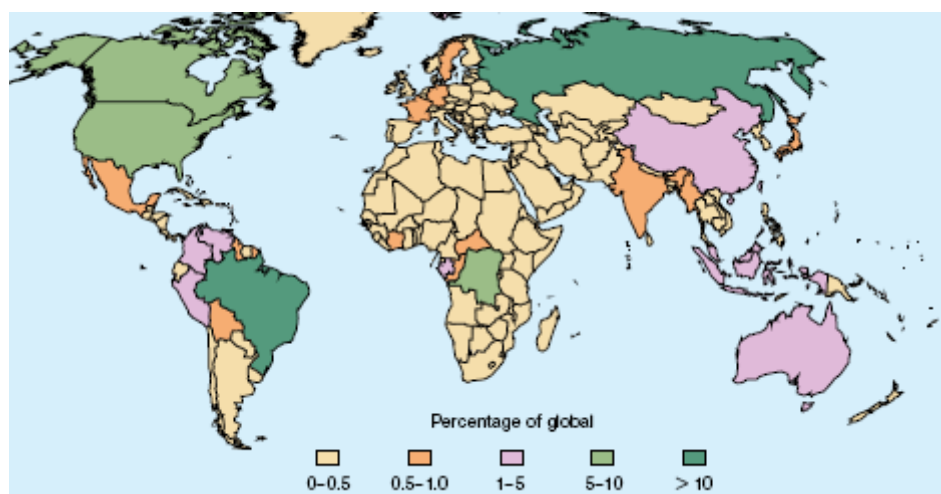


Figure 2.4: Share of global wood volume in forests 2005 (Source: FAO 2007, Figure 70)

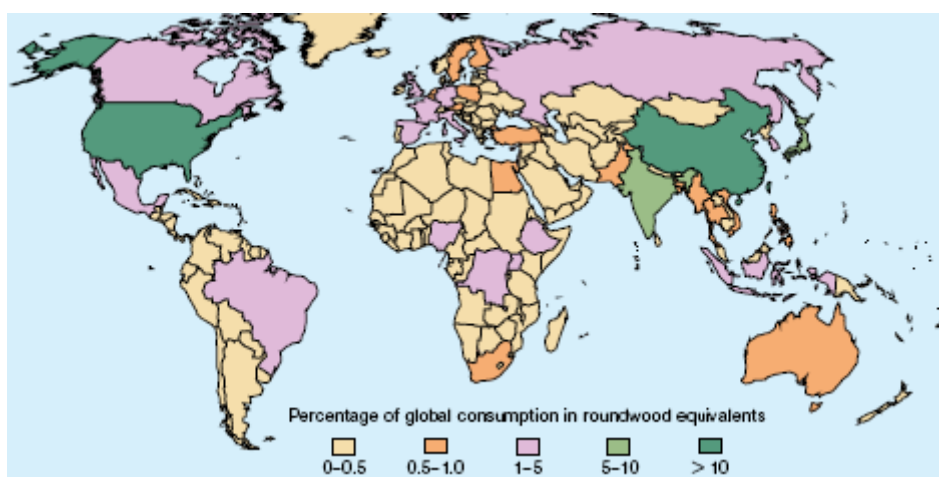


Figure 2.5: Consumption of wood products 2003 (Source: FAO 2007, Figure 74)

However, growing stock per hectare varied among regions, owing primarily to climatic and other ecological differences. In 2005, the area of forest designated for wood production as one of the management objectives was 50 percent. About 34 percent of the world's forest area was designated primarily for production.

The area of productive forest plantations increased by 2.5 million hectares between 2000 and 2005, indicating that a larger portion of wood removals may come from forest plantations in the future.

Primary forest products (roundwood, including industrial roundwood and fuelwood) represent a relatively large share of the forest sector's value in Africa, Asia and the Pacific and Latin America and the Caribbean. In contrast, wood-processing industries and pulp and paper account for the lion's share of the value of the sector in more developed regions.

Trade of forest products among countries is increasing. A positive net trade balance indicates that the value of exports exceeds the value of imports. Throughout the period from 1990 to 2004, Asia and the Pacific continued to be the major net importer of forest products. North America was, for many years, a net exporter, but in recent years it has become a net importer as well. The trend for Europe is the opposite of that for North America; today Europe is the leading net exporter of forest products (Figure 2.5).

2.3 Regional tropical forest resources

The following chapters are largely based on FAO (2007). They cover:

- Latin America and the Caribbean
- Africa
- Asia and the Pacific region

It should be noted that these regions do also cover non-tropical areas; a detailed assessment on tropical forest resources is not available as the respective ITTO assessment is limited to ITTO member countries.

2.3.1 Latin America and the Caribbean

The Latin America (also covering non-tropical areas) and the Caribbean region boasts abundant forest resources – about 47 percent of the land – and accounts for 22 percent of the world's forest area. The annual rate of change of forest area from 2000 to 2005 was -0.51 percent, compared with -0.46 percent

during the 1990s. From 1990 to 2005, Latin America and the Caribbean lost about 64 million hectares of forest. During this period, forest area increased by 11 percent in the Caribbean and declined by 19 percent in Central America and 7 percent in South America. Forest area declined from 51 to 47 percent of the total land area in Latin America and the Caribbean during 1990–2005. The total area of other wooded land was stable, accounting for 6 percent of total land area.

In the Latin America and the Caribbean region, plantations account for 1.4 percent of total forest area¹. Although this is a relatively small figure, plantations are increasing at the rate of about 1.6 percent per year.

From 2000 to 2005, net forest area continued to decline in Central and South America. The leading cause of deforestation was the conversion of forest land to agriculture. Within the region, the largest area loss was in South America, while the largest percentage loss of forest area was in Central America. Forest area increased in Chile, Costa Rica, Cuba and Uruguay, and forest plantations increased throughout the region.

Primary forests account for 70 percent of the region's forest area and 56 percent of the world's primary forests.

About 12 percent of all forest area in the region is designated primarily for timber production, compared with a global average of 32 percent. While the difference is significant, not all countries interpret this designation in the same way. Brazil reported only 5.5 percent of its forests in this category, bringing down the regional average. In contrast, Uruguay reports 60 percent of its forest area as designated for production, Chile 45 percent, Honduras 42 percent, the Bolivarian Republic of Venezuela 38 percent and Peru 37 percent.

At the other extreme, Bolivia, Costa Rica and Nicaragua reported zero percent forest area designated for production; these countries included their production forests under the category "multiple purpose".

Growing stock as a whole is declining, which is to be expected given declining total forest area. However, in the Caribbean it is increasing, along with forest area. Growing stock per hectare is relatively stable in Central and South America, and increasing in the Caribbean.

For the region as a whole, growing stock is about 30 percent of the global total (compared with 22 percent of the global forest area), and per hectare it is 29 percent higher than the global average. According to this parameter, Latin America and the Caribbean forests are significantly more productive than those of the world at large.

In Central America and the Caribbean, by far the majority of wood removed from the forest (Figure 38) is used for fuel (90 and 82 percent, respectively). In South America, the use of wood for fuel declined sharply in the 1990s. It continued to decline, but at a slower rate, from 2000 to 2005, while industrial roundwood continued to increase throughout the entire 15-year period. In 2005, the use of wood for industrial purposes surpassed the use of wood for fuel for the first time. It will be interesting to see if this trend continues: there are reports that the use of wood for fuel (including biofuels for motor vehicles) is increasing in response to the rising cost of fossil fuels.

Non-wood forest products (NWFPs) are also significant, but the lack of available data at the regional level is such that it is not possible to draw meaningful conclusions regarding trends.

2.3.2 Africa

The estimated forest area for Africa in 2005 is 635 million hectares, accounting for about 16 percent of global forest area. Net annual forest loss is about 4 million hectares for the period 2000–2005. This

¹ globally, forest plantations represent about 4 percent of total forest area.

amounts to almost 55 percent of the global reduction in forest area. However, the reported forest cover is distributed unevenly among the different sub-regions and countries.

A significant share of net forest loss is reported from those countries with the greatest extent of forests. For example, Angola, the United Republic of Tanzania and Zambia together account for a majority of the forest loss in East and Southern Africa. Available information also indicates a high rate of forest loss in Zimbabwe, estimated at 1.7 percent per year, far above the average of 0.7 percent for all Southern Africa. In Northern Africa, the Sudan alone accounts for most of the forest cover and for 60 percent of the forest reduction. In West and Central Africa, Cameroon, the Democratic Republic of the Congo and Nigeria together account for most of the loss.

Africa also has more than 400 million hectares of “other wooded land”, with scattered trees but not enough to be defined as “forest”. Data on the extent and growing stock of other wooded land are weak, but the extent continues to decline.

Africa’s total area of forest plantations – a subset of planted forests defined as those consisting primarily of introduced species – is about 13.0 million hectares. Approximately 2.4 million hectares (18 percent) forest plantations are planted for protective purposes; the remainder are planted to produce wood, particularly industrial roundwood and fuelwood. Most forest plantations are in Northern Africa, which is dependent on plantations because of the scarcity of natural forests. Southern Africa has developed a globally competitive forest industry almost entirely based on planted forests.

Since 1990, forest cover in Africa has been declining at one of the highest rates in the world (together with Latin America and the Caribbean). However, the rate of loss has shown signs of declining slightly in the past five years.

There is evidence of an overall decline in the area of primary forests in the region, but some of the most important forested countries were not able to report on this parameter, especially in Central Africa. Thus, it is not possible to make a definitive statement regarding the magnitude of this trend.

Unfortunately, there are only a few countries in which forest cover is increasing or marginally improving, and most of these are the “low forest cover” countries of Northern Africa in which substantial efforts have been made to establish planted forests. Improvement has been reported in Rwanda and Swaziland as well, also largely resulting from increased planting.

The production of wood and non-wood forest products (NWFPs) is a very important function of African forests and woodlands and has great impact on socio-economic development. Some 30 percent of total forest area is designated primarily for production, compared with a global average of 34 percent.

Country data suggest a significant decline in total growing stock in almost all countries, although a few countries have registered an increase resulting from an increase in the area of forest plantations.

Another key issue for the productive functions of forests – given the declining trend in growing stock in most countries – is whether the level of wood removals exceeds the annual allowable cut. Almost 90 percent of the wood removals in Africa are used for fuel, compared with less than 40 percent in the world at large. For Africa as a whole, wood removals in 2005 were about 1 percent of growing stock. However there is considerable variation among regions, largely resulting from disparities in access to forest resources and the proportion of commercial species. For example, in West and Central Africa, the removal rate is about 0.06 percent of the estimated growing stock, while in Northern Africa it is over 7 percent.

In the absence of information on annual allowable harvests, it is difficult to conclude whether current removals are sustainable. The dominant use of wood in Africa is for fuel, and a large part of the demand is met from other wooded land and trees outside forests. Since market demand and forest access are key determinants of the intensity of wood removal, areas that are easily accessible are more intensively logged than remote ones.

2.3.3 Asia and the Pacific region

Forests and other wooded land together cover about one-third of the Asia and the Pacific region (including also non-tropical regions). Excluding the Russian Federation, forest area in 2005 was estimated at 734 million hectares, accounting for about 19 percent of global forest area. The region as a whole experienced a net increase in forest area of about 633'000 ha annually during 2000–2005. This is an important breakthrough, since the region had experienced a net loss of forest cover during the 1990s. The improvement was largely the result of an increase of more than 4 million hectares per year in China, which has been investing heavily in afforestation in recent years.

Bhutan, India and Viet Nam also increased their forest area from 2000 to 2005. However, most other countries experienced a net loss. Southeast Asia experienced the largest decline in forest area, with an annual net loss of forests of more than 2.8 million hectares per year, about the same rate as had occurred during the 1990s. The greatest forest loss occurred in Indonesia, almost 1.9 million hectares per year, followed by Myanmar, Cambodia, the Philippines, Malaysia and the Democratic People's Republic of Korea.

During the first five years of the twenty-first century, the variation among Asian countries in the net rate of change of forest area was dramatic; this variation is much more pronounced in Asia and the Pacific than in other regions. Several countries are losing forests at rates exceeding 1.5 percent per year; these are among the highest rates of loss in the world. On the other hand, forest area is increasing considerably in several countries, especially in China and Viet Nam.

Other wooded land is extensive, accounting for 13 percent of the land area in the region. The overall trend in other wooded land is downward, both in Asia and the Pacific and in the world, despite an increase in Southeast Asia. However, reporting on this category is not fully consistent from one country to the next, and it is difficult to monitor this category with remote sensing, so significant conclusions have not been drawn from the data.

All sub-regions within the broader Asia and the Pacific region experienced a substantial increase in forest plantations during 2000–2005, continuing the trend from the 1990s. China led the way; Viet Nam, India, Indonesia, the Republic of Korea, Myanmar and the Lao People's Democratic Republic have also made significant investments in forest plantations in recent years.

Net forest cover in Asia and the Pacific is increasing. This net increase at the regional level is built mainly on large investments in forest plantations in several countries. However, the growth in plantations does not negate the continued loss of natural forests.

In East Asia, the area of primary forests is fairly stable, having declined slightly in the 1990s and increased slightly since 2000. In South Asia, a negative trend has continued and accelerated over the past 15 years. In Southeast Asia, the negative trend is consistent and very disturbing, exceeding a 2 percent loss of primary forest per year. In Oceania, a recovery of primary forests in the 1990s has been replaced by a negative trend since 2000.

In Asia and the Pacific, 37 percent of the total forest area is designated primarily for production, as compared with the global average of 34 percent. At the regional level, the extent of forests designated for production was fairly stable in the 1990s, but has declined in the past five years. The downward trend has occurred both in Asia and the Pacific and in the world at large. However, it is difficult to ascertain whether this is a negative trend. It may be a sign that more area is excluded from productive purposes and set aside for conservation, or it may be an indication that productive forests are being cleared to convert land to non-forest uses.

Growing stock is another indicator of forest productivity. Country data suggest a decline in total growing stock in many countries, with the exception of countries with large investments in forest plantations. The net result at the regional level is a modest decline in total growing stock in cubic metres and in cubic metres per hectare.

Regarding trends in wood removals, about 40 percent of the wood in the region is used for fuel, the same as the global average. However, the importance of wood for fuel is highly variable throughout

the region: in South Asia, 89 percent of wood is used for fuel, about the same as in Africa; the figure drops to 64 percent in Southeast Asia, 33 percent in East Asia and 16 percent in Oceania.

For Asia and the Pacific as a whole, wood removals in 2005 were about 0.76 percent of growing stock, higher than the global average of 0.69 percent, but lower than Africa at 0.90 percent.

Within the region, the highest rates of wood removals as a percentage of growing stock are in East Asia and Oceania, 0.87 percent. The lowest rate is in Southeast Asia, 0.61 percent, whereas South Asia is at 0.76 percent, equal to the regional average. Variations among regions and sub-regions result from such factors as access, proportion of commercial species, effectiveness of management controls and the supply and demand for wood.

In consonance with global trends, productive forests in the Asia and the Pacific region have declined in the recent past. This trend is further reflected in terms of growing stock and wood removals, both industrial roundwood and fuelwood. However, in the absence of information on annual allowable harvests, it is difficult to establish if current removals are sustainable. Because market demand and forest access are key determinants of the intensity of removals, areas that are easily accessible are more intensively logged than remote ones.

2.4 Socio-economic, biological and cultural importance of tropical forests

The socio-economic, biological and cultural importance of tropical forests can only be sketched very roughly within the framework of this report (based on The World Bank 2001):

- For many developing countries and their populations, tropical forests are an important and potentially sustainable source of revenue, thus contributing to poverty reduction. While timber and its derivatives remain the main products, trade of other forest products (so called nonwood forest products) is growing in importance. Its diversity holds a big potential for the future.
- 190 million m³ of tropical timber are produced each year worldwide, of which 40 million m³ are exported. For Indonesia for example, the trade and use of tropical timber and other forest products, including exports and national consumption represents 15% of its gross national product.
- For its inhabitants, tropical forests represent not only the main means of subsistence but also a fundamental socio-cultural environment. Thus the conservation of tropical forests plays a crucial role in strengthening ethno-diversity.
- 90% of the 1.2 billion people who live in extreme poverty depend on forests to satisfy aspects of their livelihoods. 350 million people living in or close to forests obtain the major part of their revenue and food from forest resources.
- 550 billion tonnes of CO₂ are stored as biomass in tropical forest ecosystems. The deforestation of 12 million ha/year, implies an equivalent surface area of tropical soils which is subject to reduced fertility. In addition to that, most of the carbon emissions of developing countries come from deforestation contributing to 10 to 30 percent of global carbon emissions.
- Tropical forests constitute the main reserve of terrestrial biodiversity. Nearly 90 percent of terrestrial biodiversity is found in the forests of developing countries. The economic potential of conserving this biodiversity is undervalued and most threatened by inadequate management.
- Through their different environmental functions (e.g. climate regulation, CO₂ fixation, protection of water springs and soils) forests provide inestimable services, which are currently under-valued economically and insufficiently compensated.

2.5 Deforestation of tropical forests

The following section is largely based on Roper (1999).

Deforestation is the permanent loss of forests to other land uses such as agriculture, grazing, new settlements, infrastructure, and dam reservoirs (see chap. 2.5.3 for forest degradation). Tropical deforestation is now widely recognized as one of the most critical environmental problems facing the world today, with serious long-term economic and social consequences. Largely overlooked by the developed countries and the urban dwellers of the developing countries until the 1980s, deforestation has received much attention in recent years with regard to its effects on biodiversity and climate change.

2.5.1 Causes for deforestation

Deforestation is the product of the interaction of the many environmental, social, economic, cultural, and political forces at work in any given region. The mix of these forces varies from decade to decade, and from country to country. As a consequence, generalizations are dangerous. In most cases, deforestation is a process that involves a competition amongst different land users for scarce resources, a process exacerbated by counter-productive policies and weak institutions. It creates wealth for some, causes hardships for others, and almost always brings serious consequences for the environment.

This section discusses four aspects of the causes of deforestation:

- I. the predisposing conditions,
- II. the direct causes,
- III. the indirect causes, and
- IV. the role of forest exploitation and plantation development for the loss of natural forests.

I. Predisposing conditions

The predisposing conditions create an environment where deforestation can occur. The direct causes are the most visible, the most easily identified and are readily associated with the agents of deforestation. They are driven by the other less visible, socioeconomic forces -- the indirect causes.

- **Population growth:** Without a doubt, one of the most important predisposing conditions that underlies tropical deforestation and many of the world's other problems related to achieving sustainable development is our growing population.

Approximately 4.5 billion people, or 75 per cent of the world's population, live in the developing countries and a 1,000 million of them live in abject poverty. Most of those countries are in the tropics where deforestation is a serious problem (FAO 1998). Furthermore, an estimated 2.8 billion live in rural areas and are dependent on agriculture to meet their basic needs. The exact number of people who live by clearing the forest to plant subsistence crops is not known, but the accepted figure is at least 500 million people or about 1 person in every 12 on the planet.

- **Poverty:** Another predisposing condition of deforestation is poverty, particularly poverty in rural areas. Although poverty is not a "cause" of deforestation, it is a condition of life that the majority of people in this world must endure. While greed and power can be the motivations of some groups in society that deforest, survival and the desire to escape from poverty is what drives most people.

The rural poor have very few options. There are few prospects of off-farm employment in either the urban centers or the rural areas. For those opportunities that do exist, there is intense competition for the few jobs available. Illiteracy further limits the options of many because they do not have the basic tools needed to pursue other economic alternatives to subsistence

farming. In some cases, people migrate from the overpopulated, depressed regions to the forest frontier in search of a more prosperous, more secure life. Hand-in-hand with poverty comes food insecurity and chronic undernourishment. With few alternatives available to them, the rural poor look to the forests as a short-term solution to their economic problems.

- **Greed and the quest for economic and political power:** Greed and the quest for economic and political power are important underlying forces. Individual and corporate greed that seeks excessive profits at the expense of human suffering and environmental degradation can be witnessed in the actions of many of the agents of deforestation. Unregulated land uses and monopolistic national markets favour the politically influential at the expense of the majority. This can be manifested in competing land uses that favour export oriented agricultural crops or exploitative logging practices.

II. Indirect Causes

- **Fiscal and Development Policies** - Government policies outside the forest sector have profound impacts on the forest resource, as do international policies on debt repayment, structural adjustment, and trade. Structural adjustment programs have encouraged the expansion of foreign exchange-earning export crops, which have in turn encouraged the liquidation of forest capital either by accelerating timber harvesting or by converting forests to agricultural uses. The expansion of agricultural cash crops means that either forests are cleared directly for these crops or subsistence farmers are displaced for them, forcing the farmers to relocate to the forest where they practice slash-and-burn agriculture. Incentives (e.g., low interest rates or tax exemptions) to industries that would otherwise be less economical, or even uneconomical, have permitted them to prosper at the expense of forests when they couldn't otherwise.

Government-sponsored colonization schemes have been used as "development" projects by many governments. Sometimes they have been officially sanctioned by governments and sometimes they have occurred more spontaneously. They have been attractive to governments because they allowed them to avoid the politically sensitive issues of population control and land reform, relieve the pressure of overcrowded and underserved urban areas, defer otherwise needed investments in urban infrastructure, and avoid investments in agricultural research and extension to increase agricultural productivity on existing arable lands. Many countries have used colonization schemes as a way of asserting national sovereignty on their frontiers.

Economic structural adjustment and macroeconomic reform programs being implemented in many countries have the potential to be a serious threat to tropical forests. Economic reforms have aggravated unemployment in some sectors, causing greater poverty which has, in turn, motivated people to migrate to forested lands to practice slash-and-burn farming. The greater emphasis on exports has, at times, resulted in unsustainable timber exploitation and the encroachment of commercial agriculture upon forested lands. The World Bank and some of the bilateral donor agencies have been advocating the privatization of public resources in the structural adjustment programs and have made it conditional for loan approval. The privatization of state forest resources favours those management alternatives that can produce a short-term economic gain for the new owners whether they be local governments, communities, or the private sector. Protection forests or forests that are "rich" in non-monetary values like soil conservation are held in very low esteem in such a market-driven environment.

- **Land Access and Land Tenure** - In most developing countries, the arable land base cannot support the growing population. First, the amount of land suitable for farming is limited. The real arable land that can sustain long-term cropping is, for the most part, currently under cultivation. Increases in agricultural production can come from increased productivity through the use of improved technology, but they cannot come from extending the land under cultivation into forested areas because there are no large "reserves" of unused forested land suitable for

farming. Second, as the farming population grows and the land passes on from generation to generation through inheritance, the individual farm plots become too small to be economical. Third, much of the truly arable land is held by large landowners or by corporations and, therefore, is not accessible to the majority of the farming population who really need it. In many countries, particularly in Latin America; large landowners -- latifundistas -- have traditionally controlled most of the farming land, a bad situation made worse in the second half of the 20th century when many small farms were bought out to become more economically viable. The introduction of new agricultural pesticides and fertilizers and the greater mechanization of farm labour shifted the profitability in farming to those landowners who had the available capital to invest. The small farmers were displaced and often went to the forest frontier to start over again.

Land tenure has an important influence on people's attitude towards land use. The vast majority of the world's slash-and-burn farmers do not have formal land title -- at best they have customary rights, at worst no rights at all. Without some guarantee that the land will remain theirs, farmers have no incentive to invest in making it more productive. Under these circumstances, clearing the forest and planting annual crops for a few seasons before moving on to clear more land is a logical farming strategy. Governments are either unwilling to title state lands to small farmers or their land titling procedures are so complicated and so costly that small farmers find it impossible to obtain legal title. The lack of ownership excludes them from obtaining credit for much needed farm inputs and discourages any long term investment that could lead to increased productivity, prosperity, and enhanced well-being. The short term alternative is to slash-and-burn the forest.

In many countries, settlers must clear the land to exercise their tenure rights. In this case, deforestation is considered an "improvement" to the land and an expression of the occupant's good faith in developing the property.

Tree tenure systems can also discourage the planting and tending of tree crops as an economic alternative to agriculture. Some countries like the Dominican Republic and Guinea have had laws that extend state ownership to all trees and forests whether they be on private property or state land. When tree ownership rests with the state, there is no incentive for the rural population to invest their labours in forest management because the benefits derived are only enjoyed by the government. In fact, this situation has encouraged deforestation because many farmers illegally removed the trees on their property so there would be no government interference in the way they used their land.

Market Pressures - Often mentioned as causes of deforestation are the demand for forest products and the demand for other goods (mostly food) that are produced on deforested lands.

Indeed, the demand for forest products continues to rise as population and affluence grow. The national demand for forest products within the countries where deforestation is occurring is a much more important cause of deforestation than the demand for these same products on the international markets. This generalization varies from region-to-region and from country-to-country within each region.

In the agriculture sector, the importance of export crops as a driving force behind deforestation is, again, difficult to generalize. Rice is the staple food crop in Asia, but it is not an export commodity in most Asian countries that are losing their tropical forests. For example, Indonesia and Malaysia are net importers of rice, and of the important deforesting countries only Thailand is a significant exporter of about 40 per cent of its 1996 production (FAO 1998). In Central America, the single most important crop of the slash-and-burn farmers is maize. All countries in the region are net importers of maize, which indicates that all of the forest land converted into maize production is for internal consumption only, not for export. In 1996, Central America's production of 2.87 million metric tons had to be supplemented by importing 1.07 million metric tons of maize and maize products to feed its population. In Indonesia, exports of palm oil have been more or less constant at 6 to 8 per cent of total production over the

last decade, although the area under palm plantations has steadily grown for the same period. The conclusion to be drawn is that most of the demand for palm oil is coming from within Indonesia, driven by its large and still growing population.

The situation in the livestock sector is very similar.

Like the growth in forest products, the growth in agricultural production and its consequent impact on deforestation has been more a response to the growing national markets than an attempt to satisfy the international demand. While international market pressures are important factors in understanding the causes of deforestation, their importance should never be overemphasized. The causes of deforestation, like the solutions for controlling it, are to be found within the borders of each of the countries affected. International trade offers some leverage to halt or reverse deforestation in some countries, but it is far from a panacea.

- **Undervaluation of Natural Forests** - In economic terms, there is little understanding of the value of the goods and services provided by tropical forests or of the real costs of forest management being borne by resource users. As a consequence, forests are undervalued and play a less significant role than they should in the decisions affecting resource allocations, development priorities, and land use. They are, therefore, more susceptible to being converted to other land uses which are perceived to be more beneficial. Tropical forests are undervalued because:
 - they produce many different products that are consumed in many unrelated markets often outside the cash economy, thereby creating the perception that they are less important;
 - they produce many non-market goods (e.g. forest food, game, resins, fibres) and environmental services (e.g. climate control, water regulation, soil conservation) that do not enter into the national economic accounts;
 - "downstream" benefits of "upstream" conservation are enjoyed but not paid for by beneficiaries;
 - the harvest cycle (rotation) of natural forests in the tropics is very long compared to agricultural crops, even outside the realm of conventional commerce;
 - the establishment of natural forests incurs no direct costs for the exploiter hence; they are viewed as "free" commodities;
 - there is still much unknown about the potential value of the forests, a consequence of the lack of systematic research,
 - market knowledge is imprecise, except for the traditional timber products,
 - prices are often set by monopolies (government or private sector) and do not necessarily reflect society's value of forest products and services,
 - forests are important to the rural poor, a social group that has little political influence and therefore little economic influence.

Forests that are perceived to have low value will be cleared and replaced by other more attractive land uses. It is important that people's perceptions be based on as complete an understanding as possible of the true value of all the goods and services that forests provide.

In recent years, much has been said and much written about the potential of both ecotourism and pharmaceutical research as saviours of the tropical forests. While these can be important alternatives to slash-and-burn farming at a very local level, they have a limited potential to impact global land use. The magnitude of the challenge and the need for meaningful benefit-sharing with the hundreds of millions of persons involved dwarfs their limited potential to generate grassroots benefits. On the other hand, the economic potential of the forests' carbon sequestration and storage capacity could be enormous under the joint implementation agreements that are coming out of the climate change convention.

- **Weak Government Institutions** - Many institutional failures have been identified as contributing factors to deforestation. In most countries, forestry departments have a low status within governments relative to competing land uses, reflecting the economic power base in the countries. Typically, forestry departments are handicapped by poorly paid staff, inadequate budgets, lack of staff, and lack of staff training. As a consequence, the departments have been ineffective in successfully putting forth pro-forestry arguments to the political decision-makers and to the public at large. Even when there are adequate policies and legislation in place, the weakness of the departments in enforcing the law, resisting political pressures, and maintaining a field presence has bred contempt and indifference for the law.

Corruption in government has had a disastrous impact on forest conservation. It has been evident at all levels of government and includes such actions as influencing the granting of timber concessions and timber-cutting permits, giving approval to clear the forest for ranching or agriculture, undergrading the value of timber exports, condoning illegal logging, even allowing the cutting of endangered tree species. The end result has been that government decisions have been taken with an eye to the personal benefits for the bureaucrats taking the bribes rather than to sustainable forest management, a loss in government revenues that could otherwise be used to fund sustainable forestry, and a lack of incentives for the private investors to practice sustainable forest management. Corruption also undermines the respect for forestry departments at large as administrators of the law. This has had a direct impact on people's attitudes towards the forestry departments' efforts to stop deforestation.

- **Social Factors** - Faced with political decisions about urban migration, food production, agrarian reform, employment generation, national security, economic structural adjustment, and all the other issues that demand their attention; many governments have opted to ignore deforestation. Deforestation has been a safety valve that has helped to take the socioeconomic pressure off other areas, thereby avoiding political turmoil that would inevitably come. While politically expedient, this has been a very short-sighted approach that is not in the long term interest of anyone.

In many cultures, "common" resources like publicly owned forests are not looked upon as opportunities for collective management of valuable resources. They are perceived as "free" commodities to be used by anyone, free from government regulation. Rather than being managed for the common good, they are abused and neglected. Without a sense of ownership, there is no incentive to manage the resource.

In most countries, forestry development has been characterized by centralized planning and management of the resource. Government departments have been created to act as the public's custodian of the trees and the land upon which they grow. Forestry department activities like tax collection and cutting control have usually been more important than extension and cooperation with rural communities. As populations have grown and their demands on the resource increased, governments have begun to look for new, more democratic approaches to managing forests.

Many countries lack a "forest culture", an appreciation by the population of the value of forests to their society and a tradition of managing the resource for the collective benefit of all. Forests are often looked upon as impediments to development. In other societies, communities have traditionally managed their forests but recent changes in their political systems have destroyed the custom.

One of the lessons of the last 30 years of trying to contain deforestation is that the people who are meant to benefit from the forests must be full partners in the process of identifying and implementing solutions.

III. Direct Causes

- **Slash-and-Burn Farming:** By far the most important agents of deforestation globally are the slash-and-burn farmers who live in or on the margins of all of the world's tropical forests. It is estimated that small farming families account for nearly 2/3 of all deforestation (Rowe, Sharma *et al.* 1992). "Slash-and-burn" farming includes a diverse collection of farming systems from long fallow shifting cultivation to short fallow shifting cultivation to forest pioneer farming. Unlike traditional farming methods that were used in harmony with the forests' recuperative capacity, current slash-and-burn farming depletes the very soil resource upon which all agriculture and forestry depend. One of the strong commonalities of all slash-and-burn farmers is that they are among the poorest, most marginalized groups of their societies and have little or no influence on the important land use policy decisions made in their countries.

Rather than the villains of the deforestation cycle, small farmers are its victims. Prisoners of illiteracy and endemic poverty, and driven by the lack of access to arable lands and the lack of alternative employment opportunities, subsistence farming families must survive by clearing the forests to plant their crops. For most, it is a day-to-day fight for survival with their family's future dependent on the fortunes of the next uncertain crop. Their labours are rewarded by only meager cash incomes that keep them well below the poverty line.

- **Commercial Agriculture:** In contrast to subsistence farming, commercial or plantation agriculture is often agribusiness practiced by corporations. Important plantation crops in the tropics include sugar, palm oil, natural rubber, coffee, cacao, and tropical fruits (bananas, citrus, etc.). Commercial agriculture's role in deforestation is two-fold. First, agribusiness can indirectly result in deforestation. Commercial farms occupy the best, most fertile agricultural soils located in the valleys. As a consequence, this land is not available to the growing rural population that depends on agriculture for their subsistence. Without access to farmland in their immediate area, farming families have had to relocate to less fertile, less productive forested land. In Honduras in the 1970s, thousands of small farmers and ranchers were displaced from the north coast valleys to make way for the establishment of oil palm cooperatives. They were pushed onto the steep forested slopes and benchlands and proceeded to clear them for farms and pasture.

Second, agribusiness can be a direct cause of deforestation. Through a concession agreement, land purchase, or an informal land occupation; companies take possession of forested land with the intention of converting it to another use. For example, the area of oil palm plantations in Indonesia has skyrocketed in the last 20 years. This has been at the expense of the natural forests and of the fallow brush that comes in after slash-and -burn farming. The Indonesian experience with oil palm has been replicated in many other tropical countries in recent years. Examples of other important agricultural tree crops that are cultivated on forest lands include coffee, cacao, citrus, and rubber.

- **Cattle Ranching and Livestock Grazing:** Cattle ranching, particularly in Latin America, is a major cause of deforestation. Ranchers either occupy large tracts of forests and clear the land themselves or they buy the "improvements" made by small farmers. Traditionally, ranchers favoured the more easily managed range and pasture lands of the dry forest zones, but for the last four decades there has been intensive clearing of the moist tropical forests in both South America and Central America. Open-range grazing as is practiced in the dry woodlands and savannas of Africa can be a major contributor to deforestation when herd populations exceed the carrying capacity of the range. It can also seriously degrade the composition and quality of the forest when practiced too intensively.

In conclusion, cattle ranching has been an important direct cause of deforestation in the latter half of the last century, particularly in Latin America. The expansion of cattle pasture is closely linked to slash-and-burn agriculture through land speculation in forest land.

- **Mining and Petroleum Exploration:** Mining and oil exploration are locally important to deforestation. Large mines like those of Carajás in Brazil and the Copperbelt of Zambia consumed vast quantities of indigenous woodlands to supply fuel to their smelting operations before plantations of fast-growing species were established. The impact of gold mining has been widely publicized, particularly placer mining in the Amazon, but its negative impacts have affected the indigenous peoples and the quality of the water more than the adjacent forests. Oil exploration activities, such as the clearing of the seismic lines in the forests of eastern Ecuador, not only destroy the forests but also open them up to colonization by subsistence farmers who follow the exploration crews.
- **Infrastructure Development:** The construction of new roads has a profound impact on the forest. The Trans-Amazonian highway opened up millions of square kilometres of previously inaccessible forest to colonization and expansion of the cattle industry. Main arteries were soon followed by secondary roads that penetrate deeper into the forest, eventually producing a wide swath of deforested land on either side of the road. All roads that are constructed with the purpose of providing better access to less developed regions within a country tend to push up real estate values for non-forest uses and encourage land speculation and deforestation.

Logging roads are among the most important types of access roads that facilitate deforestation. Hydroelectric development is another important factor in deforestation. Reservoirs flood forest lands and transmission line right-of-ways are cut out of the forest to carry the energy to consumers, causing permanent losses of forest cover. Forests are also encroached upon by industrial and residential development as populations grow and cities extend outward.

IV. Role of Forest Exploitation and Plantation Development

- **Fuelwood Collection and Charcoal Making:** Fuelwood is the most important wood product in developing countries where it accounts for 80 per cent of all wood used. Even with predicted fuel substitution from electricity, kerosine, and propane, it is not expected that this dependence on fuelwood will change significantly before the end of the 21st century. Worldwide, nearly 3,000 million people use fuelwood as their principal source of energy, particularly in rural areas and particularly among the least privileged groups in society.

In many places, fuelwood collection, while not completely destroying the forests and woodlands, significantly impoverishes them and alters the habitat by the selective removal of preferred species. However, around urban areas there is often a ring of denuded land that has been stripped of all its combustible material by people trying to meet their basic energy needs. With very intensive collection over a long period of time, the original trees and shrubs lose their ability to coppice and die out, giving way to a different mix of plant species. Fuelwood is collected mostly from unregulated commons and, as a consequence, is very sensitive to overexploitation as the population increases.

- **Logging:** FAO (2006) reports that more than six million hectares logged annually in the tropics and that the rate of logging has doubled in the last 30 years. Like most forest sector statistics, these estimates are not precise due to the poor record keeping and lack of field supervision of logging operations. The greatest increase in activity can be found in Asia and Latin America while the annual area logged in Africa has remained somewhat constant.

Very few natural forests in the tropics are managed professionally. Logging in the tropics does not in any way resemble scientific forestry and is often characterized by a "cut-and-get-out" mentality in logging companies. Forestry uses science and management skills to manipulate the natural vegetation to favor the long term production of a selected number of goods and services. In contrast, most tropical logging involves the short term exploitation of only industrial wood products with no eye to the future of the forests. This approach has led many observers to the conclusion that sustainable forest management is not possible in tropical forests.

Although the intensity of logging is low in most tropical forests with most of the original timber being left standing, there is high felling damage and residual waste, no long-term regulation of the harvest, and poor natural regeneration of commercially useful species. The intensity of logging in South East Asia is much higher than in the tropical forests found in Latin America or Africa. The removal of high volumes per hectare has led to serious degradation of these forests, even causing their destruction in extreme cases where clear-cutting has been used. Poorly designed logging roads damage watercourses and cause severe soil erosion. The intrusion of men and logging machinery with the resulting changes in the forest ecosystem, displaces many forms of animal life, particularly birds and larger mammals. Environmentally appropriate silviculture systems have failed in the tropics, not for ecological reasons, but because they lack the appropriate policy and strong institutional frameworks in which to operate.

Without question, logging continues to be the principal cause of forest degradation in the tropics but not one of the principal causes of deforestation. However, there are examples of logging being the direct cause of deforestation. Intensive logging in South East Asia has resulted in the invasion of Imperata grass -- a noxious weed that excludes most other vegetation -- on thousands of hectares of once forested land. In this example, deforestation has occurred without the intervention of one of the competing land uses like farming or grazing.

In terms of its contribution to deforestation, the single most important failing of governments and forest products companies has been their inability to maintain a permanent forest estate. Sustainable forest management assumes that once the forest has been logged, it will remain a forest until the end of the rotation or to the end of the next cutting cycle and beyond. In most cases, this does not happen. When the logging is finished, the farmers, agribusiness agents, ranchers, and fuelwood collectors move in to clear the land for other economic uses. Previously inaccessible tracts of frontier forest are opened up by logging companies when they build new haul roads, open new skid trails, and remove a portion of the forest biomass, making it easy for the other land users to clear the remaining trees. In short, logging provides them access to the forests. Throughout the tropics, production forests are inadequately protected from this type of encroachment, despite the fact that management plans and concession agreements oblige both government and industry to do so.

Although usually well meant, many government policies in the forest sector are counter-productive in that they produce undesirable, unforeseen impacts that are detrimental to the sustainable development of tropical forests. For example, logging concession agreements are meant to be a tool to regulate the commercial exploitation of forests but they can have negative impacts on the resource and increase the susceptibility to deforestation. Concessions are given out for timber extraction without consideration to the other goods and services produced by the forest nor the impact that exploitation can have on local people. Concessions are usually short term, often less than 10 years in duration and always less than the rotation of the timber crop. Without a long term commitment, the concessionaire has no incentive to protect the forest from encroachment or invest in forest management.

Stumpage, the tax the government charges loggers for buying public timber, is almost always lower than the real cost of forest management. This type of depressed pricing undervalues the resource and makes it appear less economically attractive to other land uses; i.e. it is an incentive to deforest. Low pricing encourages waste which in turn results in degradation of the forest and subsequent deforestation.

- **Tree Plantations:** Planted forest are of increasing importance for wood supply in the tropics. For the most part they are even-aged, single-species plantations. There has been a growing interest in Indonesia and Brazil in establishing plantations to produce fast-growing fibre of Eucalyptus and Acacia for the global pulp and paper industry. There are many issues concerning tree plantations that are not related to deforestation that will not be touched on here - e.g. sustainability, genetic impoverishment, soil depletion, danger of insect and disease.

Large tracts of heterogenous natural forests have been cut down in the past to plant more uniform, more easily managed monocultures. Tree plantations have the potential to produce substantial benefits in terms of supplying wood and fibre and sequestering atmospheric carbon that could potentially outweigh the costs of losing natural forest.

Carbon sequestration and storage is still an emerging issue and the values that will be assigned to carbon-sink plantations are still not known. In theory, the sequestration and storage values could substantially exceed those of the values of the wood and non-wood forest products. Universal standards will have to be developed to guide plantation establishment to ensure that natural forests are not destroyed in the name of ameliorating global warming. The vast areas of unproductive forest fallow found throughout the tropics could be made available for new plantations, making it unnecessary to disturb the remaining natural forests.

2.5.2 Agents of deforestation

In all geographic areas, slash-and-burn farmers rank high as the most important agents of deforestation. They occupy forest land to clear the trees to plant food crops. Other major agents of deforestation in the agriculture sector are ranchers who remove the forest to seed new pasture to graze their livestock and the commercial farmers who establish estate crops like rubber and oil palm. Secondary agents of deforestation include loggers, commercial tree planters, firewood collectors, mining and petroleum industrialists, and infrastructure developers.

There is considerable variation region to region and country to country as to which groups are the most important agents of deforestation. In Latin America, slash-and-burn farmers and cattle ranchers are at the front line of deforestation. In contrast, in Southeast Asia, commercial farming, logging, and oil palm plantations play a more significant role. The situation in Africa is a complex mixture of overgrazing in the dry forest zones, with slash-and-burn farming and high-grade logging in the moist forests of West and Central Africa (see Table 2.1).

However, due to the poor statistical basis, care must be taken in attempting to apply such generalizations to any specific country or geographic area.

Table 2.1: Main agents of deforestation (source: <http://www.rcfa-cfan.org/english/issues.12-4.html>)

Africa	Asia – Oceania	Latin America and Caribbean
1. slash-and-burn farmers	1. commercial farmers	1. slash-and-burn farmers
2. commercial farmers	2. slash-and-burn farmers	2. cattle ranchers
3. loggers	3. loggers	3. commercial farmers
4. livestock herders	4. commercial tree planters	4. loggers
5. refugees from civil disturbances	5. infrastructure developers	6. infrastructure developers

2.5.3 Forest degradation

It is not within the scope of this report to elaborate on tropical forest degradation. Unlike deforestation, degradation does not involve a change in land use. Nevertheless, it is a serious problem in the tropics. Probably more than 10 million hectares are degraded each year by the action of loggers, firewood collectors, and livestock herders. Although the land remains under forest use, its composition and biological functions are compromised by human intervention. The chief cause of forest degradation is exploitative logging, with its selective removal of tree species, felling damage to the residual forest, soil damage from road construction and log skidding, and the displacement of bird and mammal species. Logging roads provide new access for farmers and ranchers who come to clear and burn the

logged-over forests once the skidders and chainsaws have left. Forest degradation also occurs as a consequence of intensive fuelwood collection and overgrazing by livestock and wildlife. More research is required on the scale and impact of degradation on all types of forests.

2.5.4 Strategies to combat deforestation

Strategies to combat deforestation are as manifold as its causes and actors. Strategies include:

Protection and Sustainable Management of Remaining Tropical Forests

- Adopt sustainable forest management practices as defined by FAO; sustainable forest management aims to ensure that the goods and services derived from the forest meet present-day needs while at the same time securing their continued availability and contribution to long-term development. In its broadest sense, forest management encompasses the administrative, legal, technical, economic, social and environmental aspects of the conservation and use of forests. It implies various degrees of deliberate human intervention, ranging from actions aimed at safeguarding and maintaining the forest ecosystem and its functions, to favouring specific socially or economically valuable species or groups of species for the improved production of goods and services
- Improve the quality of resource information base on national and operational level, based on a concerted set of criteria and indicators for sustainable forest management as a sound basis for decision-making¹
- Support for protected area networks

Socioeconomic Development

- Improve productivity of subsistence agriculture and extensive ranching to mitigate the need for further forest clearances
- Promote the use of agroforestry for the same reasons
- Investigate and promote the use of non-timber forest products to expand the economic revenues from existing forests while incentivating its sustainable use
- Valorate forest services via payment schemes for environmental services such as carbon storage, hydrological services, soil conservation, biodiversity conservation, scenic beauty, etc. for the same reason.
- Expand the area of tree plantations to extend the basis for timber and fibre supply basis
- Restore degraded lands

Policy and Institutional Reform

- Enhance participation in decision-making and benefit sharing
- Reform government policies and strengthen institutions
- Support education and awareness building about the forest sector issues

Certification of sustainably managed forest has also been proposed as a means to combat deforestation. For tropical timber, the most important label to document the sustainably managed source of

¹ see, e.g. <http://www.cifor.cgiar.org/acm/methods/candi.html>

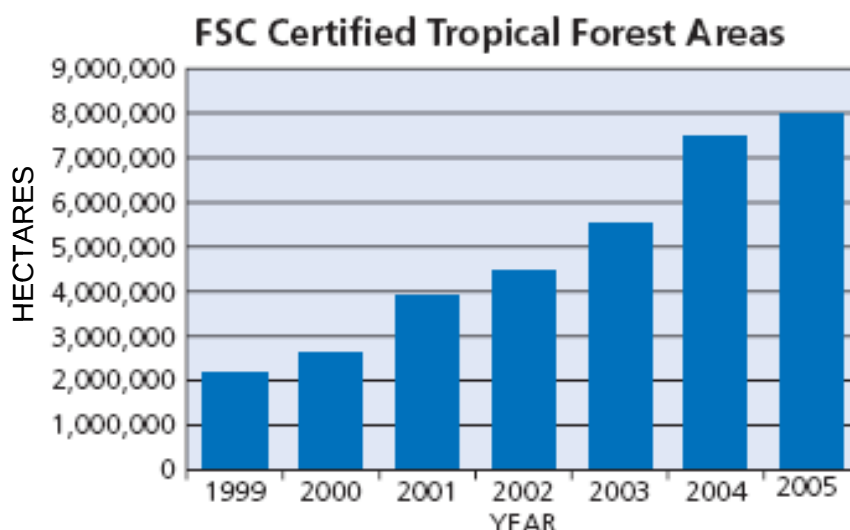


Figure 2.6: Tropical forest areas certified according to the Forest Stewardship Council's Criteria and Principles (source: http://www.fsc.org/keepout/en/content_areas/45/2/files/fs_south_web.pdf)

wood has been developed by the Forest Stewardship Council (FSC). FSC-labeled products guarantee that wood material originates from socially and environmentally responsible forest operations that fulfill the FSC Principles and Criteria¹.

FSC controlled wood standards allow manufacturers and traders to avoid wood that has been:

1. Illegally harvested.
2. Harvested in violation of traditional and civil rights.
3. Harvested in forests in which High Conservation Values are threatened by management activities.
4. Harvested from areas being converted from forests and other wooded ecosystems to plantations or non-forest uses.
5. Harvested from Forest Management Units in which genetically modified trees are planted.

Tropical forest areas certified according to the Forest Stewardship Council's Criteria and Principles have been increasing steadily over the last years (Figure 2.6).

Tropical timber has still a very small share of the global market for certified wood: in 2002 certified forest area in Asia, Latin America and Africa added up to only 8 % of globally certified forests (Eba'a Atyi and Simula 2002).

Yet, the original objective of combating deforestation in the tropics has not been widely realized. In addition to the very limited area of certified forests in the tropics, the problem is fairly straightforward: most deforestation in the tropics is caused by conversion to other land uses rather than by logging; certification is not cheap; and its successful implementation requires a solid institutional and governance platform. Moreover, it is yet to be seen whether consumers are willing to pay more for certified products on a large scale.

¹ http://www.fsc.org/en/about/policy_standards/princ_criteria

3 Tropical wood: production and global trade

3.1 Global production

Commonly harvested commercial tropical species per geographical region include¹:

- *Africa*: Iroko (African teak), Afara (Korina), Sapelli, Utile, Ayan, Ebony, Sipo, Okoumè (Gaboon), Moabi, Ayous, African mahogany (Khaya), Wengwe, Ozingo
- *Asia*: Ironwood, Amboyna, Kapur, Sepetir, White Seraya, Jelutong, Ramin, Meranti varieties, Teak
- *Latin America*: Mahogany, Virola, Balsa, Greenheart, Crabwood, Mora, Ekki, Brazil Nut

In addition to the material properties of wood in general (see Werner, et al. 2007), tropical wood is imported to industrialized countries for several reasons:

- Particular technical material properties, such as high fire resistance or high resistance to biological or photo-chemical degradation.
- Aesthetic attractiveness, as tropical wood is usually dark and of particular grain due to a lack or low formation of annual rings
- Attractive price compared to alternative materials.

The following text is largely based on the annual statistics by the International Timber Trade Organisation (ITTO 2006), which brings together the major producer and consumer countries of tropical wood².

Logs

Production of tropical industrial roundwood (logs) in ITTO producer countries³ totalled over 128 million m³ in 2004, a 1.5% decrease from 2003. Log production increased to 131 million m³ in 2005. The proportion of logs domestically processed in Africa increased slightly from 80% in 2004 to 82% in 2005. The Asian figure for domestic log processing averaged 92% over the same period. This reflects larger populations/domestic demand, growing economies and the emphasis on producing and exporting value-added products in this region. Latin American countries continued to process virtually all tropical logs harvested in 2004-2005.

Sawnwood

Tropical sawnwood production totalled nearly 41 million m³ in 2004, a 4% increase from 2003 levels. In 2005 sawnwood production rose to 41.3 million m³.

Veneer

Tropical hardwood veneer production in producer countries dropped by 14% to 2.9 million m³ in 2004. This was due mainly to sharp declines by Côte d'Ivoire, Gabon and Indonesia. Production recovered by nearly 5% to 3 million m³ in 2005, due primarily to production increases in the Philippines and Gabon.

Plywood

ITTO producer countries' tropical plywood production decreased by 5.2% in 2004 to 14.4 million m³. This was due mainly to a decrease in Indonesia, which in 2004 ceded its position as the world's largest producer of tropical plywood to Malaysia. Plywood production in producer countries was 14.5 million m³ in 2005.

¹ <http://www.itto.or.jp/policy/pds10/contents.html> for indicators on national level;
<http://www.itto.or.jp/policy/pds10/contents.html> for criteria and indicators of forest management unit level

² see for details: <http://www.itto.or.jp/live/PageDisplayHandler?pageId=233&id=224>

³ not including tropical timber production in ITTO consumer countries; see further down this chapter

Tropical wood from ITTO consumer countries

ITTO consumer countries¹ also produced substantial quantities of tropical timber products in 2004. China (2.7 million m³) and Australia (80'000 m³) together produced an estimated total of 2.8 million m³ of logs from their tropical regions. Consumer countries produced just over 1.8 million m³ of sawnwood, nearly 0.9 million m³ of veneer and 6.3 million m³ of plywood in 2004, all (with the exception of China and Australia) from imported tropical logs. ITTO consumer countries' production levels of all tropical timber products but plywood were stable or increased in 2005.

3.2 Global imports

Logs

Tropical log imports by ITTO consumer countries declined by 6% to 11.9 million m³ in 2004 due to a 9% reduction in China's imports, which fell for the first time in over a decade. However, log imports increased by 1.2% in 2005 to nearly 12 million m³ due to an increase in Japanese tropical log imports. China (still the world's top tropical log importer) maintained imports at 7.3 million m³ in 2005.

If imports by producing members are taken into account, total 2004 tropical log imports by ITTO members were 15.6 million m³, 3.1% less than in 2003.

The 2004 total log import figure is nearly 3.5 million m³ higher than total ITTO exports, with this gap decreasing to around 3.4 million m³ in 2005. This balance is presumably provided by non-ITTO log suppliers, although under-reporting of log exports, misclassification of imports, smuggling and/or statistical errors can also contribute to such gaps.

Major non-ITTO tropical log suppliers include Equatorial Guinea and the Solomon Islands, with exports estimated to average over 400'000 m³ per year each.

Japan's imports of tropical logs decreased 9% to 1.6 million m³ in 2004, but increased 5% in 2005. Despite this increase, Japan's imports have still nearly halved in the past decade due to its contracting economy, reduced supplies from Malaysia, competition from China for available log supplies, and its increasing reliance on softwood logs for plywood manufacture.

India, on the other hand, is now ITTO's second largest importer of tropical logs with imports up by 9% to over 3 million m³ in 2004. India, Thailand and the Philippines are the major ITTO producer country log importers, accounting for 94% of total producer imports of 3.8 million m³ in 2004. Both Thailand and the Philippines increased log imports in 2005, while India's were estimated to have remained stable.

Sawnwood

China also continued as the world's largest tropical sawnwood importer in 2004, up 4% to nearly 3 million m³. Thailand's imports (which more than halved in the Asian crisis of 1998) also increased by 30% to 1.8 million m³ in 2004, returning to pre-crisis levels. Japan's imports of tropical sawnwood decreased 23% to 378'000 m³ in 2004, but increased 15% to 434'000 m³ in 2005.

Imports of tropical sawnwood by all consumer countries increased by 5.9% in 2004 to 7.7 million m³ and remained stable in 2005. Increased imports by producers led total ITTO tropical sawnwood imports upwards by 7% to 11.1 million m³ in 2004. Total imports decreased to 10.6 million m³ in 2005 due to declines in consumer country markets.

¹ including the most important wood consumers, but excluding, e.g. Russia, Eastern Europe, Near East, non-tropical countries in Latin America and Africa

Veneer

Total ITTO tropical veneer imports decreased nearly 10% to 1.3 million m³ in 2004, but increased by 12.4% to over 1.5 million m³ in 2005. Korea remained the largest ITTO tropical veneer importer in 2004, with 236'000 m³. Korea's imports were stable in 2005. Mexico became ITTO's second largest tropical veneer importer in 2004, overtaking China and even with Taiwan P.O.C. with 175'000 m³, although the sources of these imports are unclear. Mexico's imports increased sharply in 2005. Meanwhile, China's imports fell sharply by 23% in 2004 to 98'000 m³ and remained stable in 2005 as it met its veneer needs increasingly via production from imported tropical logs.

The EU absorbed 357'000 and 345'000 m³ of tropical veneer in 2004 and 2005, over one-fifth of total ITTO imports. Japan imported 44'000 m³ of tropical veneer in 2004, a 10% increase from 2003 levels, before declining by 5% in 2005 to 42'000 m³. Formerly a major tropical veneer importer, Japan is now less significant than producer countries like the Philippines and Mexico.

Plywood

Tropical plywood importers are led by Japan with nearly 4.6 million m³, a 38% increase from depressed 2003 levels which fell due to new product safety standards. Imports continue to replace domestic production of tropical plywood in Japan due to reduced availability of tropical peeler logs and relatively low prices of imported plywood. Japan's imports made up 41% of total ITTO imports of nearly 11 million m³ in 2004. Tropical plywood imports by ITTO members remained stable in 2005.

3.3 Global exports

Logs

ITTO producer countries exported nearly 12 million m³ of logs worth \$1.6 billion in 2004, with Malaysia (the largest exporter) providing just over one-third of this volume, down from almost three-quarters of the ITTO total in the early 1990s. Producer log exports in 2004 were down 8% from 2003 levels and fell further to 10.9 million m³ in 2005, well under half the level exported just over a decade ago.

Sawnwood

Sawnwood exports by producer members jumped by nearly 31% to 10.5 million m³ (worth \$2.2 billion) in 2004, remaining stable in 2005.

Exports from the Latin American and Asia-Pacific regions fluctuated in 2004 and 2005, with African exports following a steady upward trend. Sawnwood exports from Malaysia increased by 10% to 2.8 million m³, benefiting from an Indonesian export ban.

Veneer

Veneer exports from ITTO producer countries increased by 8.1% in 2004 to slightly over 1 million m³, increasing a further 25.1% in 2005.

Plywood

Tropical plywood exports by producer members in 2004 declined by 1.8% to just under 9 million m³, worth nearly \$3.1 billion, with Malaysia (4.3 million m³) and Indonesia (3.1 million m³) accounting for 71% of this total. As for production, 2004 marked the first year when Malaysia's plywood exports exceeded those of its neighbour. Indonesia's exports remained stable in 2005, but Malaysia's increased to almost 4.7 million m³.

Exports by ITTO consumer countries

ITTO consumer countries also exported or re-exported substantial quantities of tropical timber in 2004, led by sawnwood and plywood 1.5 million m³ respectively. Log and veneer exports were smaller (126'000 m³ and 130'000 m³ respectively in 2004). Exports of tropical plywood by consumers

increased slightly in 2005, while log, sawnwood and veneer exports declined. Growth of China's tropical plywood exports has been rapid and notable, reaching 959'000 m³ in 2004 (up 69% from 2003 levels) and increasing a further 4% in 2005 to 1 million m³. Brazil remained the third largest exporter of tropical plywood in 2004, but China almost certainly overtook it in 2005.

3.4 Secondary processed wood products

Exports of secondary processed wood products (SPWP) by ITTO producers continued their upward trend in 2004. Except 2001, when exports contracted 11% due mainly to a slowdown in the US market, exports of SPWP by these countries have been expanding steadily since ITTO started regularly tracking them in the mid-1990s. Exports jumped by 23% in 2004 to reach almost \$9.9 billion, led by increases in Indonesia's, Malaysia's, Brazil's and Thailand's exports.

Indonesia and Malaysia were the only ITTO producer countries among the world's top SPWP exporters. The top five ITTO producer country exporters of SPWP in 2004 (Indonesia, Malaysia, Brazil, Thailand and Mexico) accounted for 88% of total ITTO producers' SPWP exports.

Japan and the USA remained the two largest markets for SPWP from ITTO producers, with such products making up 31% and 22% of their total SPWP markets respectively in 2004. However, these shares have declined (from 35% in Japan and 25% in the USA) since 2000, primarily due to competition from China.

The USA is the main partner of ITTO producers in value terms (\$4.8 billion in 2004) and its market continues to be the engine driving SPWP (mainly furniture) trade, growing almost four-fold in the last decade and up by 52% in the five years to 2004.

Although ITTO producer countries accounted for only 11% of the total EU market for SPWP in 2004, the magnitude of this huge market meant that the value of this share (\$2.9 billion) was 1.5 times the value of their Japanese market share and 61% of the value of their share of the US market.

In 2004, imports of SPWP by ITTO consumers from ITTO producers were worth a record \$9.4 billion, equivalent to an estimated 93% of the value of their imports of primary tropical timber products from these countries.

3.5 Tropical wood in Switzerland

The international timber trade organisation ITTO quantifies world-wide exports of tropical wood to add up to at least 34 Mio. m³ in 2005 (see chap. 2.2). Swiss imports of tropical wood in 2006 of 21'000 tons, roughly converted to 32'000 m³ contributes less than 1 per mill to the globally traded tropical wood. These quantities cover round- and sawn wood, plywood (at least the outer layers being made of tropical wood) and veneer. Other semi-finished and finished fabricates cannot be extracted unambiguously from the Swiss foreign trade statistics.

Figure illustrates the main products imported as tropical timber, as detailed in the Swiss foreign trade statistics.

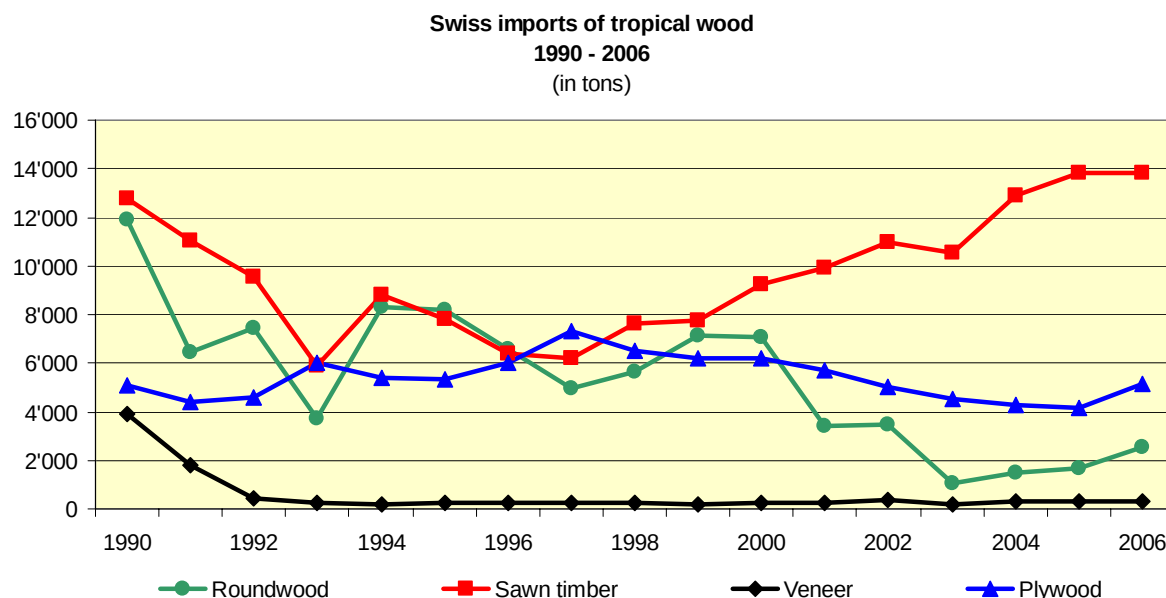


Figure 3.1: Development of Swiss imports of tropical wood (without semi-finished and finished products) (SHHZ 2007)

Several trends can be observed in Figure 3.1 and the underlying Swiss Foreign Trade statistics (SHHZ 2007, referring to Eidg. Zollverwaltung / Abt. Aussenhandelsstatistik):

- In 2006, imports from tropical roundwood added up to 2'500 t. Compared to 1990 (11'878 t) this corresponds to a decrease of 79 %; compared to the imports in 1965 (82'200 t), tropical roundwood imports have decreased about 97 %.
- Tropical sawn timber imports have almost doubled over the last 10 years, now ranging around 14'000 t per year. Tropical sawn timber was mainly imported via the transit countries Germany, France, Belgium and the Netherlands. Direct imports came mainly from the originating countries Cameroun and Malaysia.
- Tropical plywood imports and panels with at least the outer tropical veneering were imported at around 5'500 t/year. This implies that some wood from non-tropical sources is integrated in this data.
- Imports of tropical veneer, which had added up to 3'930 t in 1990, now are practically insignificant with imports of 294 t per year.
- Of the total amount of tropical roundwood of 2'500 t to imported to Switzerland, the main amount was directly imported from Cameroon (8'300 t equalling to 33 %) and from Congo (770 t equalling to 30 %). The most important transit countries Germany, France and Belgium supplied 670 t or 26% to Switzerland.
- Veneer from tropical species imported to Switzerland in 2006 was mainly produced in France (2'800 t).
- Imports of tropical wood have been reduced to one fifth over the last 40 years.

Compared to the calculated wood consumption of 5.3 Mio. m³ (including use as biofuel and for pulp-production) in Switzerland for the year 2005, tropical wood imports (roundwood, sawn timber, plywood and veneer) of 32'000 m³ cover only about 0.6%. Imported tropical wood contributes only 1.3% of total Swiss wood imports of 2.4 Mio m³ in 2005 (BAFU 2006).

Tropical woods imported today are mainly precious woods for special applications. Almost half of the imported tropical wood is used in the production of doors. Due to their high fire resistance, e.g. Sipo or Sapelli are mainly used for fire protection doors. A further important application is the production of parquet floorings with an estimated share of 30% of tropical wood. The remaining wood is used for

furniture, plywood or outdoor applications such as planks for terrasses, garden furniture or in boat construction (NZZ 2007).

Market penetration of FSC-certified tropical wood is still very low in Switzerland and still will take some years to expand to considerable levels. Only in October 2006, FSC-certified wood stemming from Kongo was imported to Switzerland for the first time. These imports were due to a voluntary commitment of the Association of Swiss Door Producers (VST) to use no more illegally harvested wood and to restrict their wood imports to sustainably produced timber on the medium term (NZZ 2007).

4 Life cycle inventories: methodological background

4.1 Data compilation

The main source of information on LCI/LCA data of tropical timber is an international survey by Murphy (2004). This survey lists four studies on tropical timber, which have been considered all in Chapter 5 of this report.

In May 2007, a complementary survey was made by contacting relevant research institutes, universities and LCA consultants in the field of forestry/LCA across the world.

Several commercial databases were found to already provide some LCI data on tropical timber. The IVAM LCA 4.0 database¹ contains data on meranti production and processing; references for the dataset, however, could not be found. The product development tool IdeMat 2005 by the TU Delft² provides LCA data of several tropical timber species; the environmental data however is the same for all species, assuming equal production and processing processes for all tropical timber species.

However, no further LCA studies on tropical timber could be found.

4.2 General modelling principles and allocation

The available data on tropical timber is implemented into the ecoinvent database according to the 'scope and modelling principles for processes' as outlined in Frischknecht et al. (2007a).

Forestry processes are usually allocated on a mass-basis, following the original literature reference.

Sawmill processes are allocated to sawn timber and industrial wood using economic allocation. Revenue-based allocation poses a specific problem for the material inherent characteristics of the biomass stored carbon, energy content and wood as a resource, three parameters associated with the (oven dry) mass of wood (de Feyter 1995). As a consequence of the revenue-based allocation and given the large price differences for co-products, carbon bound in wood and its energy content would be attributed to the main product in a much higher proportion than what corresponds to its mass. Thus, the main product would 'profit' from a proportionally much more negative GWP (i.e. storing of carbon) and mathematically produce much more thermal energy when incinerated than it actually does. On the other hand, the co-product with a much lower price would appear to carry neither the benefit of bound carbon nor to have any energy content. It is obvious that such an allocation would not lead to a consistent model of the wood chain. In ecoinvent, this issue is solved by allocating wood as a resource from the ecosphere and the carbon and energy content of wood based on volume (under bark). In a system of subsequent unit processes, this can only be done by using allocation correction factors. The modules used for allocation correction contain resource use, solar energy uptake and carbon uptake from atmosphere per m³ wood over bark (u=0%). Economic allocation is corrected by multiplying the respective module with the amount of the co-product *i* and the following weighting factor: (see Werner, et al. 2007, Hirschier et al. 2004).

$$\text{Weighting factor for product } i = \left(\frac{\text{Volume}(i)}{\text{Volume}(\text{tot})} \frac{\text{Revenue}(i)}{\text{Revenue}(\text{tot})} \right) \times \text{Volume}(\text{tot})$$

Specific assumptions underlying the datasets are described in the respective chapters for each species.

¹ <http://www.ivam.uva.nl/uk/producten/product5.htm> available in Simapro
(http://www.pre.nl/simapro/inventory_databases.htm)

² http://www.io.tudelft.nl/research/dfs/ideimat/Onl_db/od_frame.htm

4.3 Inventoring land use

Land occupation and land transformation are inventorised. Where data is available, land use is allocated to all commercial timber extractable from one hectare. If no such data was available in literature, commercial timber quantities per hectare were estimated from average above-ground biomass minus branches by applying a biomass expansion factor. This data was taken from IPCC (2003).

The total forest area is allocated to the wood production despite the fact that forests have many other functions e.g. as room for recreational activities or as protection against avalanches or floods.

Due to the lack of an appropriate CORINE land use class in ecovinent, tropical plantations forests are inventorised as either 'forest plantations, intense' or 'forest plantations, intense, short-rotation'. This reflects the expected decrease in biodiversity compared to natural (tropical) forest, but is geographically not representative.

The land occupied by forest roads is not considered due to the lack of respective data.

4.4 Inventoring carbon storage and release in tropical timber

Growth of natural wood, i.e. the physiological processes of trees, is normally not covered in LCA. An exception is usually the consideration of the CO₂-uptake during photosynthesis and the embodiment of solar energy (de Feyter 1995). In this project the embodiment of solar energy is defined to equal the calorific gross value of the wood. The carbon dioxide uptake is calculated based on the carbon content of wood which for this project is 49.4% (w/w) of dry wood matter.

Revenue-based allocation requires the correction of material-related flows, such as biogenic carbon and energy content; in multi-output processes with revenue-based allocation, an allocation correction is applied (see chap. 4.2).

5 Life cycle inventories of tropical wood

Disclaimer

Natural site conditions and forest management practices in tropical forests, socio-economic circumstances as well as species-specific material properties of tropical wood can vary considerably. These factors can have decisive influence on data inventories of tropical wood. As LCI data availability on tropical forestry processes and tropical wood is very limited, the following data inventories should be considered as case-specific data. Any specific environmental impact from land use change following timber harvesting needs to be considered on a case-specific basis.

This chapter contains LCI data for:

- Azobe from managed forests in Central Africa
- Meranti from managed forests in Malaysia
- Paraná pine from plantations/managed natural forests in Brazil
- Eucalyptus ssp. from plantations in Thailand

5.1 Azobe from managed forests in Central Africa

5.1.1 Scope of the data set

The LCI data on Azobe from natural forests in Central Africa is taken from ICL/SHR (2002), representing (close to) sustainable forest management (SFM). The data set comprises natural wood production, harvesting from one data source, transportation to Europa and production of plained sawn timber in one of the main importing countries (BE/FE/DE). The data set is structured into the following processes:

- azobe, standing, under bark, in rain forest
- azobe (SFM), under bark, at forest road (u=30%)
- roundwood azobe (SFM), debarked, at maritime harbour Europe (BE/NL/DE/FR) (u=30%)
- sawing/planing timber, azobe, air dried, at sawmill (BE/NL/DE/FR) (u=15%), yielding:
 - sawn timber (SFM), azobe, planed, air dried, at plant (BE/NL/DE/FR) (u=15%)
 - industrial residual wood, azobe (SFM), at sawmill (BE/NL/DE/FR) (u=15%)

The authors of ICL/SHR (2002) declare to have covered at least 95 % of all material and energy flows; if no data was available for flows that were considered to potentially contribute more than 1 % of any impact category (referring to Ecoindicator 99; Goedkoop, Effting *et al.* (2000)), respective process data was estimated based on closely related processes, for which data was available.

The forestry related dataset is based on data collected in Camaroon, West Africa.

5.1.2 Natural wood production, Azobe

Azobe (*Lophira alata*; also know as Ekki or Bongossi) is a large to very large tree, up to over 50 m height and 150 cm diameter, fairly cylindrical almost down to the ground, without buttresses. It prefers sandy to sandy-loam moist soil, in a near fresh water swamps and riverbanks.

Table 5.1: Some properties and characteristics of Azobe timber (Sell 1997)

Raw density (u=0%); g/cm ³	Raw density (u=15%); g/cm ³	Dimensional stability	Processability	Potential health aspects
0.94 – 1.05	1.02 – 1.15	poor to medium	High tendency to form cracks during drying; hardly nail- or screwable without pre-drilling; strong dimensional changes with changes in moisture content; polishable; hardly glueable	skin irritations

Azobe is often found growing in clusters of roughly the same age group and quite abundant in wet, evergreen, littoral rain forests of West and Central Africa. Azobe trees grow in dense clumps in these forests in comparison with other species. The timber is also of much lower value than of many co-extracted species. The timber is not yet produced in plantations, but is all derived from natural, mixed forest concessions.

The timber is dark red, chocolate brown, or purple brown with the vessels showing conspicuous white deposits. The sapwood (which cannot be used for timber) is up to 3 cm wide, pale pink, and well defined. The heartwood is classified by CEN EN 350-1 as 'very durable', class 1.

Table 5.1 summarizes some properties and characteristics of Azobe timber. Azobe is used as construction wood for exterior works with high strains that do not impose high requirements on dimensional stability, such as harbour construction, bridges, pontoons, pilings in saltwater or brackwater, cooling towers, sewage plants, noise protection walls and blinds, fences, or floorings in industrial buildings.

The solar energy uptake is calculated as the upper heating value (24.3 MJ/kg at u=30%¹) according to Frischknecht et al. (2007a, Eq. 1) based on a lower heating value² of 18.35 MJ/kg.

A carbon content of 49.4% of oven-dried wood is assumed in line with Frischknecht et al. (2007a).

The background information to the processes are given in Table 5.2, flows are shown in Figure 5.1:

5.1.3 Forestry processes and direct forest products

The typical overall average Azobe yield in littoral forest concessions in Cameroon is about 6 m³/ha (~1/2 tree/ha). The legal minimum diameter at breast height (DHB) for felling is 60 cm, but loggers will not fell any Azobe trees below 90 – 100 cm DHB. Smaller logs are uneconomical, as the thick, useless sapwood lowers the yield below acceptable level. Many small trees, often whole populations, are left standing to be harvested in another cycle, 35 – 40 years afterwards (total rotation length can thus be assumed to be 70 to 80 years).

Other species are felled at 60 cm DBH, because of their higher value and their thinner, or absent sapwood. The types and quantities of other timbers harvested with Azobe vary greatly from site to site, and there is no information available concerning co-harvested materials.

The datasets have been developed by Dutch consultants in Cameroon. The data stems from one large logging company and represents good forest management practices: individual trees are selected for felling and marked on a pre-harvest map. The felling and crosscutting of the logs is closely supervised, to ensure compliance with licencing restrictions and maximum utilisation of the log.

¹ <http://www.holzwissen.net/azobe-bongossi.html>

² <http://www.matbase.com/material/wood/class2-15-25-years/azobe/properties>

Technology involves large modern motorised forestry equipment. Felling is by hand chainsaw, and cross-cutting takes place in the forest prior to extraction; extraction from the stump to the siding is by bulldozer or wheel-skidder.

Tops and branches are burned; the quantity of tops and branches is estimated based on a biomass expansion factor for azobe from IPCC (2003); emissions from biomass burning are estimated from IPCC (2003) and EPA (1995).

The forest area is left to regenerate naturally as part of a polycyclical forest management.

The background information to the processes are given in Table 5.2, flows are shown in Figure 5.2.

5.1.4 Debarking and transportation to main maritime port in Europe.

A transport distance of 200 km by a lorry 32 t is assumed for the transportation of the log to a main naval port in Central Africa (estimated from ICL/SHR 2002).

No data on debarking is available but takes place before shipping the logs to Europe. Average European data for debarking pine in the forest (Werner, et al. 2007) is used, taking into account the modern equipment of the logging company. The resulting bark is left to rot.

A transport distance of 7000 km by a maritime freight ship is assumed for the overseas transportation of the log to a main naval port in Europe such as Rotterdam, Antwerpen, Bremerhafen or Hamburg (estimated from ICL/SHR 2002).

The background information to the processes are given in Table 5.2, flows are shown in Figure 5.3.

5.1.5 Azobe panel production

Azobe panel production data stems from a sawmill in the Netherlands, using modern equipment (representative for Europe).

The dataset contains transportation to a sawmill by lorry as well as sawing and planing of the azobe log in 20 mm sawn timber boards; natural drying is assumed; 30% of log input (including sapwood) is generated as co-product that is sold or is used for energy recovery outside this process.

Revenue-based allocation is applied to allocate the sawing processes to the resulting sawn timber and industrial residual wood.

The background information to the processes are given in Table 5.2, flows are shown in Figure 5.4.

5.1.6 Allocation correction for azobe sawmill processes

Revenue-based allocation requires the application of allocation correction factors for material-inherent properties (carbon content and energy content. See Werner et al. (2007) and Hischer et al. (2004) for details.

The background information to the allocation correction are given in Table 5.3, flows are shown in Figure 5.5.

Table 5.2: Ecoinvent meta information to the datasets for "azobe (SFM), standing, under bark, in rain forest", "roundwood, azobe (SFM), under bark, u=30%, at forest road", "roundwood, azobe (SFM), debarked, u=30%, CM, at maritime harbour" and "sawing and planing, timber, azobe, air dried, u=15%, CM"

Name	azobe (SFM), standing, under bark, in rain forest	roundwood, azobe (SFM), under bark, u=30%, at forest road	roundwood, azobe (SFM), debarked, u=30%, CM, at maritime harbour	sawing and planing, timber, azobe, air dried, u=15%, CM
Location	CM	CM	RER	RER
Infrastructure Process	0	0	0	0
Unit	m3	m3	m3	m3
Data Set Version	2	2	2	2
Included Processes	includes the CO2 absorption from air and the land use for the trees to grow (including 9% bark). Forest road area is not included in the land useDensity: 1000 kg/m3 (u=0)	includes harvesting and hauling operations as well as fuel used for forest road construction, including burning of branchings; land use of forest roads and gravel are disregarded; branchings are left to rot on the ground. logging impacts on further vegetation and environmental impacts from post-harvest processes (potential forest degradation/deforestation) are not covered.	includes debarking and natural decomposition of the bark, transportation of the log to a main harbour in Europe by truck and freight ship	includes transportation to sawmill by lorry; sawing and planing of azobe log in 20 mm sawn timber boards; natural drying is assumed; 30% of log input (including sapwood) is generated as co-product that is sold or is used for energy recovery used outside this process.
Amount	1	1	1	1
Local Name	Azobe (NWB), stehend, Festkubikmeter, in Regenwald	Rundholz Azobe (NWB), Festkubikmeter, u=30%, ab Waldstrasse	Rundholz Azobe (NWB), entrindet, u=30%, CM, ab Meerhafen Europa	Sägen-Hobeln, Schnittholz, Azobe, luftgetrocknet, u=15%, CM
Synonyms	Bongassi/Ekki/Lophira alata	Bongassi/Ekki/Lophira alata	Bongassi/Ekki/Lophira alata	Bongassi/Ekki/Lophira alata
General Comment to reference function	data refers to wood 'under bark'. CO2 assimilation is based on 49.4% carbon in the wood including the bark (19%). Biomass energy equals gross calorific value including bark.	dataset represents well-documented, legally operating logging company with modern equipment; emissions from biomass burning according to IPCC 2003	data refers to wood from Central Africa, used in Europe. Main ports in Europe include Rotterdam, Hamburg, Bremerhafen or Antwerpen.	data refers to wood from Central Africa, used in Europe, the multi-output process "sawing and planing, timber, azobe, air dried, u=15%, CM" delivers the two coproducts "sawn timber (SFM), azobe, planed, air dried, u=15%, CM, at sawmill" and "industrial residue wood, azobe (SFM), u=15%, CM, at sawmill". Allocation is based on the over-all proceeds of the process; resource use of natural wood growth is corrected via a mass-based
Start Date	2000	2000	2000	2000
End Date	2005	2005	2005	2005
Data Valid For Entire Period	1	1	1	1
Other Period Text				
Geography text			data for debarking in Africa is not available - instead, European processes are used, taking into account of the modern equipment used for harvesting.	Wood from Central Africa, sawing in Europe
Technology text	natural growth; relates to the forest area and CO2 from air needed to produce azobe	modern machinery and equipment; high technological standard and legal operations cannot be assumed to be representative for azobe harvesting	high technological standard for debarking; average processes for transportation to Europe; transport distance however can vary. high technological standard and legal operations cannot be assumed to be representative for azobe harvesting.	data extrapolated from Meranti window production in the Netherlands. Depending on sawing technology, specific wood density of a particular log and type of sawn timber produced, data may vary
Representativeness [%]				
Production Volume	unknown	unknown	unknown	unknown
Sampling Procedure	literature	literature	literature	literature
Extrapolations	none	none	see Geography	see Technology
Uncertainty Adjustments	none	none	none	none

Table 5.3: Ecoinvent meta information to the datasets for "azobe, allocation correction 1" and "azobe, allocation correction 2"

Name	azobe, allocation correction 1	azobe, allocation correction 2
Location	GLO	GLO
Infrastructure Process	0	0
Unit	m3	m3
Data Set Version	2	2
Included Processes	corrects economically allocated resource use, feedstock energy and carbon uptake to fit with physical flows (=mass-based allocation)	corrects economically allocated resource use, feedstock energy and carbon uptake to fit with physical flows (=mass-based allocation)
Amount	1	1
Local Name	Azobe, Allokationskorrektur 1	Azobe, Allokationskorrektur 2
Synonyms		
General Comment to reference function	auxiliary module	auxiliary module
Start Date	2000	2000
End Date	2005	2005
Data Valid For Entire Period	1	1
Other Period Text		
Geography text		
Technology text		
Representativeness [%]		
Production Volume	none	none
Sampling Procedure	none	none
Extrapolations	none	none
Uncertainty Adjustments	none	none

General Flow Information						Representation in ecoinvent								Uncertainty		
Input		Process Name		Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
azobe	➔	azobe (SFM), standing, under bark, in rain forest			including 9 % bark (95 cm diameter stemwood plus 2 x 2 cm bark)	resource	biotic			Wood, hard, standing	1.09E+00	m3	calculated	1	1.26	(3,4,1,3,1,5)
solar energy	➔				equals to gross calorific value	resource	biotic			Energy, gross calorific value, in biomass	2.65E+04	MJ	lower heating value: 18.35 MJ/kg (http://www.matbase.com/material/wood/class2-15-25-years/azobe/properties); upper heating value calculated according to Frischknecht et al. (2003), Eq. 1: 24.3 MJ/kg	1	1.26	(3,4,1,3,1,5)
CO2 regenerativ	➔				dried matter =1000 kg/m3, C content of dried matter = 49.4% excluding forest roads: ICL/SHR 2002: yield: 6 m3/ha (excl. bark and forest roads) besides other more valuable commercial species; rotation length: 75 years	resource	in air			Carbon dioxide, in air	1.97E+03	kg	calculated	1	1.26	(3,4,1,3,1,5)
transformation, from rain forest, unknown	➔					resource	land			Transformation, from tropical rain forest	1.20E+02	m2		1	2.00	(,,,,)
transformation, to rain forest, extensive	➔				rotation length: 75 years	resource	land			Transformation, to tropical rain forest	1.20E+02	m2		1	2.00	(,,,,)
land occupation	➔				Total above-ground biomass for African wet tropical forest (IPCC 2003, Table 3A.1.2.): 310 t d.m./ha; BEF2 (IPCC 2003, Table 3A.1.10): 3.4; rotation length: 75 years	resource	land			Occupation, tropical rain forest	8.97E+03	m2a		1	1.50	(,,,,)
			➔	azobe (SFM), standing, under bark, in rain forest	under bark (Inputs refer to 1.09 m3 oven-dried azobe (u=0%) including bark); moisture content according to: http://www.holzwissen.net/azobe-bongossi.html	wooden materials	extraction	No	CM	azobe (SFM), standing, under bark, in rain forest	1.00E+00	m3				

Figure 5.1: Flows for “azobe (SFM), standing, under bark, in rain forest” and its representation in the ecoinvent database

General Flow Information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Module name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
azobe, standing, under bark, in rainforest (u=30%)	→			wooden materials	extraction	No	CM	azobe (SFM), standing, under bark, in rain forest	2.00E+00	m³	calculated from IPCC (2003) GPG for LULUCF	1	1.26	(3,4,1,3,1,5)
diesel for harvesting operations	→			construction processes	machinery	No	GLO	diesel, burned in building machine	5.10E+02	MJ	IPU/SHR (2002)	1	1.26	(3,4,1,3,1,5)
diesel for road construction	→			construction processes	machinery	No	GLO	diesel, burned in building machine	4.14E+01	MJ	IPU/SHR (2002)	1	1.26	(3,4,1,3,1,5)
hydraulic oil for harvesting operations	→			chemicals	organics	No	RER	lubricating oil, at plant	7.70E-01	kg	IPU/SHR (2002)	1	1.26	(3,4,1,3,1,5)
hydraulic oil/tube oil for road construction	→			chemicals	organics	No	RER	lubricating oil, at plant	1.40E-01	kg	IPU/SHR (2002)	1	1.26	(3,4,1,3,1,5)
chainsawing, carburetor without catalytic converter	→		assumed to cover tube oil consumption inventoried in ICL/SHR 2002	construction processes	machinery	No	RER	power sawing, without catalytic converter	1.00E-01	h	estimated based on IPU/SHR (2002)	1	1.26	(3,4,1,3,1,5)
land transformation from forest	→		unknown n; not inventorised									1		
land transformation to forest roads	→		unknown n; not inventorised											
land occupation of forest roads	→		unknown n; not inventorised											
transportation of gravel (lorry 16 t)	→		unknown n; not inventorised											
gravel for forest roads	→		unknown n; not inventorised											
	→	roundwood azobe, at forest road (u=30%)	100% allocation to roundwood production	wooden materials	extraction	No	CM	roundwood, azobe (SFM), under bark, u=30%, at forest road	1.00E+00	m³				
	→	branchings, burned	volume under bark; biomass expansion factor =2 (IPCC 2003); not inventorised						1.00E+00	m³	calculated from IPCC (2003) GPG for LULUCF	1	0.00	
	→		volume under bark	resource	biotic			Energy, gross calorific value, in biomass	-2.65E+04	MJ	ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
	→	CO2 regenerativ released from branchings	volume under bark	air	unspecified			Carbon dioxide, biogenic	1.97E+03	kg	ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
	→	N2O from biomass burning	volume under bark	air	low population density			Dinitrogen monoxide	1.18E-02	kg	calculated based on IPCC 2003	1	1.59	(3,4,1,3,1,5)
	→	CH4 from biomass burning	volume under bark	air	low population density			Methane, biogenic	1.72E+00	kg	calculated based on IPCC 2003	1	1.59	(3,4,1,3,1,5)
	→	PM <2.5	4 g/kg	air	low population density			Particulates, > 2.5 µm, and < 10µm	5.00E+00	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
	→	PM >2.5>10	5 g/kg	air	low population density			Particulates, < 2.5 µm	6.00E+00	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
	→	PM >10	6 g/kg	air	low population density			Particulates, > 10 µm	4.00E+00	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
Azobe	→		resource correction for branchings	resource	biotic			Wood, hard, standing	-1.09E+00	m³	calculated	1	1.26	(3,4,1,3,1,5)

Figure 5.2: Flows for “roundwood, azobe (SFM), under bark, u=30%, at forest road” and its representation in the ecoinvent database

General Flow Information					Representation in ecoinvent								Uncertainty			
Input		Process Name		Output	Remarks	Category	Subcategory	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
roundwood azobe, at forest road (u=30%)	→	roundwood, azobe (SFM), debarked, u=30%, CM, at maritime harbour				wooden materials	extraction	No	CM	roundwood, azobe (SFM), under bark, u=30%, at forest road	1.00E+00	m3	ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
transportation to maritime harbour; diesel truck	→				200 km; including bark	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.83E+02	tkm	calculated based on ICL/SHR (2002)	1	2.07	(3,4,1,3,1,5)
freight ship to main European harbour	→				7000 km from Central Africa; excluding bark	transport systems	ship	No	OCE	transport, transoceanic freight ship	9.10E+03	tkm	ICL/SHR (2002)	1	2.07	(3,4,1,3,1,5)
diesel used for debarking	→				debarking before maritime transport; estimated based on European data	construction processes	machinery	No	GLO	diesel, burned in building machine	2.84E+00	MJ	Werner et al. 2003	1	1.26	(3,4,1,3,1,5)
lubricant used for debarking	→				debarking before maritime transport; estimated based on European data	chemicals	organics	No	RER	lubricating oil, at plant	2.52E-02	kg	Werner et al. 2003	1	1.26	(3,4,1,3,1,5)
			→	roundwood azobe, debarked, at maritime harbour Europe (u=30%)	100% allocation to roundwood	wooden materials	extraction	No	RER	roundwood, azobe (SFM), debarked, u=30%, CM, at maritime harbour	1.00E+00	m3	ICL/SHR (2002)			
			→	bark from debarking	net volume; not inventorised						9.00E-02	m3	ICL/SHR (2002)			
solar energy released from bark	→					resource	biotic			Energy, gross calorific value, in biomass	-1.31E+03	MJ	IPL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
			→	CO2 regenerativ released from bark		air	unspecified			Carbon dioxide, biogenic	1.63E+02	kg	IPL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
Azobe	→				resource correction for bark	resource	biotic			Wood, hard, standing	-9.00E-02	m3	calculated	1	1.26	(3,4,1,3,1,5)

Figure 5.3: Flows for "roundwood, azobe (SFM), debarked, u=30%, CM, at maritime harbour" and its representation in the ecoinvent database

General Flow Information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
roundwood azobe, debarked, at maritime harbour Europe (u=30%)	→			wooden materials	extraction	No	RER	roundwood, azobe (SPM), debarked, u=30%, CM, at maritime harbour	1.30E+00	m³	calculated based on ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
transportation from harbour to saw mill, truck 32 t	→	250 km		transport systems	road	No	RER	transport, lorry >16t, fleet average	4.23E+02	tkm	calculated based on ICL/SHR (2002)	1	2.07	(3,4,1,3,1,5)
electricity	→			electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.06E+02	kWh	calculated based on ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
diesel	→	usage unclear		construction processes	machinery	No	GLO	diesel, burned in building machine	2.20E+01	MJ	calculated based on ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
gasoline	→	usage unclear; possibly maintenance		oil	fuels	No	RER	petrol, low-sulphur, at regional storage	5.52E-02	kg	calculated based on ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
propane	→			natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.36E+01	MJ	calculated based on ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
utility water	→			water supply	production	No	RER	tap water, at user	2.11E+02	kg	calculated based on ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
saw mill	→			wooden materials	extraction	Yes	RER	saw mill	4.00E-07	unit	estimated based on Werner/Althaus 2003	1	3.07	(3,4,1,3,1,5)
	→	sawn timber, azobe, planed, air dried, u=15%, at plant (BENL/DEFRR)	20mm	wooden materials	processing	No	RER	sawn timber (SPM), azobe, planed, air dried, u=15%, CM, at saw mill	1.00E+00	m³	calculated based on ICL/SHR (2002)			
	→	industrial residual wood, azobe (u=15%)		wooden materials	processing	No	RER	industrial residual wood, azobe (SPM), u=15%, CM, at saw mill	3.00E-01	m³	calculated based on ICL/SHR (2002)			
Azobe, sawn timber, allocation correction	→			wooden materials	processing	No	GLO	azobe, allocation correction 1	-2.48E-01	m³	calculated based on ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
Azobe, industrial residual wood, allocation correction	→			wooden materials	processing	No	GLO	azobe, allocation correction 2	2.48E-01	m³	calculated based on ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
	→	waste heat		air	unspecified			Heat, waste	3.81E+02	MJ	calculated based on ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)
	→	VOC from machine maintenance	Potential emissions from maintenance using gasoline	air	unspecified			NM/VOC, non-methane volatile organic compounds, unspecified origin	5.52E-02	kg	extrapolated	1	1.59	(3,4,1,3,1,5)
	→	BOD		water	unspecified			BOD5, Biological Oxygen Demand	4.22E-07	kg	calculated based on ICL/SHR (2002)	1	1.59	(3,4,1,3,1,5)
	→	suspended solids		water	unspecified			Suspended solids, unspecified	4.32E-06	kg	calculated based on ICL/SHR (2002)	1	1.59	(3,4,1,3,1,5)
	→	HC (oil and grease) into water		water	unspecified			Hydrocarbons, aliphatic, unsaturated	1.88E-07	kg	calculated based on ICL/SHR (2002)	1	3.07	(3,4,1,3,1,5)
	→	phenol into water		water	unspecified			Phenol	8.46E-06	kg	calculated based on ICL/SHR (2002)	1	3.07	(3,4,1,3,1,5)
	→	solid waste		waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.44E+00	kg	calculated based on ICL/SHR (2002)	1	1.26	(3,4,1,3,1,5)

Figure 5.4: Flows for “sawing and planing, timber, azobe, air dried, u=15%, CM” and its representation in the ecoinvent database

General Flow Information					Representation in ecoinvent								Uncertainty				
Input		Process Name		Output	Remarks	Category	Subcategory	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment	
azobe	➔	Azobe, allocation correction, 1			over bark	resource	biotic			Wood, hard, standing	1.00E+00	m3	calculated	1	1.26	(3,4,1,3,1,5)	
solar energy	➔					equals to gross calorific value	resource	biotic			Energy, gross calorific value, in biomass	2.43E+04	MJ	lower heating value: 18.35 MJ/kg (http://www.matbase.com/material/wood/class2-15-25-years/azobe/properties); upper heating value calculated according to Frischknecht et al. (2003), Eq. 1: 24.3 MJ/kg	1	1.26	(3,4,1,3,1,5)
CO2 regenerativ	➔						resource	in air			Carbon dioxide, in air	1.81E+03	kg	calculated	1	1.26	(3,4,1,3,1,5)
			➔	Azobe, allocation correction, 1			wooden materials	processing	No	GLO	azobe, allocation correction 1	1.00E+00	m3				
azobe	➔	Azobe, allocation correction, 2			over bark	resource	biotic			Wood, hard, standing	1.00E+00	m3	calculated	1	1.26	(3,4,1,3,1,5)	
solar energy	➔					equals to gross calorific value	resource	biotic			Energy, gross calorific value, in biomass	2.43E+04	MJ	lower heating value: 18.35 MJ/kg (http://www.matbase.com/material/wood/class2-15-25-years/azobe/properties); upper heating value calculated according to Frischknecht et al. (2003), Eq. 1: 24.3 MJ/kg	1	1.26	(3,4,1,3,1,5)
CO2 regenerativ	➔						resource	in air			Carbon dioxide, in air	1.81E+03	kg	calculated	1	1.26	(3,4,1,3,1,5)
			➔	Azobe, allocation correction, 2			wooden materials	processing	No	GLO	azobe, allocation correction 2	1.00E+00	m3				

Figure 5.5: Flows for “azobe, allocation correction 1” and “azobe, allocation correction 2” and their representation in the ecoinvent database

5.1.7 Data quality

Data quality is assessed using the pedigree matrix as described in Frischknecht et al. (2007a).

Data for CO₂-assimilation and for the energy embodied in the biomass are calculated from the carbon and energy content. Wood density and other material-characteristic parameters can vary considerably within a species.

Land use data can vary considerably depending on how intensive the forest is used. Since the data used are valid for one specific company and region, the uncertainty is rather high.

It should be taken into account that the dataset stems from a legally operating, apparently well-documented logging company with modern equipment. Unfortunately, this cannot be assumed to be the standard case for azobe production.

5.2 Meranti from managed natural forests in Malaysia

5.2.1 Scope of the data set

The LCI data on Meranti from managed natural forests in Malaysia is based on Rugge (1999) and complemented with Collins, Fisher *et al.* (2005), representing (close to) sustainable forest management (SFM). The data set comprises natural wood production, harvesting from one data source and transportation to Europa (BE/FE/DE). The data set is structured into the following processes:

- meranti, standing, under bark, in rain forest
- roundwood meranti (SFM), under bark, at forest road (u=70%)
- roundwood meranti (SFM, debarked, at maritime harbour Europe (BE/NL/DE/FR) (u=70%)

Inventory data was obtained from company reports, scientific journals, the internet, and a field visit to a logging site as well as consultation with forestry experts, in the fields of silviculture, economics, hydrology and soil science in Malaysia. The forest area is part of the state of Terengganu and is managed under the selective management system. The logging company professes that its timber is produced sustainably.

5.2.2 Natural wood production, Meranti

Meranti (*Shorea ssp.*; also known as Seraya or Lauan) is a major internationally traded group of species¹. The group consists of red, white, and yellow species. There are many species in the red group and they range in density and weight, which are broad guides to performance.

The trade divides them into groups of Dark Red, Red and Light Red. The shoreas are of even growth and texture, easy to work and widely used for all types of joinery, framing, doors and fittings. They are the principal ingredients in tropical Far Eastern plywood.

Exploitation of forests in peninsular Malaysia became confined to the hill forests about 20 years ago, where timber yield is often variable. These forests are not as well stocked as low-land forests, except for Seraya (*Shorea curtisii*), which is part of the dark red Meranti group, concentrations on ridge tops. The selective management system is applied here, whilst the Malaysian uniform system is applied in the well stocked low lying areas. These areas have now mostly been logged over, or have been converted to other land use.

Table 5.4 summarizes some properties and characteristics of Meranti timber. The dataset is elaborated for the harvesting of dark red meranti.

¹ Major species include *S. dolichocarpa*, *S. faguetiana*, *S. gibbosa*, *S. hopeifolia*, *S. longisperma*, *S. maxima* and *S. multiflora*

The solar energy uptake is calculated as the upper heating value (28.0 MJ/kg at $u=70\%$ ¹) according to Frischknecht et al. (2007a, Eq. 1) based on a lower heating value² of 18.47 MJ/kg. A carbon content of 49.4% of oven-dried wood is assumed in line with Frischknecht et al. (2007a) assuming an average density of 600 kg/m³ ($u=0\%$).

The background information to the processes are given in Table 5.5, flows are shown in Figure 5.6:

Table 5.4: Some properties and characteristics of Meranti timber (Sell 1997)

Raw density ($u=0\%$); g/cm ³	Raw density ($u=15\%$); g/cm ³	Dimensional stability	Processability	Potential health aspects
0.45 – 0.70	0.48 – 0.76	medium to good	Tendency to form cracks during drying; easily workable; no problems with glueing or surface treatment (after leaking resins have been removed); well polishable	Singular skin irritations

5.2.3 Forestry processes and direct forest products

Silvicultural management for this dataset relies on selective management as a polycyclical system. A rotation period of 60 years (derived from reproduction time) is assumed, allowing for two cutting cycles of 30 years.

Harvesting practices are predominantly manual. Trees are felled using a chainsaw. Lianas on the trees to be felled are cut about six months previously, which prevents connected trees to be pulled down with the felling of one tree. After the crown is lopped off the trees are crosscut into logs below the first branches and these logs are extracted with the use of a cable and powerwinch by a crawler tractor (Komatsu) from the felling position to the skidding track.

Heavy crawler tractors are used to haul trees from their felling location to the temporary log landings. Tractors are restricted to skidding tracks. The temporary log landings are located within the forest area and are traversed by heavy machinery, such as the tractors and San-tai-wong (6x6 trucks). These carry logs from the temporary log landings in the forest along permanent forest roads to the permanent logyards next to the main roads.

Fuel consumption for chain saws, tractors, skidders and building machinery for forest road construction is estimated based on Collins, Fisher *et al.* (2005) as being less intensive than the Central African azobe harvesting but more intense than European wood harvesting.

Although the forest canopy remains fairly closed, logging damages due to falling trees, preparation of skidding tracks, log landings etc. as well as impacts on top soil can be considerable. According to Rugge (1999), the damage to residual stands reported, range from 39 to 69% in Peninsular Malaysia. Damage due to felling, skidding and road construction usually affects 15-35 % of all trees in a polycyclic harvesting system (harvesting modest volume of timber every 20-40 years) compared to 40-60 % in a monocyclic system (harvesting all saleable trees every 70-80 years).

Branchings are assumed to be burned on site; the amount of branchings and respective emissions from biomass burning are estimated based on IPCC (2003) and EPA (1995).

¹ <http://woodwizard.mtc.com.my:8888/report.asp?AttrID=5&ItemID=123>

² <http://www.matbase.com/material/wood/class2-15-25-years/meranti/properties>

The average yield in Jengai forest reserve in Terengganu is 46.0 m³/ha. Dark red Meranti, mostly consisting of *Shorea curtisii*, make 33% of the total volume above the cutting limit of 65 cm diameter breast height (dbh) and 20.8% of the number of commercial trees. The average volume per meranti tree is 4.24 m³; the number of dark red Meranti trees harvested per ha is 2.6 trees or 11.14 m³/ha.

However, other sources are reported to assume 40-60 m³/ha as an average yield, of which Meranti makes up 10%.

Rugge (1999) assumes that the total area damaged as a result of the logging action is about 30%, because of the relatively low volumes harvested, supervision and the company's implementation of low impact harvesting techniques (reduction of skidding areas, directional felling and cutting of lianas). The area damaged by the extraction is therefore estimated at 30% of 0.3 ha, which amounts to 0.01 ha. If Dark red Meranti only represents 20% of the harvested volume, it should theoretically only be allocated 20% of the damage to the area, which amounts to 0.002 ha. This would estimate the damage caused by the felling of one tree (approximately 4.23 m³ at this site), to be 0.1325 ha.

The background information to the processes are given in Table 5.5, flows are shown in Figure 5.7.

5.2.4 Debarking and transportation to main maritime port in Europe.

A transport distance of 200 km by a lorry 32 t is assumed for the transportation of the log to a main naval port in Central Africa (own estimate).

No data on debarking is available which takes place before shipping the logs to Europe. Average European data for debarking pine in the forest (Werner, et al. 2007) is used, taking into account the modern equipment of the logging company. The resulting bark is left to rot.

A transport distance of 15'000 km by a maritime freight ship is assumed for the overseas transportation of the log to a main naval port in Europe such as Rotterdam, Antwerpen, Bremerhafen or Hamburg (own estimate).

The background information to the processes are given in Table 5.5, flows are shown in Figure 5.8.

It should be noted however, that Malaysia (and other Southeast-Asian countries) have built up considerable sawmilling and plywood processing facilities to process Meranti locally.

5.2.5 Data quality

Data quality is assessed using the pedigree matrix as described in Frischknecht et al. (2007a).

Data for CO₂-assimilation and for the energy embodied in the biomass are calculated from the carbon and energy content. Wood density and other material-characteristic parameters can vary considerably within a species.

Land use data can vary considerably depending on how intensive the forest is used. Since the data used are valid for one specific company and region, the uncertainty is rather high.

It should be taken into account that the dataset stems from a legally operating, apparently well-documented logging company. Unfortunately, this cannot be assumed to be the standard case for meranti in Malaysia production.

Fuel consumption for forestry operations is taken from Collins, Fisher *et al.* (2005) referring to teak plantations in Indonesia. This can be considered an expert's guess.

Location and technology used for debarking is unknown; mobile debarking is assumed.

Table 5.5: Ecoinvent meta information to the datasets for "meranti, standing, under bark, u=70%, in rain forest", "roundwood, meranti (SFM), under bark, u=70%, at forest road" and "roundwood, meranti (SFM), debarked, u=70%, MY, at maritime harbour"

Name	meranti, standing, under bark, u=70%, in rainforest	roundwood, meranti (SFM), under bark, u=70%, at forest road	roundwood, meranti (SFM), debarked, u=70%, MY, at maritime harbour
Location	MY	MY	RER
Infrastructure Process	0	0	0
Unit	m3	m3	m3
Data Set Version	2	2	2
Included Processes	includes the CO2 absorption from air and the land use for the trees to grow (including 11% bark). Forest road area is not included in the land use; dataset refers to dark red meranti; Density: 600 kg/m3 (u=0)	includes harvesting and hauling operations as well as fuel used for forest road construction, including burning of branchings; land use of forest roads and gravel are disregarded; branchings are left to rot on the ground. logging impacts on further vegetation and environmental impacts from post-harvest processes (potential forest degradation/deforestation) are not covered.	includes debarking and natural decomposition of the bark, transportation of the log to a main harbour in Europe by truck and freight ship
Amount	1	1	1
Local Name	Meranti, stehend, Festkubikmeter, u=70%, in Regenwald	Rundholz Meranti (NWB), Festkubikmeter, u=70%, ab Waldstrasse	Rundholz Meranti (NWB), entrindet, u=70%, MY, ab Meerhafen Europa
Synonyms	Shorea ssp./Seraya/Lauan	Shorea ssp./Seraya/Lauan	Shorea ssp./Seraya/Lauan
General Comment to reference function	data refers to wood 'under bark'. CO2 assimilation is based on 49.4% carbon in the wood including the bark (11%). Biomass energy equals gross calorific value including bark.	dataset represents well-documented, legally operating logging company with modern equipment; emissions from biomass burning according to IPCC 2003	data refers to wood from Malaysia, used in Europe. main ports in Europe include Rotterdam, Hamburg, Bremerhafen or Antwerpen.
Start Date	2000	2000	2000
End Date	2005	2005	2005
Data Valid For Entire Period	1	1	1
Other Period Text			
Geography text			
Technology text	natural growth; relates to the forest area and CO2 from air needed to produce meranti	average technology; forest operations largely based on manual labor. Fuel consumption is interpolated by Collins et al 2002 assumed to lie between azobe dataset and European average consumption	data for debarking is not available; manual deparking is assumed. average processes for transportation to Europe; transport distance however can vary.
Representativeness [%]			
Production Volume	unknown	unknown	unknown
Sampling Procedure	literature	literature, partly relying on estimates (fuel consumption)	literature
Extrapolations	none	none	see Technology
Uncertainty Adjustments	none	none	none

General Flow Information					Representation in ecoinvent									Uncertainty		
Input		Process Name		Output	Remarks	Category	Subcategory	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
meranti	➔	meranti, standing, under bark, u=70%, in rainforest			including 11 % bark (70 cm diameter stemwood plus 2 x 2 cm bark)	resource	biotic			Wood, hard, standing	1.11E+00	m3	calculated	1	1.26	(3,4,1,3,1,5)
solar energy	➔				equals to gross calorific value	resource	biotic			Energy, gross calorific value, in biomass	1.87E+04	MJ	lower heating value: 18.47 MJ/kg (http://www.matbase.com/material/wood/class2-15-25-years/meranti/properties); upper heating value calculated according to Frischknecht et al. (2003), Eq. 1: 28.03 MJ/kg	1	1.26	(3,4,1,3,1,5)
CO2 regenerativ	➔				dried matter =600 kg/m3, C content of dried matter = 49.4%	resource	in air			Carbon dioxide, in air	1.20E+03	kg	calculated	1	1.26	(3,4,1,3,1,5)
transformation, from rain forest, unknown	➔				excluding forest roads: Rugge (1999): yields: 11.14 m3/ha (excl. bark and forest roads) besides other more valuable commercial species: total yield: 46 m3/ha; rotation length: 75 years	resource	land			Transformation, from tropical rain forest	2.71E+02	m2		1	2.00	(,,,,)
transformation, to rain forest, extensive	➔					resource	land			Transformation, to tropical rain forest	2.71E+02	m2		1	2.00	(,,,,)
land occupation	➔					resource	land			Occupation, tropical rain forest	1.63E+04	m2a		1	1.50	(,,,,)
			➔	meranti, standing, under bark, u=70%, in rainforest	under bark (Inputs refer to 1.11 m3 oven-dried meranti (u=0%) including bark); moisture content according to: http://woodwizard.mtc.com.my:8888/report.asp?AttrID=5&ItemID=123	wooden materials	extraction	No	MY	meranti, standing, under bark, u=70%, in rainforest	1.00E+00	m3				

Figure 5.6: Flows for "meranti, standing, under bark, u=70%, in rain forest" and its representation in the ecoinvent database

General Flow Information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Subcategory	Infrastructure	Location	Module name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
meranti, standing, under bark, in rainforest (u=70%)	→			wooden materials	extraction	No	MY	meranti, standing, under bark, u=70%, in rainforest	2.00E+00	m³	calculated from IPCC (2003) GPG for LULUCF	1	1.26	(3,4,1,3,1,5)
diesel for harvesting operations	→			construction processes	machinery	No	GLO	diesel, burned in building machine	2.86E+02	MJ	estimated by Collins et al 2005 based on IPL/SHR (2002)	1	1.32	(4,4,1,3,1,5)
diesel for road construction	→			construction processes	machinery	No	GLO	diesel, burned in building machine	2.32E+01	MJ	estimated by Collins et al 2005 based on IPL/SHR (2002)	1	1.32	(4,4,1,3,1,5)
hydraulic oil for harvesting operations	→			chemicals	organics	No	RER	lubricating oil, at plant	4.32E-01	kg	estimated by Collins et al 2005 based on IPL/SHR (2002)	1	1.32	(4,4,1,3,1,5)
hydraulic oil/lube oil for road construction	→			chemicals	organics	No	RER	lubricating oil, at plant	7.86E-02	kg	estimated by Collins et al 2005 based on IPL/SHR (2002)	1	1.32	(4,4,1,3,1,5)
chainsawing, carburetor without catalytic converter	→		assumed to cover lube oil consumption inventoried in IPL/SHR 2002	construction processes	machinery	No	RER	power sawing, without catalytic converter	5.62E-02	h	estimated by Collins et al 2005 based on IPL/SHR (2002)	1	1.32	(4,4,1,3,1,5)
land transformation from forest	→		unknown n; not inventorised											
land transformation to forest roads	→		unknown n; not inventorised											
land occupation of forest roads	→		unknown n; not inventorised											
transportation of gravel (lorry 16 t)	→		unknown n; not inventorised											
gravel for forest roads	→		unknown n; not inventorised											
	→	roundwood meranti, at forest road (u=70%)	volume under bark; 100% allocation to roundwood production	wooden materials	extraction	No	MY	roundwood, meranti (SFM), under bark, u=70%, at forest road	1.00E+00	m³				
	→	branchings, burned	volume under bark; biomass expansion factor =2 (IPCC 2003); not inventorised						1.00E+00	m³	calculated from IPCC (2003) GPG for LULUCF			
solar energy released from branchings	→		volume under bark	resource	biotic			Energy, gross calorific value, in biomass	-1.83E+04	MJ	calculated from IPCC (2003) GPG for LULUCF	1	1.26	(3,4,1,3,1,5)
	→	CO2 regenerativ released from branchings	volume under bark	air	unspecified			Carbon dioxide, biogenic	1.18E+03	kg	calculated from IPCC (2003) GPG for LULUCF	1	1.26	(3,4,1,3,1,5)
	→	N2O from biomass burning	volume under bark	air	low population density			Dinitrogen monoxide	7.11E-03	kg	calculated based on IPCC 2003	1	1.59	(3,4,1,3,1,5)
	→	CH4 from biomass burning	volume under bark	air	low population density			Methane, biogenic	1.03E+00	kg	calculated based on IPCC 2003	1	1.59	(3,4,1,3,1,5)
		PM <2.5	4 g/kg	air	low population density			Particulates, > 2.5 um, and < 10um	3.00E+00	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
		PM >2.5>10	5 g/kg	air	low population density			Particulates, < 2.5 um	3.60E+00	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
		PM >10	6 g/kg	air	low population density			Particulates, > 10 um	2.40E+00	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
Meranti	→		resource correction for branchings	resource	biotic			Wood, hard, standing	-1.11E+00	m³	calculated	1	1.26	(3,4,1,3,1,5)

Figure 5.7: Flows for "roundwood, meranti (SFM), under bark, u=70%, at forest road" and its representation in the ecoinvent database

General Flow Information					Representation in ecoinvent								Uncertainty				
Input		Process Name		Output	Remarks	Category	Subcategory	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment	
roundwood meranti, at forest road (u=70%)	→	roundwood, meranti (SFM), debarked, u=70%, MY, at maritime harbour				wooden materials	extraction	No	MY	roundwood, meranti (SFM), under bark, u=70%, at forest road	1.00E+00	m3		1	1.26	(3,4,1,3,1,5)	
transportation to maritime harbour; diesel truck	→				200 km; including bark	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.26E+02	tkm	estimate	1	2.32	(5,5,1,3,1,5)	
freight ship to main European harbour	→				15000 km from Malaysia, via Singapur; excluding bark	transport systems	ship	No	OCE	transport, transoceanic freight ship	1.53E+04	tkm	estimate	1	2.32	(5,5,1,3,1,5)	
diesel used for debarking	→				manual debarking	construction processes	machinery	No	GLO	diesel, burned in building machine	0.00E+00	MJ	assumed based on Collins et al. 2005	1	1.26	(3,4,1,3,1,5)	
Infrastructure for debarking					unknown; not inventorised												
lubricant used for debarking	→				debarking before maritime transport; estimated based on European data	chemicals	organics	No	RER	lubricating oil, at plant	0.00E+00	kg	assumed based on Collins et al. 2005	1	1.26	(3,4,1,3,1,5)	
			→	roundwood meranti, debarked, at maritime harbour Europe (u=70%)	100% allocation to roundwood	wooden materials	processing	No	RER	roundwood, meranti (SFM), debarked, u=70%, MY, at maritime harbour	1.00E+00	m3					
			→	bark from debarking	net volume; not inventorised						1.10E-01	m3	calculated				
solar energy released from bark	→					resource	biotic			Energy, gross calorific value, in biomass	-1.85E+03	MJ	calculated	1	1.26	(3,4,1,3,1,5)	
			→	CO2 regenerativ released from bark		air	unspecified			Carbon dioxide, biogenic	1.19E+02	kg	calculated	1	1.26	(3,4,1,3,1,5)	
Meranti	→			resource correction for bark	resource	biotic			Wood, hard, standing	-1.10E-01	m3	calculated	1	1.26	(3,4,1,3,1,5)		

Figure 5.8: Flows for "roundwood, meranti (SFM), debarked, u=70%, MY, at maritime harbour" and its representation in the ecoinvent database

5.3 Paraná pine from plantations/managed natural forests in Brazil

5.3.1 Scope of the data set

The LCI data on Paraná pine from plantations/managed natural forests in Brazil is based on Alexandrino (2000), representing (close to) sustainable forest management (SFM). The data set comprises natural wood production, plantation establishment and management and harvesting from two data sources. The share of timber coming from plantations and managed natural forests is not known.

Data on debarking, sawmilling and technical drying stems from one of the companies.

It is assumed that the sawn timber is transported with an overseas freight ship to one of the main European harbours (BE/FE/DE).

The data set is structured into the following processes:

- Paraná pine standing, under bark, in rain forest
- Paraná pine (SFM), under bark, at forest road (u=50%)
- sawing/planing timber, Paraná pine, kiln dried, at Brazilian sawmill (u=15%), yielding:
 - sawn timber, Paraná pine (SFM), kiln dried, at Brazilian sawmill
 - industrial residual wood, Paraná pine (SFM), at Brazilian sawmill (u=15%)
- sawn timber, Paraná pine (SFM), at maritime harbour Europe (BE/NL/DE/FR) (u=15%)

Complementary data to company data was obtained from scientific journals, the internet, and a field visit to a logging site as well as consultation with forestry experts, in the fields of silviculture, economics, hydrology and soil science in Brazil.

The process data is assumed to represent best-practice close to certification for sustainable forest management.

5.3.2 Natural wood production, Paraná pine

Paraná pine is the common name of *Araucaria angustifolia*¹, a softwood tree species whose main population is found in Southern Brazil in the states of Paraná, Santa Catarina and Rio Grande do Sul. The tree is also found in smaller populations in Northern Argentina and Paraguay. Other common names of the tree are pinheiro-do-Paraná, cori, curiúva, pinhão, pinheiro-branco or pinheiro-brasileiro, while in Argentina and Paraguay it is known as kuri'y.

Paraná pine grows best on rich, brown-red soils, often called 'jungle' soil. This soil is argyllic, has a good depth and permeability, medium acidity and medium organic matter holding capacity.

Paraná pine is the dominant species of the moist subtropical forest, or paraná pine forest, both in abundance and height, forming the top layer of the forest canopy. The number of other tree species occurring in paraná pine forest varies with location, from 51 to 145. The most abundant of these species belong to the families of Lauraceae and Myrtaceae. Paraná pine is also associated with the occurrence of Imbuia (*Ocotea porosa*), and Erva-mate (*Ilex paraguariensis*).

Paraná pine occurs naturally at elevations between 500 to 1,800 m in Brazil, and up to 2,300m in Argentina and Paraguay. It occurs naturally between 19°15' south latitude and 30° south of the equator,

¹ There are 19 species in the *Araucaria* family. Two of them, *Araucaria araucana* and *Araucaria angustifolia* or *Araucaria brasiliana*, are found in South America, while the rest are distributed in Australia, New Caledonia, Papua-New Guinea and Norfolk Island. Despite their wide geographical and ecological variation, the *Araucarias* share similar wood properties and utilisation potential

and 41°20' and 54° west of Greenwich, in moderate temperatures of 15-18° C. The eastern limit of parana pine in Brazil is 30 to 60 km from the sea coast.

Table 5.6: Some properties and characteristics of Paraná pine timber
(<http://www.holzwissen.net/.html>)

Raw density (u=0%); g/cm ³	Raw density (u=15%); g/cm ³	Dimensional stability	Processability	Potential health aspects
0.5	0.575	medium to good	Tendency to form cracks during drying; easily workable; no problems with glueing or surface treatment; well polishable; easy cleavable; good machinability	Singular skin irritations

¹⁾ (Basic) wood density does not differ markedly between male and female trees.

Parana pine is a wind pollinated, dioecious species. Paraná pine seed dispersers are small animals, like birds, squirrels and many Brazilian rat species. However, natural regeneration problems have occurred as parana pine forests have been fragmented.

Until recently, Paraná pine was the most important species in the economy of Southern Brazil. The decimation of the natural forest and the introduction of incentives by the government in the 1970s to plant exotic Eucalyptus and Pinus species plantations reduced its economical importance.

However, in good quality areas, Paraná pine has the same incremental growth as exotic Pinus species. Genetic improvement and zoning to plant Paraná pine at appropriate edaphic and climatic conditions could increase the competitiveness of the species in the market. This potential has attracted the attention of the pulp and paper companies who are interested in initiating cultivation of the species.

The commercially important Araucarias in natural stands have long, branchless boles, giving high yields of wood free of knots. Parana pine maintains this property even when grown in a plantation, unlike the other Araucarias. Parana pine sheds its lower branches early in its development. The absence of knots improves the quality of pulp and particleboard. The residual branches of Paraná pine have numerous large knots which are live and tight. These properties of knots are preferable to dead knots that chip easier and increase compression.

Almost all parts of the trees including its nust, branches, resins etc.. are used. The timber is used for a variety of applications, including sawn timber, boards, furniture, mouldings, etc.

Table 5.6 summarizes some properties and characteristics of Paraná pine timber.

The solar energy uptake is calculated as the upper heating value (26.25 MJ/kg at u=70%¹) according to Frischknecht et al. (2007a, Eq. 1) based on a lower heating value² of 18.26 MJ/kg.

A carbon content of 49.4% of oven-dried wood is assumed in line with Frischknecht et al. (2007a) assuming an average density of 500 kg/m³ (u=0%).

The background information to the processes are given in Table 5.7, flows are shown in Figure 5.9.

¹ Alexandrino 2000

² <http://www.matbase.com/material/wood/class4-5-10-years/parana-pine/properties>

Parana pine has been classified as ‘Critically endangered’ species according to the IUCN categories¹. Since 2001 there has been an official Brazilian ban on log exports of this species. The Brazilian Government is also promoting several initiatives to protect *Araucaria* genetic resources.

5.3.3 Forestry processes and direct forest products

Data on forestry processes and direct forest products stems from two vertically integrated large-scale companies in the Departement Paraná, Brazil². Both companies are involved in forestry research on parana pine in collaboration with other institutes (universities and government agencies) and in conservation and environmental education. The companies have obtained environmental certification, or are very near the award of an FSC environmental certificate. Their forestry practices can thus be considered as best practice.

Company 1 is located in the South-West of Parana. The company has 24'000 ha of natural Paraná pine forest, some of which have been managed in the past, and 12'000 ha of parana pine plantations. The natural forest areas are designated as conservation areas. In Company 1 the parana pine plantations are young, at 15 years of age, and the production comes either from thinnings or from an area of older trees, planted by the previous owner. Earlier thinnings (dbh of 6-25 cm) are used in paper packaging material. There is a new line of panel and moulding products and parana pine is planned to be included in it.). The total annual volume of timber processed is 13'000 m³. Of this 1'500 m³ is Paraná pine, with the prospect of increasing to 5'000-6'000 m³ after certification. The annual production of Paraná pine sawn timber is 700 m³, with an output of 300 m³ of final product. The orientation of the company is towards the external market.

Company 1 plants Paraná pine in the forest with some scarification, when necessary, and use of pesticide and herbicide. The last planting was in 1993 (236 ha), until this year, when 150 ha were planted. A harvester/forwarder system is currently being introduced, a transition from the manual system used before. Debarking is done mechanically at the sawmill, while brushings are removed in the forest. There is some research regarding the ecologically optimal amount of offcuts left on site. Carrying ways are opened for harvesting. Permanent roads are covered with gravel from the company's own mine.

Company 2 is situated in the North-East of Curitiba. The plantation area of Paraná pine is nearly 9'000 ha. This company also holds reserve areas with natural parana pine forests of 1'000 ha. Company 2 already holds certification for its pine, eucalyptus and parana pine plantations. It is possible that commercial exploitation of the species will cease in the future, and it will be kept only for conservation purposes. Company 2 has worked only with parana pine until the 1970s. The company has an annual average parana pine production of 13.6 m³/ha over the 45 years of a Paraná pine plantation rotation. Only the timber harvested after the 4th thinning, at 28 years of age and onwards, and the final clear-cut are used for the production of sawn wood. Earlier thinnings are destined for the production of high quality paper in the company's sawmill. All production is directed to the internal market. However, it might be possible that products are exported indirectly via national traders.

Company 2 plants parana pine seeds in the forest. Mechanical scarification is used. Planting is manual and apart from early applications of herbicide and pesticide, there is no other tending of the seedlings. Planting is infrequent, with the last two plantings of 100 ha each happening two and five years ago. Before felling, the plantation is thinned five times. Cutting of trees is manual, as the trees are too big for machinery. A skidder removes the trees. The undergrowth is cleared before cutting. The logs are

¹ <http://www.iucnredlist.org/search/details.php/32975/all>

² a third company mainly obtaining their logs from (managed) natural forest is disregarded as the data provided in Alexandridou (2000) does not provide credible data related to forest operations

sold with bark, and the offcuts are left on site, and often burned. Normally, new roads are not required for harvesting, and the existing ones are provided with gravel from an on-site mine.

The background information to the processes are given in Table 5.7, flows are shown in Figure 5.10.

5.3.4 Sawn timber production in Brazilian sawmill

An average transport distance of 90 km for the forest to a sawmill are assumed based on Bauch, Amacher *et al.* (2007). Data on debarking and sawmilling is only available from one of the two companies. The yield for sawn timber is 45%.

Due to a lack of detailed information, technical drying is assumed, using part of the residual wood; energy requirements and respective wood inputs are calculated based on average European processes for softwood, based on Werner, *et al.* (2007), referring to Ressel (1986).

Revenue-based allocation is applied to allocate the sawing processes to the resulting sawn timber and industrial residual wood.

The background information to the processes are given in Table 5.7, flows are shown in Figure 5.11.

5.3.5 Allocation correction for Paraná pine sawmill processes

Revenue-based allocation requires the application of allocation correction factors for material-inherent properties (carbon content and energy content. See Werner *et al.* (2007) and Hischier *et al.* (2004) for details.

The background information to the allocation correction are given in Table 5.8, flows are shown in Figure 5.12.

5.3.6 Transportation of sawn timber to main maritime port in Europe*

It is assumed that the sawn timber is transported to a close harbour in Brazil by a lorry. The main exportation port, where 90% of the export production is directed, is Sao Francisco in Santa Catarina. An average distance of 450 km is assumed.

It is assumed that the sawn timber is brought to one of the main ports in Europe (Rotterdam, Antwerpen, Hamburg, Bremerhafen) by an overseas freight ship (9000) km.

However, it should be noted that currently an export ban is in place on Brazilian Paraná pine (*) (Alexandrido 2000). Similar products might be imported to Europe from Argentina, implying a slightly longer overseas travel distance.

The background information to the processes are given in Table 5.7, flows are shown in Figure 5.13.

Table 5.7: Ecoinvent meta information to the datasets for "paraná pine, standing, under bark, in rain forest", "roundwood, paraná pine (SFM), under bark, u=50%, at forest road", "sawing and planing, timber, paraná pine, kiln dried, u=15%" and "sawn timber, paraná pine (SFM), u=15%, BR, at maritime harbour"

Name	paraná pine, standing, under bark, in rain forest	roundwood, paraná pine (SFM), under bark, u=50%, at forest road	sawing and planing, timber, paraná pine, kiln dried, u=15%	sawn timber, paraná pine (SFM), u=15%, BR, at maritime harbour
Location	BR	BR	BR	RER
Infrastructure Process	0	0	0	0
Unit	m3	m3	m3	m3
Data Set Version	2	2	2	2
Included Processes	includes the CO2 absorption from air and the land use for the trees to grow (including 15% bark). Forest road area is not included in the land useDensity: 500 kg/m3 (u=0)	includes plantation establishment and management, harvesting and hauling operations incl. emissions from burning of branchings; fuel consumption and gravel for forest road construction is unknown; land use of forest roads and gravel are disregarded; branchings are left to rot on the ground	includes transportation to sawmill by lorry; sawing into timber; technical drying is assumed; 45% of log input (including sapwood) is generated as co-product that is partly used for technical drying and partly sold or is used for energy recovery used outside this process.	includes transportation of the sawn timber from the sawmill to a main harbour in Europe by truck and freight ship.
Amount	1	1	1	1
Local Name	Paraná Kiefer, stehend, Festkubikmeter, in Regenwald	Rundholz, Paraná Kiefer (NWB), Festkubikmeter, u=50%, ab Waldstrasse	Sägen-Hobeln, Schnittholz, Paraná Kiefer, kammergetrocknet, u=15%	Schnittholz, Paraná Kiefer (NWB), u=15%, BR, ab Meerhafen Europa
Synonyms	Araucaria angustifolia	Araucaria angustifolia	Araucaria angustifolia the multi-output process	Araucaria angustifolia
General Comment to reference function	data refers to wood 'under bark'. CO2 assimilation is based on 49.4% carbon in the wood including the bark (15%). Biomass energy equals gross calorific value including bark.	dataset represents average of two well-documented, large-scale logging companies. both companies own also natural forests where (some) Paraná pine is extracted. Percentage of plantation trees and trees from managed natural forests as well as intensity of management are not known. Data of the two companies differs considerably. emissions from biomass burning according to IPCC 2003	"sawing and planing, timber, paraná pine, kiln dried, u=15%" delivers the two coproducts "sawn timber, paraná pine (SFM), kiln dried, u=15%, at sawmill" and "industrial residue wood, paraná pine (SFM), u=15%, at sawmill". Allocation is based on the over-all proceeds of the process; resource use of natural wood growth is corrected via a mass-based "allocation correction". Data stems from one well-documented, large, vertically	data refers to wood from Brazil, used in Europe. main ports in Europe include Rotterdam, Hamburg, Bremerhafen or Antwerpen. Currently, there is an export ban in place for Brazilian Paraná pine; products might come from Argentina with slightly longer transport distances
Start Date	2000	2000	2000	2000
End Date	2005	2005	2005	2005
Data Valid For Entire Period	1	1	1	1
Other Period Text				
Geography text			wood consumption for technical drying is estimated based on European data	
Technology text	natural growth; relates to the forest area and CO2 from air needed to produce Paraná pine	state-of-the art technology; forest operations largely based on manual labor, some mechanisation is currently undertaken	average European data. depending on sawing and drying technology, specific wood density of a particular log and type of sawn timber produced, data may vary	average processes for transportation to Europe; transport distance however can vary
Representativeness [%]				
Production Volume	unknown	unknown	unknown	unknown
Sampling Procedure	literature	literature, partly relying on estimates (fuel consumption)	literature	literature
Extrapolations	none	none	see Geography	none
Uncertainty Adjustments	none	none	none	none

Table 5.8: Ecoinvent meta information to the datasets for "paraná pine, allocation correction 1" and "paraná pine, allocation correction 2"

Name	paraná pine, allocation correction 1	paraná pine, allocation correction 2
Location	GLO	GLO
Infrastructure Process	0	0
Unit	m3	m3
Data Set Version	2	2
Included Processes	corrects economically allocated resource use, feedstock energy and carbon uptake to fit with physical flows (=mass-based allocation)	corrects economically allocated resource use, feedstock energy and carbon uptake to fit with physical flows (=mass-based allocation)
Amount	1	1
Local Name	Paraná Kiefer, Allokationskorrektur 1	Paraná Kiefer, Allokationskorrektur 2
Synonyms		
General Comment to reference function	auxiliary module	auxiliary module
Start Date	2000	2000
End Date	2005	2005
Data Valid For Entire Period	1	1
Other Period Text		
Geography text		
Technology text		
Representativeness [%]		
Production Volume	none	none
Sampling Procedure	none	none
Extrapolations	none	none
Uncertainty Adjustments	none	none

General Flow Information					Representation in ecoinvent								Uncertainty		
Input		Process Name	Output	Remarks	Category	Subcategory	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
Paraná pine	→	Paraná pine standing, under bark, in rain forest		including 20 % bark (50 cm diameter stemwood plus 2 x 2 cm bark)	resource	biotic			Wood, soft, standing	1.15E+00	m3	calculated	1	1.26	(3,4,1,3,1,5)
solar energy	→			equals to gross calorific value	resource	biotic			Energy, gross calorific value, in biomass	1.51E+04	MJ	lower heating value: 18.47 MJ/kg (http://www.matbase.com/material/wood/class4-5-10-years/parana-pine/properties); upper heating value calculated according to Frischknecht et al. (2003), Eq. 1: 26.25 MJ/kg	1	1.26	(3,4,1,3,1,5)
CO2 regenerativ	→			dried matter =500 kg/m3, C content of dried matter = 49.4%	resource	in air			Carbon dioxide, in air	1.04E+03	kg	calculated	1	1.26	(3,4,1,3,1,5)
transformation, from unknown	→			average: 13.6 m3/ha; rotation length: 45 years; quantity of co-extraction unknown	resource	land			Transformation, from forest, intensive, normal	1.60+E01	m2		1	2.00	(,,,,)
transformation, to plantation, intensive	→			geographically not representative	resource	land			Transformation, to forest, intensive, normal	1.60+E01	m2		1	2.00	(,,,,)
land occupation	→			Total above-ground biomass for American pine plantations under moist tropical climate with long dry season (IPCC 2003, Table 3A.1.2.): 110 t d.m./ha; BEF2 (IPCC 2003, Table 3A.1.10): 1.3; rotation length: 45 years; geographically not representative	resource	land			Occupation, forest, intensive	3.06E+03	m2a		1	1.50	(,,,,)
		→	Paraná pine standing, under bark, in rain forest	under bark (Inputs refer to 1.15 m3 oven-dried Paraná pine (u=0%) including bark); moisture content according to Alexandridiou (2000)	wooden materials	extraction	No	BR	paraná pine, standing, under bark, in rain forest	1.00E+00	m3				

Figure 5.9: Flows for "paraná pine, standing, under bark, in rain forest" and its representation in the ecoinvent database

General Flow Information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
Paraná pine, standing, under bark, in plantation (u=50%)	→			wooden materials	extraction	No	BR	paraná pine, standing, under bark, in rain forest	2.00E+00	m³	calculated from IPCC (2003) GPG for LULUCF	1	1.26	(3,4,1,3,1,5)
diesel for site preparation/mechanical planting/harvesting operations incl. Skidding	→		could contain also lube oil	construction processes	machinery	No	GLO	diesel, burned in building machine	1.68E+03	MJ	calculated based on Alexandridiou 2000	1	1.26	(3,4,1,3,1,5)
diesel for road construction	→		unknown n; not inventorised											
hydraulic oil for harvesting operations	→		unknown n; possibly considered under diesel consumption for forest management activities											
hydraulic oil/lube oil for road construction	→		unknown n; not inventorised											
chainsawing, carburetor without catalytic converter	→			construction processes	machinery	No	RER	power sawing, without catalytic converter	4.00E-01	h	calculated based on Alexandridiou 2000	1	1.26	(3,4,1,3,1,5)
herbicides/pesticides inorganic	→		Aluminium phosphide, Sulphuramide; covers only production of herbicides/pesticides	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	2.30E-03	kg	calculated based on Alexandridiou 2000	1	1.60	(3,4,1,3,4,5)
Herbicides/Pesticides organic	→		Glyphosphate, Oxyfluorfen; covers only production of herbicides/pesticides	chemicals	organics	No	GLO	chemicals organic, at plant	1.10E-01	kg	calculated based on Alexandridiou 2000	1	1.60	(3,4,1,3,4,5)
Water for herbicide application	→		process data not very adequate for Brazilian conditions	water supply	production	No	RER	tap water, at user	6.88E+01	kg	calculated based on Alexandridiou 2000	1	1.26	(3,4,1,3,1,5)
land transformation from forest	→		unknown n; not inventorised											
land transformation to forest roads	→		unknown n; not inventorised											
land occupation of forest roads	→		unknown n; not inventorised											
transportation of gravel (lorry 16 t gravel) for forest roads	→		unknown n; not inventorised											
			unknown n; not inventorised											
	→	roundwood Paraná pine, at forest road (u=50%)	volume under bark; 100% allocation to roundwood production	wooden materials	extraction	No	BR	roundwood, paraná pine (SFM), under bark, u=50%, at forest road	1.00E+00	m³				
	→	branchings, burned	volume under bark; biomass expansion factor =2 (IPCC 2003); not inventorised						1.00E+00	m³	calculated from IPCC (2003) GPG for LULUCF			
solar energy released from branchings	→		volume under bark	resource	biotic			Energy, gross calorific value, in biomass	-1.72E+04	MJ	calculated from IPCC (2003) GPG for LULUCF	1	1.26	(3,4,1,3,1,5)
	→	CO2 regenerativ released from branchings	volume under bark	air	unspecified			Carbon dioxide, biogenic	1.18E+03	kg	calculated from IPCC (2003) GPG for LULUCF	1	1.26	(3,4,1,3,1,5)
	→	N2O from biomass burning	volume under bark	air	low population density			Dinitrogen monoxide	7.11E-03	kg	calculated based on IPCC 2003	1	1.59	(3,4,1,3,1,5)
	→	CH4 from biomass burning	volume under bark	air	low population density			Methane, biogenic	1.03E+00	kg	calculated based on IPCC 2003	1	1.59	(3,4,1,3,1,5)
		PM <2.5	4 g/kg	air	low population density			Particulates, > 2.5 um, and < 10um	2.50E+00	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
		PM >2.5>10	5 g/kg	air	low population density			Particulates, < 2.5 um	3.00E+00	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
		PM >10	6 g/kg	air	low population density			Particulates, > 10 um	2.00E+00	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
Paraná pine	→		resource correction for branchings	resource	biotic			Wood, hard, standing	-1.15E+00	m³	calculated	1	1.26	(3,4,1,3,1,5)

Figure 5.10: Flows for "roundwood, paraná pine (SFM), under bark, u=50%, at forest road" and its representation in the ecoinvent database

General Flow Information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infrastructure	Location	Module name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	General Comment
roundwood Paraná pine, at forest road (u=50%)	→		volume under bark	w ooden materials	extraction	No	BR	roundwood, paraná pine (SFM), under bark, u=50%, at forest road	2.19E+00	m³	calculated based on Alexandridiou 2000	1	1.26	(3,4,1,3,1,5)
transportation to saw mill	→		90 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	1.70E+02	tkm	calculated based on Bauch et al 2007	1	2.07	(3,4,1,3,1,5)
electricity for debarking/ sawing	→		UCTEmtx assumed for Brazil	electricity	supply mix	No	BR	electricity, medium voltage, at grid	3.20E+02	kWh	calculated based on Alexandridiou 2000	1	1.26	(3,4,1,3,1,5)
electricity for drying	→		UCTEmtx assumed for Brazil	electricity	supply mix	No	BR	electricity, medium voltage, at grid	6.55E+02	kWh	calculated based on Alexandridiou 2000	1	1.26	(3,4,1,3,1,5)
Petrol for machine maintenance	→		0.06 l/kg w ood (u=15%); seems extremely high	oil	fuels	No	RER	petrol, low-sulphur, at regional storage	3.38E+01	kg	calculated based on Alexandridiou 2000	1	1.26	(3,4,1,3,1,5)
Dryer	→		no data available	w ooden materials	extraction	Yes	RER	technical w ood drying, infrastructure	1.83E+05	unit	calculated based on Alexandridiou 2000	1	3.30	(3,4,1,3,4,5)
Saw mill	→		no data available	w ooden materials	extraction	Yes	RER	saw mill	4.00E+07	unit	estimate based on Werner/Althaus 2003	1	3.07	(3,4,1,3,1,5)
residual w ood and bark burned for thermal energy for drying	→			w ood energy	heating systems	No	CH	w ood chips, from industry, softwood, burned in furnace 300kW	7.66E+02	MJ	calculated based on Werner/Althaus 2003, referring to Ressel 1986	1	1.60	(3,4,1,3,4,5)
correction for w ood supply	→		based on Bauer 2003, Table 9.17	w ood energy	fuels	No	RER	w ood chips, softwood, from industry, u=40%, at plant	-2.55E+01	m³	calculated based on Werner/Althaus 2003 and Bauer 2003	1	1.26	(3,4,1,3,1,5)
correction for electricity consumption of dryer	→			electricity	production mix	No	CH	electricity, low voltage, production CH, at grid	-3.19E+00	kWh	calculated based on Werner/Althaus 2003 and Bauer 2003	1	1.26	(3,4,1,3,1,5)
correction for transportation	→			transport systems	road	No	CH	transport, lorry 3.5-20t, fleet average	-1.72E+00	tkm	calculated based on Werner/Althaus 2003 and Bauer 2003	1	2.07	(3,4,1,3,1,5)
correction for ashes to municipal waste incineration	→			w aste management	municipal incineration	No	CH	disposal, w ood ash mixture, pure, 0% w ater, to municipal incineration	3.04E+05	kg	calculated based on Werner/Althaus 2003 and Bauer 2003	1	1.26	(3,4,1,3,1,5)
correction for landfilling of ash	→			w aste management	sanitary landfill	No	CH	disposal, w ood ash mixture, pure, 0% w ater, to sanitary landfill	-2.73E+01	kg	calculated based on Werner/Althaus 2003 and Bauer 2003	1	1.26	(3,4,1,3,1,5)
correction for all ashes to landfarming	→		supposed to be stored inadequately causing runoff and leakage -> approximated; total ash to landfarming = 0.365 kg	w aste management	landfarming	No	CH	disposal, w ood ash mixture, pure, 0% w ater, to landfarming	2.73E+01	kg	calculated based on Werner/Althaus 2003 and Bauer 2003	1	1.26	(3,4,1,3,1,5)
	→	saw n timber, Paraná pine, kiln dried, at Brazilian saw mill (u=12%)		w ooden materials	processing	No	BR	saw n timber, paraná pine (SFM), kiln dried, u=15%, at saw mill	1.00E+00	m³				
	→	industrial residual w ood, Paraná pine	Wood input incl. Bark minus saw n timber minus w ood required to produce required energy, assumed 80% burner efficiency, 18.264 MJ/kg lower heating value	w ooden materials	processing	No	BR	industrial residual w ood, paraná pine (SFM), u=15%, at saw mill	1.42E+00	m³	calculated based on Werner/Althaus 2003, referring to Ressel 1986			
Paraná pine, saw n timber, allocation correction	→		including bark	w ooden materials	processing	No	GLO	paraná pine, allocation correction 1	-1.32E+00	m³	calculated based on Hischer et al 2004	1	1.26	(3,4,1,3,1,5)
Paraná pine, industrial residual w ood, allocation correction	→		including bark	w ooden materials	processing	No	GLO	paraná pine, allocation correction 2	1.32E+00	m³	calculated based on Hischer et al 2004	1	1.26	(3,4,1,3,1,5)
	→	VOC from machine maintenance		air	unspecified			NM/VOC, non-methane volatile organic compounds, unspecified origin	3.38E+01	kg	extrapolated	1	1.59	(3,4,1,3,1,5)
	→	w aste heat		air	unspecified			Heat, w aste	1.68E+03	MJ		1	1.26	(3,4,1,3,1,5)
	→	correction CO2 for burning w ood for drying	minus CO2 inventorised in incineration module plus w ood required to produce thermal energy, bulk density for chips= 169 kg/m³, assumed 80% burner efficiency, converted to CO2	air	unspecified			Carbon dioxide, biogenic	1.31E+01	kg	calculated	1	1.26	(3,4,1,3,1,5)

Figure 5.11: Flows for "sawing and planing, timber, paraná pine, kiln dried, u=15%" and its representation in the ecoinvent database

General Flow Information						Representation in ecoinvent								Uncertainty		
Input		Process Name		Output	Remarks	Category	Subcategory	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
Paraná pine	→	Paraná pine standing, under bark, in rain forest			including 15 % bark (50 cm diameter stemwood plus 2 x 2 cm bark)	resource	biotic			Wood, soft, standing	1.00E+00	m3	calculated	1	1.26	(3,4,1,3,1,5)
solar energy	→				equals to gross calorific value	resource	biotic			Energy, gross calorific value, in biomass	1.31E+04	MJ	lower heating value: 18.47 MJ/kg (http://www.matbase.com/material/wood/class4-5-10-years/parana-pine/properties); upper heating value calculated according to Frischknecht et al. (2003), Eq. 1: 26.25 MJ/kg	1	1.26	(3,4,1,3,1,5)
CO2 regenerativ	→				dried matter =500 kg/m3, C content of dried matter = 49.4%	resource	in air			Carbon dioxide, in air	9.04E+02	kg	calculated	1	1.26	(3,4,1,3,1,5)
			→	Paraná pine, allocation correction 1		wooden materials	processing	No	GLO	paraná pine, allocation correction 1	1.00E+00	m3				
Paraná pine	→	Paraná pine standing, under bark, in rain forest			over bark	resource	biotic			Wood, soft, standing	1.00E+00	m3	calculated	1	1.26	(3,4,1,3,1,5)
solar energy	→				equals to gross calorific value	resource	biotic			Energy, gross calorific value, in biomass	1.31E+04	MJ	lower heating value: 18.47 MJ/kg (http://www.matbase.com/material/wood/class4-5-10-years/parana-pine/properties); upper heating value calculated according to Frischknecht et al. (2003), Eq. 1: 26.25 MJ/kg	1	1.26	(3,4,1,3,1,5)
CO2 regenerativ	→					resource	in air			Carbon dioxide, in air	9.04E+02	kg	calculated	1	1.26	(3,4,1,3,1,5)
			→	Paraná pine, allocation correction 2		wooden materials	processing	No	GLO	paraná pine, allocation correction 2	1.00E+00	m3				

Figure 5.12: Flows for "paraná pine, allocation correction 1" and "paraná pine, allocation correction 2" and their representation in the ecoinvent database

General Flow Information						Representation in ecoinvent								Uncertainty		
Input		Process Name		Output	Remarks	Category	Subcategory	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
saw n timber, Paraná pine, kiln dried, at Brazilian saw mill (u=12%)	→	saw n timber, paraná pine (SFM), u=15%, BR, at maritime harbour				w ooden materials	processing	No	BR	saw n timber, paraná pine (SFM), kiln dried, u=15%, at saw mill	1.00E+00	m3	calculated based on Alexandridiou 2000	1	1.26	(3,4,1,3,1,5)
transportation to maritime harbour; diesel truck	→				450 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.59E+02	tkm	calculated based on Alexandridiou 2000	1	2.07	(3,4,1,3,1,5)
freight ship to main European harbour	→				9000 km	transport systems	ship	No	OCE	transport, transoceanic freight ship	5.18E+03	tkm	calculated based on Alexandridiou 2000	1	2.07	(3,4,1,3,1,5)
			→	saw n timber, Paraná pine, at maritime harbour Europe* (BE/NL/DE/FR) (u=12%)	*) currently, there is an export ban in place for Paraná pine from Brazil	w ooden materials	processing	No	RER	saw n timber, paraná pine (SFM), u=15%, BR, at maritime harbour	1.00E+00	m3				

Figure 5.13: Flows for "sawn timber, paraná pine (SFM), u=15%, BR, at maritime harbour" and its representation in the ecoinvent database

5.3.7 Data quality

Data quality is assessed using the pedigree matrix as described in Frischknecht et al. (2007a).

Data for CO₂-assimilation and for the energy embodied in the biomass are calculated from the carbon and energy content. Wood density and other technical parameters vary considerably within a species.

Land use data can vary considerably depending on how intensive the forest is used. Since the data used are valid for one specific company and region, the uncertainty is rather high.

Forestry processes stem from two modern, vertically integrated timber companies applying principles of sustainable forest management. This cannot be assumed to be prevailing practice in Brazil.

Both companies own and use managed natural forests and plantations; the relative share of these forest types, its forest management interventions and its related inputs/environmental impacts cannot be distinguished from Alexandrino (2000).

Data on sawing the timber stems from one of the two companies; UCTE electricity mix is assumed for Brazilian electricity consumption, due to a lack of information on the actual Brazilian electricity mix.

5.4 Eucalyptus ssp. from plantations in Thailand

5.4.1 Scope of the data set

The LCI data on *Eucalyptus ssp.* is based on Ongmongkolkul (2001), representing (close to) sustainable forest management (SFM). The data set comprises natural wood production, plantation establishment and management and harvesting.

The roundwood is debarked and pulped.

It is assumed that the unbleached pulp is transported with an overseas freight ship to one of the main European harbours (BE/FE/DE).

The data set is structured into the following processes:

- Eucalyptus ssp., standing, under bark, in plantation (u=50%)
- roundwood eucalyptus ssp. (SFM), at forest road (u=50%)
- sulphate pulp from eucalyptus ssp. (SFM), unbleached, at pulpmill in Thailand
- sulphate pulp from eucalyptus ssp. (SFM), unbleached, from Thailand, at maritime harbour Europe (BE/NL/DE/FR)

Complementary data to company data was obtained by Ongmongkolkul (2001) from scientific journals, the internet, and a field visit to a logging site as well as consultation with forestry experts, in the fields of silviculture, economics, hydrology and soil science in Thailand.

Representativeness of these datasets are supposed to be low, as European data was used by Ongmongkolkul (2001) for almost all relevant processes due to a lack of data from Thailand.

5.4.2 Natural wood production, Eucalyptus ssp.

Eucalyptus is not a species, but a group of 20 or more commercially important species with their origin in Australia and Tasmania. Wood properties vary depending on species, site conditions, genetic material, etc¹. In general, however, eucalyptus is a very good wood and is used for construction, fine furniture and everything in between. It is common to have internal stresses in the wood (growth stres-

¹ see, e.g. <http://www.csudh.edu/oliver/chemdata/woods.htm>; Raymond (2002)

ses, not drying stresses). Accommodating these stresses can require extra effort. Some eucalyptus species are quite dense, and require special care when machining.

Eucalypts have been grown in Thailand for 100 years, but only after 1970 on plantation scale. Eucalypt advantages are: easy grown, good survival, tolerant to various climates and soil types; no proven negative effects on soil, environment, human and wildlife; positive support to farm income; withstands long dry seasons, pathogen resistant; has wide domestic and industrial use.

The dataset assumes the most widely used eucalyptus species *E. camaldulensis* in Thailand¹.

The solar energy uptake is calculated as the upper heating value (26.81 MJ/kg at $u=50\%$)² according to Frischknecht et al. (2007a, Eq. 1) based on a lower heating value³ of 18.83 MJ/kg.

A carbon content of 49.4% of oven-dried wood is assumed in line with Frischknecht et al. (2007b) assuming an average density of 825 kg/m³ ($u=0\%$)⁴.

The background information to the processes are given in Table 5.9, flows are shown in Figure 5.14.

5.4.3 Forestry processes and direct forest products

Although many species of eucalypts were introduced into Thailand, only a few species and provenances show promising survival and growth. *E. camaldulensis* seems to be the first choice, as far as survival and growth rate is concerned. It is adaptable to saline soils in the north eastern Thailand, and tolerates a variety of climatic conditions. Other species which are well adapted include *E. brassiana*, *E. citriodora*, *E. deglupta*, *E. grandis*, *E. paniculata*, *E. resinifera*, *E. saligna*, *E. tereticornis*, *E. urophylla* and *E. cloeziana*.

Domestic energy supply remains a major problem; 82% of population use wood (40 mill. tons) and charcoal (3 mill. tons) annually - a serious cause of natural forest degradation. Pulpwood is a major market; important other markets are scaffolding and piling. Ecological concern peaked in 1986-89, fading in 1990, these emphasized possible damage to water supplies and soil qualities and adverse impact on wildlife. Plantation studies in differing tree crop ages showed: no significant difference in soil texture; in 13 chemical properties including total acidity, organic matter and cation exchange the study showed similar result; a trend of increasing organic matter was noted. Credible discussion concluded there was no convincing evidence of eucalypt ill effect on soils, water and wildlife, but further studies were needed on environment impact. Farmer experience echoes this in fish pond plantings, rice field cultivation and cash intercropping; eucalypt productivity increases with compatible intercrops. Honey and edible fungi production is an increasing food crop combination with eucalypts. Eucalypt farming competes with agricultural crops, is highly acceptable to farmers, is considered as replacement for

¹ Large evergreen tree 24–40(–50) m high with stout trunk often short and crooked, to 2 m in diameter; crown open, widely spreading, irregular. Bark smoothish, white, gray, or buff. Twigs reddish, long, slender, angled, drooping. Trunk can form air roots. Root system deep and spreading. Leaves alternate, drooping, narrowly lanceolate, 8–22 cm long, 1–2 cm wide, often curved or sickle-shaped, tapering to long point, short-pointed at base, entire glabrous, dull pale green on both surfaces or occasionally grayish. Umbels single at leaf base, ca 2.5 cm long on slender stalk 6–19 mm long. Flowers 5–10, each on slender stalk 5–12 mm long from ovoid buds 6–10 mm long, 4–5 mm wide. Stamens many, threadlike, white, 5–6 mm long; anthers with small round gland. Pistil with inferior, long-pointed, 3–4-celled ovary and long, stout style. Capsules several, clustered, hemiglobose or ovoid, 7–8 mm long, 5–6 mm wide, light brown, with wide raised disk and 3–4 prominent triangular teeth almost 2 mm long. Seeds many, tiny, 1.5 mm long, light brown

Source: http://www.hort.purdue.edu/newcrop/duke_energy/Eucalyptus_camaldulensis.html#Description

² assumed

³ average between 19.42 to 18.23 MJ/kg as reported by Jenkins and Ebeling (1985)

⁴ Ongmongkolkul (2001)

crops, e.g. cassava. The farmer community is capable of value judgment in eucalypt planting which is showing a marked upturn.¹

Observations on many stands show that there is considerable variation in their performance. These variations are most likely a result of the use of wrong seed sources, i.e. seed from unsuitable provenances or from undesirable mother trees or due to inbreeding effects. Seed stands or seed production areas of eucalypt have been considered and were established by cooperation between state and private sectors. The seed from qualified land race mother trees of *E. camaldulensis* in Thailand and of sources in Australia are used, but the seed stands are still young, and only small amounts of seed can be harvested presently.

Some provenances coppice well for six or more rotations, on good sites, plantations are managed on coppice rotations.

The dataset represents a short-rotation plantation of *eucalyptus ssp.* for pulp and paper production. During plantation and growing, some fertilizers are used; however, applied quantities are unknown. Trees are harvested mechanically around year 3 at a dbh of 20 centimeters and an average height of 3 meters; 17.5% of tree is loss from delimbing. The production of 228.75 kilograms of log or 3.6 trees needs land of 4.16 m² per year.

Branchings are assumed to be burned on site; the amount of branchings and respective emissions from biomass burning are estimated based on IPCC (2003) and EPA (1995).

Since it had no information about logging in Thailand, it was assumed that raw materials use and air emissions are the same as logging in Denmark.

The background information to the processes are given in Table 5.9, flows are shown in Figure 5.15.

5.4.4 Production of pulp, unbleached, from *Eucalyptus ssp.*

The logs are transported about 50 km to a pulpmill. They are debarked and chipped before entering the Kraft process. The Kraft process (also known as Kraft pulping or sulfate process) is used in production of paper pulp and involves the use of caustic sodium hydroxide and sodium sulfide to extract the lignin from wood chips in large pressure vessels called digesters. Some digesters operate in a batch manner and some in a continuous process such as the Kamyr digester.

The spent, extracted pulping liquor, called black liquor, is concentrated by evaporation and burned in the recovery boiler to recover the inorganic chemicals for reuse in the pulping process. The recovery boiler also generates high pressure steam for the mill processes. The inorganic portion of the black liquor is used to regenerate the sodium hydroxide and sodium sulfide needed for pulping in a process called causticizing.

The high pressure steam, from the recovery boiler, is led to turbogenerators, reducing the steam pressure for the mill use and generating electricity. A modern kraft pulp mill is more than self-sufficient in its electrical generation and normally will provide a net flow of energy to the local electrical grid. Additionally: bark and wood residues are often fired in a separate power boiler to generate steam.

Specific data for Thailand is not available in Ongmongkolkul (2001). Therefore, the debarking pulping processes are modelled as described in Hischier (2007, Chap. 7.2.2), for unbleached sulphate pulp production. UCTE electricity mix is assumed for all electricity consumptions; all wastes are assumed to be landfilled.

The background information to the processes are given in Table 5.9, flows are shown in Figure 5.16:

¹ <http://www.fao.org/DOCREP/005/AC772E/ac772e0m.htm>

5.4.5 Transportation of Kraft pulp, unbleached, to main maritime port in Europe

It is assumed that the pulp is transported to a maritime harbour in Thailand by a lorry. An average distance of 450 km is assumed.

It is assumed that the pulp is brought to one of the main ports in Europe (Rotterdam, Antwerpen, Hamburg, Bremerhafen) by an overseas freight ship (15'000) km

The background information to the processes are given in Table 5.9, flows are shown in Figure 5.17.

5.4.6 Data quality

Data quality is assessed using the pedigree matrix as described in Frischknecht et al. (2007a).

Data for CO₂-assimilation and for the energy embodied in the biomass are calculated from the carbon and energy content. Wood density and other material-characteristic parameters can vary considerably within a species.

Land use data can vary considerably depending on how intensive the forest is used. Since the data used are valid for one specific company and region, the uncertainty is rather high.

For the most relevant processes such as forestry processes or debarking/pulping, European data sets are used (following Ongmongkolkul 2001) and partly adapted; thus, the reliability and representativeness of these data sets for the Thai situation is supposed to be 'best available' but rather low.

Table 5.9: Ecoinvent meta information to the datasets for "eucalyptus ssp., standing, under bark, u=50%, in plantation", "roundwood, eucalyptus ssp. (SFM), under bark, u=50%, at forest road", "sulphate pulp, from eucalyptus ssp. (SFM), unbleached, at pulpmill" and "sulphate pulp, from eucalyptus ssp. (SFM), unbleached, TH, at maritime harbour"

Name	eucalyptus ssp., standing, under bark, u=50%, in plantation	roundwood, eucalyptus ssp. (SFM), under bark, u=50%, at forest road	sulphate pulp, from eucalyptus ssp. (SFM), unbleached, at pulpmill	sulphate pulp, from eucalyptus ssp. (SFM), unbleached, TH, at maritime harbour
Location	TH	TH	TH	RER
Infrastructure Process	0	0	0	0
Unit	m3	m3	kg	kg
Data Set Version	2	2	2	2
Included Processes	includes the CO2 absorption from air and the land use for the trees to grow (including 15% bark). Forest road area is not included in the land use. Density: 825 kg/m3 (u=0)	includes plantation establishment and management, harvesting and hauling operations, incl biomass burning of branchings; fuel consumption and gravel for forest road construction is unknown; land use of forest roads and gravel are disregarded; branchings are left to rot on the ground	includes transport of logs to pulp mill, debarking and pulping using the Kraft process; bleaching excluded	includes transportation of the pulp from the pulp mill in Thailand to a main harbour in Europe by truck and freight ship. Main ports in Europe include Rotterdam, Hamburg, Bremerhafen or Antwerpen.
Amount	1	1	1	1
Local Name	Eucalyptus ssp., stehend, Festkubikmeter, u=50%, in Pflanzung	Rundholz, Eucalyptus ssp. (NWB), Festkubikmeter, u=50%, ab Waldstrasse	Sulfatzellstoff, aus Eucalyptus ssp. (NWB), ungebleicht, ab Zellstoffwerk	Sulfatzellstoff, aus Eucalyptus ssp. (NWB), ungebleicht, TH, ab Meerhafen Europa
Synonyms	Eucalyptus camaldulensis	Eucalyptus camaldulensis		
General Comment to reference function	data refers to wood 'under bark'. CO2 assimilation is based on 49.4% carbon in the wood including the bark (10%). Biomass energy equals gross calorific value including bark.	data refers to wood from sustainably managed plantation. uses Dutch data on forestry processes; emissions from biomass burning according to IPCC 2003	adjusted European data	data refers to pulp from Thailand, used in Europe.
Start Date	2000	2000	2000	2000
End Date	2005	2005	2005	2005
Data Valid For Entire Period	1	1	1	1
Other Period Text				
Geography text		source assumes Dutch process data, due to the lack of specific data	assumes average European technology; UCTE electricity mix; landfilling of all wastes	
Technology text	natural growth; relates to the forest area and CO2 from air needed to produce eucalyptus	the representativity of this data set for forestry conditions in Thailand is assumed to be low	assumes average European technology	
Representativeness [%]				
Production Volume	unknown	unknown	unknown	unknown
Sampling Procedure	literature	literature	literature	literature
Extrapolations	none	none	none	none
Uncertainty Adjustments	none	see Technology and Geography	see Technology and Geography	see Technology and Geography

General Flow Information					Representation in ecoinvent								Uncertainty			
Input		Process Name		Output	Remarks	Category	Subcategory	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
Eucalypus	➔	eucalyptus ssp., standing, under bark, u=50%, in plantation			including 10 % bark (20 cm diameter stemwood plus 0.5 x 0.5 cm bark)	resource	biotic			Wood, hard, standing	1.10E+00	m3	calculated	1	1.26	(3,4,1,3,1,5)
solar energy	➔				equals to gross calorific value	resource	biotic			Energy, gross calorific value, in biomass	2.43E+04	MJ	lower heating value: 18.83 MJ/kg (calculated from Jenkins et al 1985); upper heating value calculated according to Frischknecht et al. (2003), Eq. 1: 26.81 MJ/kg	1	1.26	(3,4,1,3,1,5)
CO2 regenerativ	➔				dried matter =825 kg/m3, C content of dried matter = 49.4%	resource	in air			Carbon dioxide, in air	1.64E+03	kg	calculated	1	1.26	(3,4,1,3,1,5)
transformation, from unknown	➔				assumed rotation length: 20 years	resource	land			Transformation, from forest, intensive, short-cycle	1.50E+01	m2	estimated	1	2.00	(,,,,)
transformation, to plantation, short-rotation	➔				assumed rotation length: 20 years; geographically not representative	resource	land			Transformation, to forest, intensive, short-cycle	1.50E+01	m2	estimated	1	2.00	(,,,,)
land occupation	➔				geographically not representative	resource	land			Occupation, forest, intensive, short-cycle	3.00E+02	m2a	Ongomongkolkul 2001	1	1.50	(,,,,)
			➔	Eucalyptus ssp., standing, under bark, in plantation (u=50%)	under bark (Inputs refer to 1.10 m3 oven-dried Eucalyptus ssp. (u=0%) including bark); moisture content assumed	wooden materials	extraction	No	TH	eucalyptus ssp., standing, under bark, u=50%, in plantation	1.00E+00	m3				

Figure 5.14: Flows for "eucalyptus ssp., standing, under bark, u=50%, in plantation" and its representation in the ecoinvent database

General Flow Information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Subcategory	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
Eucalyptus ssp., standing, under bark, in plantation (u=50%)	→			wooden materials	extraction	No	TH	eucalyptus ssp., standing, under bark, u=50%, in plantation	1.20E+00	m³	Ongomongkolkul 2001	1	1.26	(3,4,1,3,1,5)
diesel for site preparation/mechanical planting/harvesting operations incl. Skidding	→			construction processes	machinery	No	GLO	diesel, burned in building machine	6.48E+01	MJ	Ongomongkolkul 2001; assuming Dutch processes (!)	1	1.26	(3,4,1,3,1,5)
diesel for road construction	→		unknown n; not inventorised											
hydraulic oil for harvesting operations	→			chemicals	organics	No	RER	lubricating oil, at plant	3.00E-01	kg	Ongomongkolkul 2001; assuming Dutch processes (!)	1	1.26	(3,4,1,3,1,5)
hydraulic oil/tube oil for road construction	→		unknown n; not inventorised											
chainsawing, carburetor without catalytic converter	→			construction processes	machinery	No	RER	power sawing, without catalytic converter	2.50E-01	h	Ongomongkolkul 2001; assuming Dutch processes (!)	1	1.26	(3,4,1,3,1,5)
Fertilizers	→		type and quantity unknown n; not inventorised											
herbicides/pesticides inorganic	→		unknown n											
Herbicides/Pesticides organic	→		unknown n; not inventorised											
Water for herbicide application	→		unknown n; not inventorised											
land transformation from forest	→		unknown n; not inventorised											
land transformation to forest roads	→		unknown n; not inventorised											
land occupation of forest roads	→		unknown n; not inventorised											
transportation of gravel (lorry 16 t)	→		unknown n; not inventorised											
gravel for forest roads	→		unknown n; not inventorised											
	→	roundwood Eucalyptus ssp., at forest road (u=50%)	volume under bark; 100% allocation to roundwood production	wooden materials	extraction	No	TH	roundwood, eucalyptus ssp. (SFM), under bark, u=50%, at forest road	1.00E+00	m³				
	→	branchings, burned	from delimbing; not inventorised						2.00E-01	m³	calculated based on Ongomongkolkul 2001			
solar energy released from branchings	→		volume under bark	resource	biotic			Energy, gross calorific value, in biomass	-4.87E+03	MJ	calculated based on Ongomongkolkul 2001	1	1.26	(3,4,1,3,1,5)
	→	CO2 regenerativ released from branchings	volume under bark	air	unspecified			Carbon dioxide, biogenic	3.28E+02	kg	calculated based on Ongomongkolkul 2001	1	1.26	(3,4,1,3,1,5)
	→	H2O from biomass burning	volume under bark	air	low population density			Dinitrogen monoxide	1.97E-03	kg	calculated based on IPCC 2003	1	1.59	(3,4,1,3,1,5)
	→	CH4 from biomass burning	volume under bark	air	low population density			Methane, biogenic	2.87E-01	kg	calculated based on IPCC 2003	1	1.59	(3,4,1,3,1,5)
		PM <2.5	4 g/kg	air	low population density			Particulates, > 2.5 um, and < 10um	8.25E-01	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
		PM >2.5>10	5 g/kg	air	low population density			Particulates, < 2.5 um	9.90E-01	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
		PM >10	6 g/kg	air	low population density			Particulates, > 10 um	6.60E-01	kg	estimated based on US EPA 1995	1	1.59	(3,4,1,3,1,5)
Eucalyptus	→		resource correction for branchings	resource	biotic			Wood, hard, standing	-2.20E-01	m³	calculated	1	1.26	(3,4,1,3,1,5)

Figure 5.15: Flows for "roundwood, eucalyptus ssp. (SFM), under bark, u=50%, at forest road" and its representation in the ecoinvent database

General Flow Information				Representation in ecoinvent								Uncertainty		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
roundwood Eucalyptus ssp. at forest road (u=50%)	→		volume under bark; 100% allocation to roundwood production	wooden materials	extraction	No	TH	roundwood, eucalyptus ssp. (SFM), under bark, u=50%, at forest road	2.27E-03	m3		1	2.08	(3,4,1,3,5,5)
Transportation to pulpmill	→	20 km		transport systems	road	No	RER	transport, lorry >16t, fleet average	1.54E-01	tkm		1	2.74	(3,4,1,3,5,5)
debarking, chipping, pulping	→			paper & cardboard	pulps	No	RER	sulphate pulp, unbleached, at plant	1.00E+00	kg		1	2.08	(3,4,1,3,5,5)
correction for wood input 1	→		subtraction of value in dataset	wooden materials	extraction	No	RER	industrial wood, hardwood, under bark, u=80%, at forest road	-5.05E-04	m3		1	1.05	(1,1,1,1,1,1)
correction for wood input 2	→		subtraction of value in dataset	wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140%, at forest road	-7.31E-04	m3		1	1.05	(1,1,1,1,1,1)
correction for wood input 3	→		subtraction of value in dataset	wooden materials	extraction	No	RER	industrial residue wood, softwood, forest-debarked, u=70%, at plant	-3.14E-04	m3		1	1.05	(1,1,1,1,1,1)
correction for wood input 4	→		subtraction of value in dataset	wooden materials	extraction	No	NORDEL	industrial wood, Scandinavian hardwood, under bark, u=80%, at forest road	-6.78E-04	m3		1	1.05	(1,1,1,1,1,1)
correction for wood input 5	→		subtraction of value in dataset	wooden materials	extraction	No	NORDEL	industrial wood, Scandinavian softwood, under bark, u=140%, at forest road	-9.81E-04	m3		1	1.05	(1,1,1,1,1,1)
correction for wood input 6	→		subtraction of value in dataset	wooden materials	extraction	No	NORDEL	crude, Scandinavian softwood (plant-debarked), u=70%, at plant	-4.21E-04	m3		1	1.05	(1,1,1,1,1,1)
correction for electricity mix 1	→		change from NORDEL to UCTE; subtraction NORDEL	electricity	production mix	No	NORDEL	electricity, medium voltage, production NORDEL, at grid	-2.38E-01	kWh		1	1.05	(1,1,1,1,1,1)
correction for electricity mix 2	→		change from NORDEL to UCTE; addition UCTE	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.38E-01	kWh		1	2.08	(3,4,1,3,5,5)
correction for transport train	→		subtraction transportation in dataset	transport systems	train	No	RER	transport, freight, rail	-7.46E-02	tkm		1	2.00	(1,1,1,1,1,1)
correction for transport lorry 32 t	→		subtraction transportation in dataset	transport systems	road	No	RER	transport, lorry >16t, fleet average	-1.57E-01	tkm		1	2.00	(1,1,1,1,1,1)
waste to landfill instead of municipal incineration	→		correction waste disposal scenario: waste to landfill instead MWI	waste management	municipal incineration	No	CH	disposal, municipal solid waste, 22.9% water, to municipal incineration	1.16E-03	kg		1	2.08	(3,4,1,3,5,5)
waste correction 1	→		subtraction waste in dataset	waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3% water, to municipal incineration	-7.31E-04	kg		1	1.05	(1,1,1,1,1,1)
waste correction 2	→		subtraction waste in dataset	waste management	municipal incineration	No	CH	disposal, paper, 11.2% water, to municipal incineration	-2.32E-04	kg		1	1.05	(1,1,1,1,1,1)
waste correction 3	→		subtraction waste in dataset	waste management	municipal incineration	No	CH	disposal, textiles, soiled, 25% water, to municipal incineration	-7.54E-05	kg		1	1.05	(1,1,1,1,1,1)
waste correction 4	→		subtraction waste in dataset	waste management	municipal incineration	No	CH	disposal, wood untreated, 20% water, to municipal incineration	-4.06E-05	kg		1	1.05	(1,1,1,1,1,1)
waste correction 5	→		subtraction waste in dataset	waste management	municipal incineration	No	CH	disposal, steel, 0% water, to municipal incineration	-8.12E-05	kg		1	1.05	(1,1,1,1,1,1)
	→	Kraft pulp from Eucalyptus ssp., at pulpmill in Thailand	allocation 100%; u=10% according to IPCC 2003, Table 3a.1.1)	paper & cardboard	pulps	No	TH	sulphate pulp, from eucalyptus ssp. (SFM), unbleached, at pulpmill	1.00E+00	kg				
	→	tall oil	not inventorised						2.55E-02	kg				
	→	turpentine	not inventorised						3.50E-03	kg				
	→	correction of C-balance	carbon content pulp: 0.5 kg/kg o.d.m. (IPCC 2003, Table 3a.1.1); input carbon minus carbon in pulp minus biogenic carbon emissions from incineration module	air	unspecified			Carbon dioxide, biogenic	1.60E+00	kg	calculated	1	1.26	(3,4,1,3,1,5)

Figure 5.16: Flows for “sulphate pulp, from eucalyptus ssp. (SFM), unbleached, at pulpmill” and its representation in the ecoinvent database

General Flow Information					Representation in ecoinvent								Uncertainty			
Input		Process Name		Output	Remarks	Category	Subcategory	Infrastructure	Location	Modul name in Ecoinvent 2000	Mean value	Unit	Source for mean value	Type	Standard Deviation 95%	GeneralComment
Kraft pulp from Eucalyptus ssp., at pulpmill in Thailand	→	sulphate pulp, from eucalyptus ssp. (SFM), unbleached, at pulpmill				paper & cardboard	pulps	No	TH	sulphate pulp, from eucalyptus ssp. (SFM), unbleached, at pulpmill	1.00E+00	kg		1	1.26	(3,4,1,3,1,5)
transportation to maritime harbour; diesel truck	→			450 km		transport systems	road	No	RER	transport, lorry >16t, fleet average	4.50E-01	tkm	calculated based on Ongomongkolkul 2001	1	2.07	(3,4,1,3,1,5)
freight ship to main European harbour	→			15000 km		transport systems	ship	No	OCE	transport, transoceanic freight ship	1.50E+01	tkm	estimated referring to transport distance from Malaysia	1	2.07	(3,4,1,3,1,5)
			→	Kraft pulp from Eucalyptus ssp., unbleached, from Thailand, at maritime harbour Europe (BE/NL/DE/FR)		paper & cardboard	pulps	No	RER	sulphate pulp, from eucalyptus ssp. (SFM), unbleached, TH, at maritime harbour	1.00E+00	kg				

Figure 5.17: Flows for “sulphate pulp, from eucalyptus ssp. (SFM), unbleached, TH, at maritime harbour” and its representation in the ecoinvent database

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Part II

Life Cycle Inventories of Renewable Fibres

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1 Cotton production, China

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Last changes:

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Abstract

The inventories refer to the cotton production from the main producer China. The multioutput-process 'cotton production' delivers the co-products 'cotton fibres, ginned, at farm' and 'cotton seed, at farm'. The allocation is based on economic criteria and mass balance.

The inventories of Chinese cotton production refer to conventional production standards and their emissions to the air, water and soil, calculated for the use of fertilizers and pesticides. In the conventional production mainly mineral fertilizers are applied. The cotton production is related to high consumption of pesticides.

The inventory includes the cultivation, harvest, transport and separation of fibres in a ginning plant. The reference function of co-products is 1 kg cotton fibres and 1 kg cotton seed respectively.

1.1 Characteristic of Chinese cotton production

Conventional cotton production for Chinese standards is described in Tab. 3.1 and compared to the US cotton production. The average net yield for US production standards is about 960 kg fibres/ha and for China about 1100 kg fibres/ha (Karst Kooistra and Aad Termorshuizen, 2006). Net yield of raw cotton varies between about 1000-4000 kg cotton depending on region and production standards. Raw cotton consists of about 35% fibres, 62% seeds and 3% other plant components (Wiegmann K., 2002).

US cotton farms are usually large intensive monocultures using high-yielding annual cotton varieties without rotations crops. Chinese cotton farms often cultivate annual cotton in rotation with other annual crops. The US production is about 19% of the world production with about 5.3 Mha of cotton producing land, the Chinese production delivers about 24% of world production with about 5.6 Mha of cotton producing land (Karst Kooistra and Aad Termorshuizen, 2006). The cultivation time from seed to harvest is about 175 to 225 days, the overall land occupation is assumed one year for the cotton production in this inventory.

Almost all nutrients are applied as synthetic fertilizers, most of the Chinese fields and about 41% of the US cotton fields are irrigated (Karst Kooistra and Aad Termorshuizen, 2006). The cotton production is related to high pesticides need (estimation 11% of pesticide production).

Tab. 1.1 Characteristics of cotton production, US conventional standards

Multioutput process	Net yield [kg /ha*a]	Cultivation time	Comments
cotton production: - cotton fibres - cotton seed	1000-4000 kg raw cotton (35% fibres, 62% seeds, 3% rest) US average fibres yield: 965 kg fibres Chinese average fibres yield: 1100 kg fibres	Occupation arable land for about one year (natural cotton perennial) About 175 to 225 days from seed to harvest	Mainly mineral fertilizers, high need of pesticides US: Irrigation on about 41% of cotton production fields (50% sprinkler, 50% flood irrigation) China: Most fields irrigated (90% flood irrigation, 5% furrow, 3% drip and 2% sprinkler)

1.1.1 Input from Technosphere

Data of the cotton production is based mostly on working processes given in the report about cultivation and processing of cotton from the Öko-Insitut (Wiegmann K., 2002) and on irrigation and fertilization in the report about sustainability of cotton from the Wageningen University (Karst Kooistra and Aad Termorshuizen, 2006). Energy need for the separation of seed and fibres in the ginning process is about 2.3 MJ/kg ginned fibres (Schmidt, 1999; citation in Wiegmann K., 2002). For the electricity need in the ginning process the UCTE production mix was considered as standard applying the ecoinvent quality guidelines for electricity use apart from Europe.

The inputs from technosphere include the machine usage of the soil cultivation, the spreading of fertilizers and pesticides, the harvest²⁷ as well as the energy consumption for the extraction of fibres from seeds in the ginning process. For the machine usage and the transport existing working processes from the ecoinvent report No. 15 (Nemecek et al., 2004) were applied unchanged. Processes defined per units or ha were normalized for the output of one kg fibres or seed by division with the yield obtained (kg per ha and year).

For the applied mineral fertilizers existing inventories were applied unchanged (Nemecek et al., 2004). Emissions of application on the field were included. The total amount of fertilizers is from an experts survey done 2005 by national panels mainly consisting of staff from ministries of agriculture (Karst Kooistra and Aad Termorshuizen, 2006). For the N-fertilization a share of 50% liquid ammonium, 25% urea and 25% ammonium nitrate was assumed. Triplesuperphosphate and Kalimchlorid were considered for P and K fertilization.

Applied pesticides are derived from the above mentioned experts survey and statistics about average doses (Karst Kooistra and Aad Termorshuizen, 2006). A transport of 100 km with camion was included for the transport of raw cotton to the ginning plant and of 50 km with a van <3.5t for the consumed fertilizers and pesticides delivered from the next regional storehouse.

Tab. 1.2 Soil cultivation and input goods for the production of cotton

	Soil cultivation and field working processes	Fertilization	Pesticides applications
cotton production, US	ploughing, harrowing, seeding, hoeing, 2-3 fertilizations, irrigation, applications of pesticides (about 6x field, 6x tractor and 3x arial spraying), harvests (twice picker and once with stripper).	USA: 100 kg N 90 kg P 80 kg K	2- 3 herbicides (dinitroanilins) 4 - 15 insecticides (carbamates, organophosphates, pyrethroides) Ø 5 unspecific (fungicide benzoate) Defoliant on most fields (carfentrazone)
cotton production, China	ploughing, harrowing, seeding, hoeing, 2-3 fertilizations, irrigation, about 20 applications of pesticides (tractor spraying), harvests (picking manually).	China: 225-300 kg N 50-150 kg P 100-220 kg K	2- 3 herbicides (MSMA/prometryn, glyphosate, alachlor, fluometuron) 4 - 15 insecticides (carbamates, organophosphates, pyrethroides) Ø 5 unspecific Defoliant on about 5% of fields (MSMA)

²⁷ Chinese cotton picked manually

1.1.2 Emissions to the air (N₂O, NO_x, Ammonia)

Emissions were calculated with the formula for N₂O, NO_x and Ammonia as described in the ecoinvent report No. 15 (Nemecek et al. 2004, Chapter 4.4). Relevant parameters are the application of fertilizers and the nitrogen fixation of the vegetation.

1.1.3 Emissions to the water (Phosphate, Phosphorus, Nitrate)

Emissions were calculated with the formula for Phosphate leaching to the ground water, for Phosphate and Nitrate effluent to surface water and for the Phosphorus loss with erosion as in the ecoinvent report No. 15 (Nemecek et al. 2004, Chapter 4.4). The Nitrate leaching was estimated about 10% of applied kg N, due to lacking data for a more detailed calculation. Emissions depend mainly on the amount and type of fertilizers applied, on the time period of the fertilization and the cultivation of soils.

1.1.4 Emissions to the soil (heavy metals)

An Input – Output balance was calculated with background data about heavy metal contents of plants, seeds and fertilizers in ecoinvent report No. 15 (Nemecek et al. 2004). No specific data about cotton heavy metal contents was available, an average value calculated from data about different plant types was used instead.

Only inputs and outputs directly related to the agricultural production were considered. As in the ecoinvent report No. 15, output through leaching, run-off and erosion - partly due to the agricultural production - were not included, due to the allocation problems (Nemecek et al. 2004).

1.1.5 CO₂-binding and Solar Energy in Biomass of cotton

A value for CO₂-binding of 1.6 kg CO₂/kg DM and an energy content of 18 MJ/kg DM was used for cotton fibres and seed. Those values lie within the range of different crop types (Nemecek et al., 2004). Allocation to seed and fibres was carried out on mass balance.

1.1.6 Land occupation of cotton production

US cotton farms are usually large intensive monocultures using high-yielding annual varieties without rotations crops. Chinese cotton farms often cultivate annual cotton in rotation with other annual crops. The cultivation time from seed to harvest is about 175 to 225 days, as no other crop is grown in between the overall land occupation is assumed one year for the US cotton production in this inventory.

Only irrigated land was considered for China. No transformation to arable land from other land use types as for example pasture and meadows was considered.

1.1.7 Cotton ginning

The energy consumption for the extraction of cotton fibres from cotton seed in a ginning plant is about 2.3 MJ/kg ginned fibres (Wiegmann K., 2002). For the electricity need in the ginning process the UCTE production mix was considered as standard applying the ecoinvent quality guidelines for electricity use apart from Europe.

Infrastructure and size of ginning plant was considered similar than plant for packaging industry. The existing data inventory of ‘packaging box production unit’ was applied assuming 50 years life time and an output of about 50’000 t fibres annually.

1.1.8 Seed, at farm

No special inventory was defined for the seed production. The co-product of fibres production ‘cotton seed, at farm’ was used as input product without processing and storage of seed (assumed not relevant).

1.2 Allocation to co-products fibres and seed

The allocation of cotton production to co-products ‘cotton fibres, ginned, at farm’ and cotton seed, at farm’ is based mainly on economic criteria and for organic material related values on mass balance.

In the United States, which is the world's second largest cotton producer behind China and just ahead of India, about 15% of the producer's income is from the seed portion of the crop²⁸. For the allocation on economic criteria related to the income 15% for seed and 85% for fibres was used. The same factors were applied for Chinese cotton.

Only for organic material related values CO₂ binding, Solar energy in biomass as well as related balance of heavy-metals in soil the mass balance was used for allocation. The harvested raw cotton consists of about 64% seed and 36% fibres (Wiegmann K., 2002).

For the allocation by mass balance 64% for seed and 36% for fibres was applied.

1.3 Statistics of cotton production²⁹

The world cotton area is estimated at around 30-31 million hectares (about 2.4% of the arable land), the production of fibres about 20 million tons per year and co-product cotton seed about 35 million tons per year.

Cotton is produced in more than 100 countries, the most important countries being China (24% of global cotton production), the USA (19%), India (16%), Pakistan (10%), Brazil (5%) and Uzbekistan (4%). The major producers of cotton are also the major producers of cotton seed.

Conventional systems represent about 80% of cotton production internationally. Integrated Pest Management (IPM) systems represent approximately 20% and organic systems 0.04%.

1.4 Characterisation and use of cotton³⁰

Cotton is a soft fibre that grows around the seeds of the cotton plant (*Gossypium* sp.). The fiber is most often spun into yarn or thread and used to make a soft, breathable textile, which is the most widely used natural-fibre cloth in clothing today. Cotton fibre makes up roughly 50% of world fibre consumption.

Cotton seed is a by-product of the cotton plant, which is primarily grown for its fibre. Cotton seed is the second most commonly produced oilseed in the world. However, cotton seed oil falls to about fourth or fifth in the world production of oils, due to the relatively low amount of oil in the seed and to the great amount of cotton seed, which is fed unprocessed to cattle, especially dairy cattle.

1.5 Life Cycle Inventories of cotton production

Inventory refers to the production of 1 kg cotton or 1 kg fibres or 1 kg seed respectively. Inventories include the soil cultivation, fertilization, application of pesticides, irrigation, harvest, ginning and related transport to the ginning plant and farm.

²⁸ Data from http://www.mcxindia.com/products_CottonSeeds.html (June 2007).

²⁹ Data from www.wikipedia.org and www.mcxindia.com (June 2007)

³⁰ Data from www.wikipedia.org and www.mcxindia.com (June 2007)

Tab. 1.3 Unit process of cotton production, China (US not shown anymore)

	Name	Location	Infrastructure	Process	Unit	cotton production	Uncertainty type	Standard Deviation	GeneralComment	cotton fibres, ginned, at farm	cotton seed, at farm	
	Location InfrastructureProcess Unit					CN 0 kg				CN 0 kg	CN 0 kg	
allocated	cotton fibres, ginned, at farm	CN	0	kg	1					100	0	
products	cotton seed, at farm	CN	0	kg	1.76E+0					0	100	
technosphere	tillage, ploughing	CH	0	ha	9.09E-4	1	1.07	(2.2.1.1.1.1); 4h/ha working time (Baumwoll Datendokumentation, Ökoinstitut, K. Wiegmann, 2002)		85.0	15.0	
	tillage, harrowing, by spring tine harrow	CH	0	ha	9.09E-4	1	1.07	(2.2.1.1.1.1); 1h/ha working time together with seeding (Baumwoll Datendokumentation, Ökoinstitut, K. Wiegmann, 2002)		85.0	15.0	
	hoeing	CH	0	ha	9.09E-4	1	1.07	(2.2.1.1.1.1); 1.5 h/ha working time(Baumwoll Datendokumentation, Ökoinstitut, K. Wiegmann, 2002)		85.0	15.0	
	tillage, curving, by weeder	CH	0	ha	1.82E-3	1	1.07	(2.2.1.1.1.1); 2 x 1.5 h/ha working time(Baumwoll Datendokumentation, Ökoinstitut, K. Wiegmann, 2002)		85.0	15.0	
	sowing	CH	0	ha	9.09E-4	1	1.07	(2.2.1.1.1.1); 1h/ha working time together with harrowing (Baumwoll Datendokumentation, Ökoinstitut, K. Wiegmann, 2002)		85.0	15.0	
	fertilising, by broadcaster	CH	0	ha	2.73E-3	1	1.07	(2.2.1.1.1.1); Assumption Carbotech AG, 2-3 applications of fertilizer		85.0	15.0	
	application of plant protection products, by field sprayer	CH	0	ha	1.82E-2	1	1.07	(2.2.1.1.1.1); Average application of pest control, about 20 applications per year (90% tractor), additional defoliation on about 5% of areas (Baumwolle Datendokumentation, Ökoinstitut, K. Wiegmann, 2002).		85.0	15.0	
	baling	CH	0	unit	5.88E-3	1	1.22	(2.1.1.3.3.1); 150 to 240 kg per piece (Baumwoll Datendokumentation, Öko-Institut, K. Wiegmann, 2002). similar procedure as for grass baling. Same process applied.		85.0	15.0	
	resource, in water	Water, well, in ground	-	-	m3	5.93E+0	1	1.83	(5.2.2.1.4.5); Most areas irrigated, about 90% flood-irrigation, 5% furrow and 3% drip (Report 223, the sustainability of cotton). Estimation of water need about 9000 m3, with 74% ground water and 26% from rivers		85.0	15.0
	resource, in water	Water, river	-	-	m3	2.08E+0	1	1.83	(5.2.2.1.4.5); Most areas irrigated, about 90% flood-irrigation, 5% furrow and 3% drip (Report 223, the sustainability of cotton). Estimation of water need about 9000 m3, with 74% ground water and 26% from rivers		85.0	15.0
	technosphere	irrigating	US	0	m3	1.64E-1	1	1.83	(5.2.2.1.4.5); Fast alle Flächen bewässert, 2% Sprinkler (Report 223, the sustainability of cotton)		85.0	15.0
		parathion, at regional storehouse	RER	0	kg	5.58E-4	1	1.31	(4.3.2.2.3.3); Average amount insecticide 1.84 kg/ha (Baumwoll Datendokumentation, Öko-Institut), distribution on three most frequent applied pesticide types (Report 223, The sustainability of cotton)		85.0	15.0
		pyrethroid-compounds, at regional storehouse	RER	0	kg	5.58E-4	1	1.31	(4.3.2.2.3.3); Average amount insecticide 1.84 kg/ha (Baumwoll Datendokumentation, Öko-Institut), distribution on three most frequent applied pesticide types (Report 223, The sustainability of cotton)		85.0	15.0
		organophosphorus-compounds, at regional storehouse	RER	0	kg	5.58E-4	1	1.31	(4.3.2.2.3.3); Average amount insecticide 1.84 kg/ha (Baumwoll Datendokumentation, Öko-Institut), distribution on three most frequent applied pesticide types (Report 223, The sustainability of cotton)		85.0	15.0
		pesticide unspecified, at regional storehouse	RER	0	kg	2.73E-3	1	1.31	(4.3.2.2.3.3); Average amount herbicides 1-6 kg/ha (Baumwoll Datendokumentation, Öko-Institut), most frequent applied substances are MSMA/prometryn, glyphosate, alachlor und fluometuron dinitroanilin (Report 223, The sustainability of cotton)		85.0	15.0
		glyphosate, at regional storehouse	RER	0	kg	2.73E-3	1	1.31	(4.3.2.2.3.3); Average amount herbicides 1-6 kg/ha (Baumwoll Datendokumentation, Öko-Institut), most frequent applied substances are MSMA/prometryn, glyphosate, alachlor und fluometuron dinitroanilin (Report 223, The sustainability of cotton)		85.0	15.0
chemicals organic, at plant		GLO	0	kg	2.27E-5	1	1.31	(4.3.2.2.3.3); MSMA for defoliation (Report 223, The sustainability of cotton). Assumption Carbotech AG, about 0.5 kg Entlaubungsmittel pro ha, auf 5% Flächen angewendet		85.0	15.0	
urea, as N, at regional storehouse		RER	0	kg	5.91E-2	1	1.22	(4.2.2.1.1.3); N-fertilization amount from Report 223, The sustainability of cotton (assumption 50% ammonium liquid, 25 % ammonium nitrate, 25% urea and 25% Ammoniumphosphate		85.0	15.0	
ammonia, liquid, at regional storehouse		RER	0	kg	1.18E-1	1	1.22	(4.2.2.1.1.3); N-fertilization amount from Report 223, The sustainability of cotton (assumption 50% ammonium liquid, 25 % ammonium nitrate, 25% urea and 25% Ammoniumphosphate		85.0	15.0	
ammonium nitrate, as N, at regional storehouse		RER	0	kg	5.91E-2	1	1.22	(4.2.2.1.1.3); N-fertilization amount from Report 223, The sustainability of cotton (assumption 50% ammonium liquid, 25 % ammonium nitrate, 25% urea and 25% Ammoniumphosphate		85.0	15.0	
triple superphosphate, as P2O5, at regional storehouse		RER	0	kg	9.09E-2	1	1.22	(4.2.2.1.1.3); Amount from Report 223, The sustainability of cotton. Choice of fertilizer type by Carbotech AG		85.0	15.0	
potassium chloride, as K2O, at regional storehouse		RER	0	kg	1.45E-1	1	1.22	(4.2.2.1.1.3); Amount from Report 223, The sustainability of cotton. Choice of fertilizer type by Carbotech AG		85.0	15.0	
operation, van < 3.5t		RER	0	vkM	1.77E-2	1	2.35	(4.4.3.3.4.5); Assumption Carbotech AG, delivery of mineral fertilizers from storehouse to farm about 50 km		85.0	15.0	
transport, lorry 3.5-16t, fleet average		RER	0	tkM	2.86E-1	1	2.35	(4.4.3.3.4.5); Assumption Carbotech AG, Transport harvested cotton to gin plant about 100 km		85.0	15.0	
electricity, low voltage, production UCTE, at grid		UCTE	0	kWh	6.39E-1	1	1.33	(2.2.3.3.3.5); Energy need from Baumwoll Datendokumentation, Öko-Institut (K. Wiegmann, 2002)		85.0	15.0	
packaging box production unit		RER	1	unit	4.00E-10	1	3.51	(5.1.1.5.4.5); Estimation plant comparable to packaging plant (building, equipment, electronics). Assumption life time 50 years with about 2000 h work time and output of 25' 000 t of fibres per year.		85.0	15.0	
emission air, low population density	Heat, waste	-	-	MJ	2.30E+0	1	1.84	(5.1.1.5.4.5); Heat waste derived from electricity consumption		85.0	15.0	
	Dinitrogen monoxide	-	-	kg	6.03E-3	1	1.90	(4.4.3.3.4.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)		85.0	15.0	
	Ammonia	-	-	kg	2.44E-2	1	1.70	(4.4.3.3.4.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004), assumption Carbotech AG for applied mineral fertilizers 50% liquid ammonium, 25 % urea and 25% ammoniumnitrate		85.0	15.0	
	Nitrogen oxides	-	-	kg	1.27E-3	1	1.90	(4.4.3.3.4.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)		85.0	15.0	
emission water, river	Phosphate	-	-	kg	4.55E-4	1	1.90	(4.4.3.3.4.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)		85.0	15.0	
emission water, ground-	Phosphate	-	-	kg	1.46E-4	1	1.90	(4.4.3.3.4.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)		85.0	15.0	
	Nitrate	-	-	kg	1.04E-1	1	1.90	(4.4.3.3.4.5); Assumption average leaching about 10% (not sufficient data about climates, soils, application period for calculation with formula as in ecoinvent report No. 15)		85.0	15.0	
emission water, river	Phosphorus	-	-	kg	4.61E-4	1	1.90	(4.4.3.3.4.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004), lacking data assumed as given in example and land cover chosen as for beans		85.0	15.0	
emission soil, agricultural	Cadmium	-	-	kg	1.59E-6	1	1.89	(4.3.3.3.4.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)		85.0	15.0	
	Chromium	-	-	kg	1.09E-4	1	1.89	(4.3.3.3.4.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)		85.0	15.0	
	Copper	-	-	kg	-5.25E-8	1	1.89	(4.3.3.3.4.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)		85.0	15.0	
	Mercury	-	-	kg	-7.33E-8	1	1.89	(4.3.3.3.4.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)		85.0	15.0	
	Nickel	-	-	kg	3.62E-6	1	1.89	(4.3.3.3.4.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)		85.0	15.0	
	Lead	-	-	kg	3.52E-6	1	1.89	(4.3.3.3.4.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)		85.0	15.0	
	Zinc	-	-	kg	3.39E-6	1	1.58	(4.3.3.3.4.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)		85.0	15.0	
	Monocrotophos	-	-	kg	9.76E-5	1	1.69	(4.3.3.2.4.5); Assumption input pesticides = input in soil (Monocrotophos, cyfluthrin, dicolol, trichlorform, imidacoprid, chlordimeform, piperonyl as most frequent applied substances)		85.0	15.0	
	Cyfluthrin	-	-	kg	9.76E-5	1	1.69	(4.3.3.2.4.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)		85.0	15.0	
	Dicolol	-	-	kg	9.76E-5	1	1.69	(4.3.3.2.4.5); Assumption input pesticides = input in soil (Monocrotophos, cyfluthrin, dicolol, trichlorform, imidacoprid, chlordimeform, piperonyl as most frequent applied substances)		85.0	15.0	
	Trichlorfon	-	-	kg	9.76E-5	1	1.69	(4.3.3.2.4.5); Assumption input pesticides = input in soil (Monocrotophos, cyfluthrin, dicolol, trichlorform, imidacoprid, chlordimeform, piperonyl as most frequent applied substances)		85.0	15.0	
	Imidacloprid	-	-	kg	9.76E-5	1	1.69	(4.3.3.2.4.5); Assumption input pesticides = input in soil (Monocrotophos, cyfluthrin, dicolol, trichlorform, imidacoprid, chlordimeform, piperonyl as most frequent applied substances)		85.0	15.0	
	Piperonyl butoxide	-	-	kg	9.76E-5	1	1.69	(4.3.3.2.4.5); Assumption input pesticides = input in soil (Monocrotophos, cyfluthrin, dicolol, trichlorform, imidacoprid, chlordimeform, piperonyl as most frequent applied substances)		85.0	15.0	
	Prometryn	-	-	kg	2.70E-4	1	1.69	(4.3.3.2.4.5); Assumption input pesticides = input in soil (Frequent applied substances: MSMA, glyphosate, alachlor, fluometuron)		85.0	15.0	
	Glyphosate	-	-	kg	2.70E-4	1	1.69	(4.3.3.2.4.5); Assumption input pesticides = input in soil (Frequent applied substances: MSMA, glyphosate, alachlor, fluometuron)		85.0	15.0	
	Alachlor	-	-	kg	2.70E-4	1	1.69	(4.3.3.2.4.5); Assumption input pesticides = input in soil (Frequent applied substances: MSMA, glyphosate, alachlor, fluometuron)		85.0	15.0	
	Fluometuron	-	-	kg	2.70E-4	1	1.69	(4.3.3.2.4.5); Assumption input pesticides = input in soil (Frequent applied substances: MSMA, glyphosate, alachlor, fluometuron)		85.0	15.0	
	MSMA	-	-	kg	1.59E-4	1	1.69	(4.3.3.2.4.5); Assumption input defoliant = input in soil		85.0	15.0	
resource, in air	Carbon dioxide, in air	-	-	kg	4.57E+0	1	1.58	(3.na.na.3.1.5); Extrapolation for sum parameter		36.1	63.9	
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	5.14E+1	1	1.65	(4.4.3.3.4.5); Estimation for energy 18 MJ / kg		36.1	63.9	
resource, land	Occupation, arable	-	-	m2a	9.09E+0	1	1.16	(1.2.3.3.1.3); Most areas irrigated, assumptions same land use before and after cotton cultivation		85.0	15.0	
	Transformation, from arable	-	-	m2	9.09E+0	1	1.93	(4.3.5.3.4.5); Most areas irrigated, assumptions same land use before and after cotton cultivation		85.0	15.0	
	Transformation, to arable	-	-	m2	9.09E+0	1	1.93	(4.3.5.3.4.5); Most areas irrigated, assumptions same land use before and after cotton cultivation		85.0	15.0	

1.6 Data Quality Considerations

Tab. 3.3 shows data quality indicators for the inventory of cotton production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation. The inventory is based on published data, statistics and recommendations. Large uncertainties exist for emissions on the field. Nevertheless in general the data quality is quite reliable.

Abbreviations

DM Dry Matter

EcoSpold Meta Information

Tab. A. 1 EcoSpold Meta Information of cotton production and related co-products

ReferenceFunction	401	Name	cotton production
Geography	662	Location	CN
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	2.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	69
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	The inventories include the cultivation of cotton upon Chinese standards. Included steps are soil cultivation, pesticides fertilisation (mineral fertilizer), harvest, loading for transport and extraction of the fibres in a gin plant
	404	Amount	1
	490	LocalName	Baumwollproduktion
	491	Synonyms	
	492	GeneralComment	The multioutput-process "cotton production" delivers the co-products "cotton fibres, ginned, at farm", "cotton seed, at farm". The allocation is based on economic criteria and on mass balance. World production of ginned cotton was 26.2 Mt in 2005. The inventory refers to 1 kg dry fibres and 1.77 kg seed from standard China production.
	494	InfrastructureIncluded	1
	495	Category	agricultural production
	496	SubCategory	plant production
	497	LocalCategory	Landwirtschaftliche Produktion
	498	LocalSubCategory	Pflanzenbau
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	
Geography	663	Text	Data mainly from expert gremiums of respective country
Technology	692	Text	Cultivation of cotton upon Chinese conventional production standards
Representativeness	722	Percent	24
	724	ProductionVolume	World production of cotton fibres was 26.2 Mt in 2005.
	725	SamplingProcedure	Literature data.
	726	Extrapolations	
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	69
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	48
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	
ProofReading	5616	Validator	8
	5615	Details	automatic validation in Excel
	5619	OtherDetails	none

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2 Cotton processing, yarn and textile production

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Last changes:

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Abstract

The inventory of the processing steps for cotton ‘yarn production, cotton fibres’, ‘weaving, cotton’ and ‘refinement, cotton’ refer to yarn production of USA and China and the further production and refinement of textiles in China and Eastern Europe.

The inventory of the cotton processing include data about the transport to the factories, infrastructure of plant and process specific energy and material need apart from the processed cotton itself. The inventory for the resulting products ‘cotton yarn, at plant’ and ‘cotton textile, at plant’ include additionally the production of the required cotton itself considering material loss of processing.

The reference function of the processing is 1 kg of related input good cotton lint or yarn. The reference function of resulting products are 1 kg yarn or 1 kg woven textile respectively.

2.1 Cotton products and processing

The inventories of processing include data of processing steps apart from the cotton production itself. The inventories of output products ‘cotton yarn, at plant’ and ‘cotton textile, at plant’ include the required cotton input good inclusive material loss of processing. Data about different process steps are listed in Tab. 3.1. The data is based on the report of the Öko-Insitut (Wiegmann, 2002). USA and China are the most important producers of yarn from lint cotton. Regarding textiles and refined textiles China is the main producer, followed by the Eastern Europe textile production with refinement mainly in Italy.

Average transport distances of Chinese and US yarn production are from the report of the Öko-Insitut (Wiegmann, 2002). For the further processing equal inland transports were considered and for the use of US cotton yarn in Eastern Europe and China textile production an additional transport overseas included.

The total amount of 168 g auxiliary products of textile refinement were split on 10 g washing, 10 g bleaching, 20 g finishing agents and 130 g organic compounds for dyeing. Regarding the refinement mainly carried out in water, a waste water treatment was included for the total declared water amount.

Lacking data for infrastructure was added for consistency with ecoinvent processes. The infrastructure of mechanical steps spinning and weaving was considered to be similar as in the packaging industry. Textile refinement was considered comparable to a small sized chemical plant. Generally infrastructure was assumed to contribute less than 5% of impacts.

2.1.1 Input from technosphere

Existing ecoinvent data inventories were applied for the energy and material consumption (water, salt, auxiliary products) and for transports. Assumption has been taken for the infrastructure and waste water treatment in the cotton processing.

2.1.2 Emissions, CO₂-binding and Solar Energy in Biomass of cotton

Emissions, CO₂-binding and solar energy are included in the used data inventories. No direct emissions were considered in the processing.

Tab. 2.1 Characteristics of cotton processing (Wiegmann, 2002)

Process	Material loss	Energy need	Materials	Transport
yarn production, cotton fibres	10-15 %	Ø 30.6 MJ/kg (electricity)	-	250 km China (lorry) 750 km USA (lorry)
weaving, cotton	1-4%	Ø 36.4MJ/kg (electricity)	-	250 km China (lorry) USA to Eastern Europe (over-seas)
textile refinement, cotton	No data, assumption 1-4%	Ø 40 MJ/kg (90% heat ³¹ , 10% electricity)	Water, Ø 140 l/kg Salt, Ø 547 g/kg Auxiliary prod. Ø 165 g/kg	250 km China (lorry) Eastern Europe - Italy (lorry)

2.1.3 Land occupation

Land occupation is included in the used data inventories of cotton cultivation and infrastructure of processing (ass 'packaging box production unit' and 'chemical plant, organic').

2.2 Statistics of yarn and textile production

Data for world production of yarn 17'275'000 t (29% China, 11% USA, 10% India) and of textiles 83'007 Mill. m² (37% China, 1.5% GUS) are from the report of the Öko-Insitut (Wiegmann, 2002).

2.3 Characterisation and Use of cotton yarn and textiles

Cotton is a soft fibre that grows around the seeds of the cotton plant (*Gossypium* sp.). The fibre is most often spun into yarn or thread and used to make a soft, breathable textile, which is the most widely used natural-fibre cloth in clothing today. Cotton fibres make up roughly 50% of world fibre consumption.

2.4 Life Cycle Inventories of cotton products and processing

The inventories of processing refer to the spinning of yarn, weaving and refinement of cotton. The inventory of processing steps include data about the transport to the factories, the energy and material need apart from cotton input good as well as estimations about the related infrastructure an waste water treatment. The reference function refers to the processing of 1 kg input good. The inventory of the resulting products yarn and textile include the production of cotton itself. The reference function is 1 kg yarn or textile respectively.

³¹ Reported are 100% heat from oil combustion in China and 50% from oil and gas in Italy (Wiegmann, 2002)

Tab. 2.2 Unit process of 'yarn production, cotton fibres'

	Name	Location	Unit	yarn production, cotton fibres	UncertaintyType	Standard Deviation 95%	GeneralComment
	Location InfrastructureProcess Unit			GLO 0 kg			
product	yarn production, cotton fibres	GLO	kg	1			
technosphere	electricity, low voltage, at grid	CN	kWh	5.10E+0	1	1.16	(2,2,3,1,1,4); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002)
technosphere	electricity, low voltage, at grid	US	kWh	3.40E+0	1	1.24	(2,2,3,1,1,5); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002)
emission air, unspecified	Heat, waste	-	MJ	3.06E+1	1	1.24	(2,2,3,1,1,5); Heat waste derived from electricity consumption
technosphere	transport, lorry 16-32t, EURO3	RER	tkm	4.50E-1	1	2.03	(2,2,3,3,1,4); Distance from report Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002). Assumption vehicle type Carbotech AG
technosphere	packaging box production unit	RER	unit	1.00E-9	1	3.50	(5,1,1,3,4,5); Estimation infrastructure comparable to packaging plant (building, equipment, electronics). Assumption life time 50 years with output of 20' 000 t of yarn per year. Totally max. 5% of impacts.

Tab. 2.3 Unit process of 'weaving, cotton'

	Name	Location	Unit	weaving, cotton	UncertaintyType	Standard Deviation 95%	GeneralComment
	Location InfrastructureProcess Unit			GLO 0 kg			
product	weaving, cotton	GLO	kg	1			
technosphere	electricity, low voltage, at grid	CN	kWh	7.08E+0	1	1.16	(2,2,3,1,1,4); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002)
technosphere	electricity, low voltage, production at grid	RER	kWh	3.03E+0	1	1.24	(2,2,3,1,1,5); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002), Czech Republic electricity mix considered representing GUS.
emission air, unspecified	Heat, waste	-	MJ	3.64E+1	1	1.24	(2,2,3,1,1,5); Heat waste derived from electricity consumption
technosphere	transport, lorry 16-32t, EURO3	RER	tkm	3.50E-1	1	2.03	(2,2,3,3,1,4); Assumption Carbotech AG, all cotton at least transport of about 250 km to next factory or harbour for shipping to China/Eastern Europe. Distance overseas transport counted twice as inecoinvent guidelines
technosphere	transport, transoceanic freight ship	OCE	tkm	4.80E+0	1	2.03	(2,2,3,3,1,4); Assumption Öko-Institut Baumwolle Datendokumentation (Wiegmann K., 2002), USA 40% of production for processing transport of 10% to China and 30% to Eastern Europe, Assumption average distance 12'000 km by Carbotech AG
technosphere	packaging box production unit	RER	unit	1.00E-9	1	3.50	(5,1,1,3,4,5); Estimation infrastructure comparable to packaging plant (building, equipment, electronics). Assumption life time 50 years with an output of 20' 000 t of yarn per year. Totally max. 5% of impacts.

Tab. 2.4 Unit process of 'textile refinement, cotton'

	Name	Location	Infrastructure Process	Unit	textile refinement, cotton	Uncertainty Type	Standard Deviation 95%	GeneralComment
	Location InfrastructureProcess Unit				GLO 0 kg			
product	textile refinement, cotton	GLO	0	kg	1			
technosphere	electricity, low voltage, at grid	CN	0	kWh	7.76E-1	1	1.25	(2,3,3,1,1,5); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002)
	electricity, low voltage, production IT, at grid	IT	0	kWh	3.33E-1	1	1.25	(2,3,3,1,1,5); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002)
emission air, unspecified	Heat, waste	-	-	MJ	3.99E+0	1	1.25	(2,3,3,1,1,5); Heat waste derived from electricity consumption
technosphere	heat, light fuel oil, at industrial furnace 1MW	RER	0	MJ	3.05E+1	1	1.25	(2,3,3,1,1,5); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002). China heat production from 100% oil and Italy 50%
	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	5.39E+0	1	1.25	(2,3,3,1,1,5); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002). Italy heat production from 50% gas
	transport, lorry >16t, fleet average	RER	0	tkm	2.50E-1	1	2.07	(2,3,3,1,1,5); Transport distance from to Italy between processing steps from Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002)
	tap water, at user	RER	0	kg	1.38E+2	1	1.25	(2,3,3,1,1,5); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002)
	sodium chloride, powder, at plant	RER	0	kg	5.47E-1	1	1.25	(2,3,3,1,1,5); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002)
	chemicals organic, at plant	GLO	0	kg	1.30E-1	1	1.25	(2,3,3,1,1,5); Amount of auxiliaries from Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002). Assumption Carbotech AG regarding mixture about 120 g organic compounds for dyeing
	fatty alcohol sulfonate, mix, at plant	RER	0	kg	1.00E-2	1	1.25	(2,3,3,1,1,5); Amount of auxiliaries from Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002). Assumption Carbotech AG regarding mixture about 10 g washing agent
	sodium perborate, tetrahydrate, powder, at plant	RER	0	kg	1.00E-2	1	1.25	(2,3,3,1,1,5); Amount of auxiliaries from Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002). Assumption Carbotech AG regarding mixture about 10 g bleaching agent
	alkylbenzene sulfonate, linear, petrochemical, at plant	RER	0	kg	1.00E-2	1	1.25	(2,3,3,1,1,5); Amount of auxiliaries from Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002). Assumption Carbotech AG regarding mixture about 10 g finishing agent
	carboxymethyl cellulose, powder, at plant	RER	0	kg	1.00E-2	1	1.25	(2,3,3,1,1,5); Amount of auxiliaries from Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002). Assumption Carbotech AG regarding mixture about 10 g finishing agent
	chemical plant, organics	RER	1	unit	1.00E-10	1	3.52	(5,3,3,1,4,5); Estimation infrastructure comparable to small sized chemical plant (10% of building, equipment). Assumption life time 50 years with about 20' 000 t of refined yarn or textile per year. Totally max 5% of impacts.
	treatment, sewage, to wastewater treatment, class 5	CH	0	m3	1.38E-1	1	1.85	(5,3,3,1,4,5); Assumptions Carbotech AG, total water consumption in waste water treatment.

Tab. 2.5 Unit process of 'yarn, cotton, at plant'

	Name	Location	Unit	yarn, cotton, at plant	Uncertainty Type	Standard Deviation 95%	GeneralComment
	Location InfrastructureProcess Unit			GLO 0 kg			
product	yarn, cotton, at plant	GLO	kg	1			
technosphere	cotton fibres, ginned, at farm	CN	kg	6.60E-1	1	1.16	(2,2,3,1,1,4); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002), assumption 10% fibres loss
technosphere	cotton fibres, at farm	US	kg	4.40E-1	1	1.16	(2,2,3,1,1,4); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002), assumption 10% fibres loss
technosphere	yarn production, cotton fibres	GLO	kg	1.00E+0	1	1.17	(2,2,3,3,1,4); Öko-Institut, Baumwolle Datendokumentation (Wiegmann K., 2002), assumption 10% fibres loss
technosphere	disposal, paper, 11.2% water, to sanitary landfill	CH	kg	1.00E-1	1	1.17	(2,2,3,3,1,4); the fibre residues goes to landfill

Tab. 2.6 Unit process of 'textile, woven cotton, at plant'

	Name	Location	Unit	textile, woven cotton, at plant	UncertaintyType	Standard Deviation 95%	GeneralComment
	Location InfrastructureProcess Unit			GLO 0 kg			
product	textile, woven cotton, at plant	GLO	kg	1			
technosphere	yarn, cotton, at plant	GLO	kg	1.02E+0	1	1.17	(2,2,3,3,1,4); Öko-Institut, Baumwoll Datendokumentation (Wiegmann K., 2002)
	weaving, cotton	GLO	kg	1.00E+0	1	1.17	(2,2,3,3,1,4); Öko-Institut, Baumwoll Datendokumentation (Wiegmann K., 2002)
	disposal, paper, 11.2% water, to sanitary landfill	CH	kg	2.00E-2	1	1.17	(2,2,3,3,1,4); the fibre residues goes to landfill

2.5 Data Quality Considerations

Tab. 2.2 to Tab. 2.6 show the standard deviation for the inventory of cotton processing and resulting products yarn and textiles. The simplified approach with a pedigree matrix has been used for calculating the standard deviation. The inventory is based on published data and completed by assumptions for infrastructure, transport distances and waste water treatment. Nevertheless in general the data quality is quite reliable.

EcoSpold Meta Information

Tab. A. 2 EcoSpold Meta Information of cotton processing

Name	yarn production, cotton fibres	textile refinement, cotton	weaving, cotton
Location	GLO	GLO	GLO
InfrastructureProcesses	0	0	0
Unit	kg	kg	kg
IncludedProcesses	The inventories include energy consumption, transport and infrastructure related to the processing of lint cotton to yarn (cleaning, carding, spinning).	The inventories include energy consumption and material need for the refinement of lint cotton, yarn or textiles including estimations about infrastructure need and waste water treatment.	The inventories include energy consumption, transports and infrastructure for the weaving of cotton yarn into a textile.
Amount	1	1	1
LocalName	Garnherstellung, Baumwolle, Spinnerei	Textilveredelung, Baumwolle	Weben, Baumwolle
Synonyms			
GeneralComment	Inventory refers to the processing of 1 kg lint cotton only, without the production of the cotton itself.	Inventory refers to textile refinement of 1 kg textile or yarn	Inventory refers to the weaving of 1 kg yarn into textile, the production of the yarn itself is not included.
InfrastructureIncluded	1	1	1
Category	textiles	textiles	textiles
SubCategory	production	processing	processing
LocalCategory	Textile	Textile	Textile
LocalSubCategory	Gewinnung	Verarbeitung	Verarbeitung
Formula			
CASNumber			
StartDate	1998	1998	1995
EndDate	1999	1999	2007
DataValidForEntirePeriod	1	1	1
OtherPeriodText			
Text	About 40% of cotton is processed in USA and about 60% in Asia (China, India and Pakistan). China and USA were considered for the processing of cotton in this study.	Most textiles are refined in China, the textiles produced in Eastern Europe mainly in Italy	Most important producer of textiles is China followed by Eastern Europe
Text	Carding and spinning are the main processes of the yarn production. Mechanical cleaning was assumed for this study.	Bleaching, washing, dying and drying are the main processes of the refinement which might be applied on lint cotton, yarn or textile.	Weaving was considered for the textile production in the present study.
Percent			
ProductionVolume			
SamplingProcedure	Data were compiled from literature.	Data were compiled from literature.	Data were compiled from literature.
Extrapolations			
UncertaintyAdjustments	none	none	none

Tab. A. 3 EcoSpold Meta Information of cotton products

Name	yarn, cotton, at plant	textile, woven cotton, at plant
Location	GLO	GLO
InfrastructureProcess	0	0
Unit	kg	kg
IncludedProcesses	The inventories include the cultivation of cotton and its processing to yarn (cleaning, carding, spinning).	The inventory includes the cultivation of cotton and its further spinning to yarn and weaving to textile.
Amount	1	1
LocalName	Garn, Baumwolle, ab Werk	Gewebe, Baumwolle, ab Werk
Synonyms		
GeneralComment	Inventory refers to 1 kg yarn produced from cotton lint	Inventory refers to 1 kg woven textile
InfrastructureIncluded	1	1
Category	textiles	textiles
SubCategory	production	production
LocalCategory	Textile	Textile
LocalSubCategory	Gewinnung	Gewinnung
Formula		
StatisticalClassification		
CASNumber		
StartDate	1995	1995
EndDate	1999	2007
DataValidForEntirePeriod	1	1
OtherPeriodText		
Text	About 40% of yarn is produced from cotton lint in USA and about 60% in Asia (China, India and Pakistan). China and USA were considered for the processing of cotton in this study.	Most important producer of textiles is China followed by Eastern Europe
Text	Cultivation, harvest and ginning of cotton and the further carding and spinning are the main processes of the yarn production.	Weaving was considered for the textile production in the present study.
Percent		
ProductionVolume		
SamplingProcedure	Data were compiled from literature.	Data were compiled from literature.
Extrapolations		
UncertaintyAdjustments	none	none

References

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3 Jute and kenaf production, India

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Abstract

The inventories refer to the jute and kenaf production from the main producer India. One important difference in the cultivation practice concerns the irrigation. It is why two data sets have been created for jute. The multi output-processes 'jute production, irrigated system' and 'jute production, rainfed system' delivers the co-products 'jute fibres, irrigated system, at farm' and 'jute stalks, from fibre production, irrigated system, at farm' respectively 'jute fibres, rainfed system, at farm' and 'jute stalks, from fibre production, rainfed system, at farm'. Another dataset has been developed for kenaf. The multi output process 'kenaf fibres, at farm' delivers the processes 'kenaf fibres, at farm' and 'kenaf stalks, from fibre production, at farm'. The allocation for all processes is based on economic criteria and mass balance.

The inventories refer to conventional production standards and their emissions to the air, water and soil, calculated for the use of fertilizers and pesticides. In the conventional production mainly organic fertilizer and a smaller amount of mineral fertilizers are applied. The jute and kenaf production is related to small consumption of pesticides.

The inventory includes the cultivation, harvest, transport and separation of fibres. The reference function of co-products is 1 kg jute fibres and 1 kg jute stalks respectively.

3.1 Characteristic of jute and Kenaf production

According to B.K. Banerjee Jute cultivation is primarily restricted to India and Bangladesh, accounting approximately for 66% and 25% respectively of the world production of about 3 million metric tons. Jute occupies the most important place among the bast fibre crops in India. It is grown on around 1 million-hectare land. About 4 million farmer families - mostly marginal - of eastern and north-eastern regions generate principal income from jute cultivation, leading to paid employment to the tune of 10 million man-days per season, earning in the process on an average 32 Rs /day /head during the season of cultivation and fibre extraction. The land area under jute cultivation has grown two-fold since independence while the production has gone up by 3.5 times. The world production of kenaf is about 20% of the jute production. The main producers are China (44%), India (39%) and Thailand (12%).

After being allowed to grow for four months (April onwards) the plant is harvested, leaving the root portion in the soil. The harvested jute plants are then allowed to dry up, shedding leaves in the process (defoliation). The roots left behind in the soil as also these dry leaves form important nutrition for the subsequent cultivation.

The dried jute plant yields two important products namely the jute fibre and the jute stick or stalk. The jute fibre and the stick are extracted from the plant by a controlled rotting process, known as retting. The jute stick forms an important fuel for the farmers during the rainy season and it is claimed that even if jute cultivation becomes uneconomical for the farmers they would continue to grow some amount just for the sake of the fuel it yields. Incidentally it has been estimated that about 1.8 kg of CO₂ is absorbed from the atmosphere per every kg of fibre produced.

Tab. 3.1 Characteristics of jute and kenaf production

Multi output process	Net yield kg /ha*a	Cultivation time	Comments
jute production: - jute fibres - jute sticks	1800 kg DM 3065 kg DM	Occupation arable land for about four month.	Mainly manure and mineral fertilizers, need of pesticides not very high
kenaf production - kenaf fibres - kenaf sticks	2100 kg DM 3576 kg DM	Seeding in April – harvesting in August	Nearly 40% is irrigated

3.1.1 Input from Techno sphere

Data of the jute production is based mostly on works done by Indian jute research institutes and FAO. The inputs from techno sphere include the use of fertilizer, pesticides and transports. In India and Bangladesh the use of machines for soil cultivation and harvesting is not very common, normally it is done by hand. The amount of fertilizers given is shown in the following table.

Applied pesticides are derived from the above mentioned experts survey and statistics about average doses (FAO, 2006 and Aimin). A transport of 50 km with a van <3.5t to the next regional storehouse was assumed.

Tab. 3.2 Inputs for the production of jute and kenaf [FAO]

	Fertilizer kg per ha				Pesticides	Water	
	Mineral			Organic	kg / ha	Irrigation	retting
	N	P ₂ O ₅	K ₂ O	manure		m ³	m ³
Jute irrigated	55.9	22.4	10.2	2500	2	2000	100
Jute rainfed	28.9	6	2.3	5000	2		100
Kenaf	39.7	12.6	5.5	4000	2	1000	115

3.1.2 Emissions to the air (N₂O, NO_x, Ammonia)

Emissions were calculated with the formula for N₂O, NO_x and Ammonia as described in the ecoinvent report No. 15 (Nemecek et al. 2004, Chapter 4.4). Relevant parameters are the application of fertilizers and the nitrogen fixation of the vegetation.

3.1.3 Emissions to the water (Phosphate, Phosphorus, Nitrate)

Emissions were calculated with the formula for Phosphate leaching to the ground water, for Phosphate and Nitrate effluent to surface water and for the Phosphorus loss with erosion as in the ecoinvent report No. 15 (Nemecek et al. 2004, Chapter 4.4). The Nitrate leaching was estimated about 10% of applied kg N, due to lacking data for a more detailed calculation. Emissions depend mainly on the amount and type of fertilizers applied, on the time period of the fertilization and the cultivation of soils.

3.1.4 Emissions to the soil (heavy metals)

An Input – Output balance was calculated with background data about heavy metal contents of plants, seeds and fertilizers in ecoinvent report No. 15 (Nemecek et al. 2004). No specific data about jute heavy metal contents was available; an average value calculated from data about different plant types was used instead.

Only inputs and outputs directly related to the agricultural production were considered. As in the ecoinvent report No. 15, output through leaching, run-off and erosion - partly due to the agricultural production - were not included, due to the allocation problems (Nemecek et al. 2004).

3.1.5 CO₂-binding and Solar Energy in Biomass of jute

A value for CO₂-binding of 1.8 kg CO₂/kg DM and an energy content of 18 MJ/kg DM was used for jute fibres. Those values lie within the range of different crop types (Nemecek et al., 2004). Allocation to stick and fibres was carried out on mass balance.

3.1.6 Land occupation of jute and kenaf production

Jute and Kenaf are cultivated in crop rotation, e.g. jute – paddy – potato. Because of the short vegetation time of four months in these regions normally there are three crops per year.

3.1.7 Retting

The traditional retting of jute plant involves submerging the plants in water. As a result the bark of the stem swells resulting in cracks and crevices in the cortex. Micro organisms, present on the plant as also in the surrounding water and soil system, then enter through these openings and start attacking the pectinous binding matter loosening in the process the long fibres located along an annular ring between the outer periderm and the inner cambium of the bark. In addition to the solubilised binding matters, other soluble matters such as Tannin etc. from the plants are also released in water during this process of retting. Objectionable gases such as methane are also released into the atmosphere. As this process is carried out in batches over a period of a month or so, the water bodies surrounding the farmer's field as also the surrounding air gets fouled up during this activity. Moreover the quality of fibre too suffers considerably as a result of this uncontrolled batch process. Subsequently the recuperative process of nature brings the water bodies and air back to equilibrium over a period of time.

The retting process is connected with methane emissions to the air and oxygen demand (DOC and BOD) in the water.

3.2 Allocation to co-products fibres and seed

The allocation of jute and kenaf to their co-products 'jute fibres, irrigated system, at farm' and 'jute stalks, from fibre production, irrigated system, at farm' respectively 'jute fibres, rain fed system, at farm' and 'jute stalks, from fibre production, rain fed system, at farm' as well as 'kenaf fibres, at farm' and 'kenaf stalks, from fibre production, at farm' is based mainly on economic criteria. For organic material, carbon dioxide uptake and heavy metals to the soil related values on mass balance have been used.

3.3 Life Cycle Inventories of jute and kenaf production

Inventory refers to the production of 1 kg jute fibres or 1 kg sticks respectively. Inventories include the soil cultivation, fertilization, application of pesticides, irrigation, harvest, retting and related transports.

Tab. 3.3 Unit process of jute production irrigated system

	Name	Location	Infrastructure	Process	Unit	jute production, irrigated system	Uncertainty type	Standard Deviation 65%	General Comment	jute fibres, irrigated system, at farm	jute stalks, from fibre production, irrigated system, at farm
	Location Infrastructure Unit					IN 0 kg				IN 0 kg	IN 0 kg
allocated products	jute fibres, irrigated system, at farm	IN	0	kg	1					100	0
	jute stalks, from fibre production, irrigated system, at farm	IN	0	kg	1.70E+0					0	100
resource, in water	Water, river	-	-	m3	1.17E+0		1	1.24	(2.2.3.1.1.5); Water need for retting 1:20 resulting in about 100 m3, Water for irrigation assumption about 2000 m3/ha	95.0	5.0
technosphere	chemicals organic, at plant	GLO	0	kg	1.11E-3		1	1.24	(2.2.3.1.1.5); Central Research Institute for Jute & allied Fibres, Interview with Sustena	95.0	5.0
technosphere	transport, tractor and trailer	CH	0	tkm	8.89E-3		1	2.07	(3.2.3.1.1.5); rough estimation	95.0	5.0
technosphere	solid manure loading and spreading, by hydraulic loader and spreader	CH	0	kg	2.78E-1		1	1.26	(3.2.3.1.1.5); rough estimation	95.0	5.0
	tillage, ploughing	CH	0	ha	1.11E-4		1	1.26	(3.2.3.1.1.5); rough estimation	95.0	5.0
	urea, as N, at regional storehouse	RER	0	kg	3.11E-2		1	1.24	(2.2.3.1.1.5); Amount of mineral fertilizer from FAO. Urea mainly, information from Central Research Institute for Jute & allied Fibres, expert interview	95.0	5.0
	single superphosphate, as P2O5, at regional storehouse	RER	0	kg	1.24E-2		1	1.24	(2.2.3.1.1.5); Amount of mineral fertilizer from FAO. Assumption for fertilizer type by Carbotech AG	95.0	5.0
	potassium chloride, as K2O, at regional storehouse	RER	0	kg	5.67E-3		1	1.24	(2.2.3.1.1.5); Amount of mineral fertilizer from FAO. Assumption for fertilizer type by Carbotech AG	95.0	5.0
	transport, van <3.5t	RER	0	tkm	2.51E-3		1	2.06	(2.2.3.1.1.5); Assumption Carbotech AG 50 kg to local storehouse and transport medium	95.0	5.0
emission air, low population density	Dinitrogen monoxide	-	-	kg	1.27E-3		1	1.62	(2.1.2.3.3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
	Ammonia	-	-	kg	4.52E-3		1	1.38	(2.1.2.3.3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
	Nitrogen oxides	-	-	kg	2.67E-4		1	1.62	(2.1.2.3.3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
	Methane, biogenic	-	-	kg	4.86E-3		1	1.62	(2.1.2.3.3.5); Assumption about 0.1% of total CO2 uptake	95.0	5.0
emission water, river	Phosphate	-	-	kg	2.44E-4		1	1.62	(2.1.2.3.3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
emission water, ground-	Phosphate	-	-	kg	8.91E-5		1	1.62	(2.1.2.3.3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
emission water, ground-	Nitrate	-	-	kg	1.67E-2		1	1.63	(3.1.2.3.3.5); Annahme 10% geht ins Grundwasser (keine spezifischen Angaben zu Zeitpunkt, Niederschläge, ...)	95.0	5.0
emission water, river	Phosphorus	-	-	kg	2.82E-4		1	1.62	(2.1.2.3.3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004), assumption land cover similar than beans	95.0	5.0
	COD, Chemical Oxygen Demand	-	-	kg	1.35E-1		1	1.65	(3.1.3.3.3.5); Assumption about 5 % of total harvested jute	95.0	5.0
	BOD5, Biological Oxygen Demand	-	-	kg	1.19E-1		1	1.65	(3.1.3.3.3.5); Assumption about 4 % of total harvested jute	95.0	5.0
emission soil, agricultural	Cadmium	-	-	kg	3.34E-7		1	1.58	(2.1.3.3.1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Chromium	-	-	kg	4.19E-5		1	1.58	(2.1.3.3.1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Copper	-	-	kg	-7.47E-6		1	1.58	(2.1.3.3.1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Mercury	-	-	kg	-9.26E-8		1	1.58	(2.1.3.3.1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Nickel	-	-	kg	2.70E-7		1	1.58	(2.1.3.3.1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Lead	-	-	kg	-3.79E-6		1	1.58	(2.1.3.3.1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Zinc	-	-	kg	-3.50E-5		1	1.58	(2.1.3.3.1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Oilis, unspecified	-	-	kg	1.00E-3		1	1.58	(2.1.3.3.1.5); Assumption input pesticides = input in soil, oil derivate with aktive substances (10%)	95.0	5.0
	Parathion	-	-	kg	1.11E-4		1	1.32	(2.1.3.3.1.5); Assumption input pesticides = input in soil, oil derivate with aktive substances (10%)	95.0	5.0
resource, in air	Carbon dioxide, in air	-	-	kg	4.86E+0		1	1.24	(3.1.1.3.1.5); Estimation for CO2, 1.8 kg CO2 per kg Jute	37.0	63.0
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	4.86E+1		1	1.24	(3.1.1.3.1.5); Estimation for energy 18 MJ / kg Jute	37.0	63.0
resource, land	Occupation, arable	-	-	m2a	2.08E+0		1	1.26	(2.2.3.2.1.5); Cultivation of one ha for the time of 4-5 months (bewässert)	95.0	5.0
	Transformation, from arable	-	-	m2	5.56E+0		1	1.32	(2.2.3.3.1.5); Assumption transformation from arable land to arable land (irrigated)	95.0	5.0
emission resource, land	Transformation, to arable	-	-	m2	5.56E+0		1	1.32	(2.2.3.3.1.5); Assumption transformation from arable land to arable land (irrigated)	95.0	5.0

Tab. 3.4 Unit process of jute production rain fed system

	Name	Location	Infrastructure	Process	Unit	jute production, rainfed system	Uncertainty type	Standard Deviation 95%	GeneralComment	jute fibres, rainfed system, at farm	jute stalks, from fibre production, rainfed system, at farm
	Location InfrastructureProcess Unit					IN 0 kg				IN 0 kg	IN 0 kg
allocated	jute fibres, rainfed system, at farm	IN	0	kg	1					100	0
products	jute stalks, from fibre production, rainfed system, at farm	IN	0	kg	1.70E+0					0	100
resource, in water	Water, river	-	-	m3	5.41E-2		1	1.24	(2,2,3,1,1.5); Water need for retting 1:20	95.0	5.0
technosphere	chemicals organic, at plant	GLO	0	kg	1.11E-3		1	1.24	(2,2,3,1,1.5); Central Research Institute for Jute & allied Fibres, expert interview	95.0	5.0
technosphere	fertilising, by broadcaster	CH	0	ha	4.44E-3		1	1.26	(3,2,3,1,1.5); rough estimation	95.0	5.0
technosphere	solid manure loading and spreading, by hydraulic loader and spreader	CH	0	kg	1.39E-1		1	1.26	(3,2,3,1,1.5); rough estimation	95.0	5.0
	tillage, ploughing	CH	0	ha	5.56E-5		1	1.26	(3,2,3,1,1.5); rough estimation	95.0	5.0
	urea, as N, at regional storehouse	RER	0	kg	1.61E-2		1	1.24	(2,2,3,1,1.5); Amount of mineral fertilizer from FAO. Urea mainly, information from Central Research Institute for Jute & allied Fibres, expert interview	95.0	5.0
	single superphosphate, as P2O5, at regional storehouse	RER	0	kg	3.33E-3		1	1.24	(2,2,3,1,1.5); Amount of mineral fertilizer from FAO. Assumption for fertilizer type by Carbotech AG	95.0	5.0
	potassium chloride, as K2O, at regional storehouse	RER	0	kg	1.28E-3		1	1.24	(2,2,3,1,1.5); Amount of mineral fertilizer from FAO. Assumption for fertilizer type by Carbotech AG	95.0	5.0
	transport, van <3.5t	RER	0	tkm	1.09E-3		1	2.06	(2,2,3,1,1.5); 50 km for transport from and to storehouse	95.0	5.0
emission air, low population density	Dinitrogen monoxide	-	-	kg	9.87E-4		1	1.62	(2,1,2,3,3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
	Ammonia	-	-	kg	4.73E-3		1	1.38	(2,1,2,3,3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
	Nitrogen oxides	-	-	kg	2.07E-4		1	1.62	(2,1,2,3,3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
	Methane, biogenic	-	-	kg	4.86E-3		1	1.62	(2,1,2,3,3.5); Assumption about 0.1% of total CO2 uptake	95.0	5.0
emission water, river	Phosphate	-	-	kg	2.44E-4		1	1.62	(2,1,2,3,3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
emission water, ground-	Phosphate	-	-	kg	8.91E-5		1	1.62	(2,1,2,3,3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
emission water, ground-	Nitrate	-	-	kg	1.31E-2		1	1.63	(3,1,2,3,3.5); Annahme 10% geht ins Grundwasser (keine spezifischen Angaben zu Zeitpunkt, Niederschläge, ...)	95.0	5.0
emission water, river	Phosphorus	-	-	kg	2.82E-4		1	1.62	(2,1,2,3,3.5); Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004); assumption land cover similar than beans	95.0	5.0
	COD, Chemical Oxygen Demand	-	-	kg	1.35E-1		1	1.65	(3,1,3,3,3.5); Assumption about 5% of total harvested jute	95.0	5.0
	BOD5, Biological Oxygen Demand	-	-	kg	1.19E-1		1	1.65	(3,1,3,3,3.5); Assumption about 4% of total harvested jute	95.0	5.0
emission soil, agricultural	Cadmium	-	-	kg	-1.10E-7		1	1.58	(2,1,3,3,1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Chromium	-	-	kg	1.16E-5		1	1.58	(2,1,3,3,1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Copper	-	-	kg	-2.68E-6		1	1.58	(2,1,3,3,1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Mercury	-	-	kg	1.29E-8		1	1.58	(2,1,3,3,1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Nickel	-	-	kg	4.23E-7		1	1.58	(2,1,3,3,1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Lead	-	-	kg	-4.39E-6		1	1.58	(2,1,3,3,1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Zinc	-	-	kg	-1.18E-5		1	1.58	(2,1,3,3,1.5); Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Oils, unspecified	-	-	kg	1.00E-3		1	1.58	(2,1,3,3,1.5); Assumption input pesticides = input in soil, oil derivate with aktive substances (10%)	95.0	5.0
	Parathion	-	-	kg	1.11E-4		1	1.32	(2,1,3,3,1.5); Assumption input pesticides = input in soil, oil derivate with aktive substances (10%)	95.0	5.0
resource, in air	Carbon dioxide, in air	-	-	kg	4.86E+0		1	1.24	(3,1,1,3,1.5); Estimation for CO2, 1.8 kg CO2 per kg Jute	37.0	63.0
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	4.86E+1		1	1.24	(3,1,1,3,1.5); Estimation for energy 18 MJ / kg Jute	37.0	63.0
resource, land	Occupation, arable, non-irrigated	-	-	m2a	2.08E+0		1	1.26	(2,2,3,2,1.5); Cultivation of one ha for the time of 4-5 months (nicht bewässert)	95.0	5.0
	Transformation, from arable, non-irrigated	-	-	m2	5.56E+0		1	1.32	(2,2,3,3,1.5); Assumption transformation from arable land to arable land, (not irrigated)	95.0	5.0
	Transformation, to arable, non-irrigated	-	-	m2	5.56E+0		1	1.32	(2,2,3,3,1.5); Assumption transformation from arable land to arable land, (not irrigated)	95.0	5.0

Tab. 3.5 Unit process of kenaf production

	Name	Location	Infrastructure	Process	Unit	kenaf production	Uncertainty	type	Standard deviation 95%	GeneralComment	kenaf fibres, at farm	kenaf stalks, from fibre production, at farm
	Location InfrastructureProcess Unit					IN 0 kg					IN 0 kg	IN 0 kg
allocated products	kenaf fibres, at farm	IN	0	kg	1						100	0
	kenaf stalks, from fibre production, at farm	IN	0	kg	1.70E+0						0	100
resource, in water	Water, river	-	-	m3	5.30E-1		1	1.24	(2,2,3,1,1,5);	Water need for retting 1:20	95.0	5.0
technosphere	chemicals organic, at plant	GLO	0	kg	9.52E-4		1	1.24	(2,2,3,1,1,5);	Central Research Institute for Jute & allied Fibres, expert interview	95.0	5.0
technosphere	fertilising, by broadcaster	CH	0	ha	3.81E-3		1	1.24	(2,2,3,1,1,5);	rough estimation	95.0	5.0
technosphere	solid manure loading and spreading, by hydraulic loader and spreader	CH	0	kg	1.19E-1		1	1.24	(2,2,3,1,1,5);	rough estimation	95.0	5.0
	tillage, ploughing	CH	0	ha	4.76E-5		1	1.24	(2,2,3,1,1,5);	rough estimation	95.0	5.0
	urea, as N, at regional storehouse	RER	0	kg	1.89E-2		1	1.24	(2,2,3,1,1,5);	Amount of mineral fertilizer from FAO. Urea mainly, information from Central Research Institute for Jute & allied Fibres, expert interview	95.0	5.0
	single superphosphate, as P2O5, at regional storehouse	RER	0	kg	5.98E-3		1	1.24	(2,2,3,1,1,5);	Amount of mineral fertilizer from FAO. Assumption for fertilizer type by Carbotech AG	95.0	5.0
	potassium chloride, as K2O, at regional storehouse	RER	0	kg	2.60E-3		1	1.24	(2,2,3,1,1,5);	Amount of mineral fertilizer from FAO. Assumption for fertilizer type by Carbotech AG	95.0	5.0
	transport, van <3.5t	RER	0	tkm	1.42E-3		1	2.06	(2,2,3,1,1,5);	50 km for transport from and to storehouse	95.0	5.0
emission air, low population density	Dinitrogen monoxide	-	-	kg	1.01E-3		1	1.63	(2,3,2,3,3,5);	Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
	Ammonia	-	-	kg	3.98E-3		1	1.38	(2,3,2,3,3,5);	Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
	Nitrogen oxides	-	-	kg	2.12E-4		1	1.63	(2,3,2,3,3,5);	Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
	Methane, biogenic	-	-	kg	4.86E-3		1	1.63	(2,3,2,3,3,5);	Assumption about 0.1% of total CO2 uptake	95.0	5.0
emission water, river	Phosphate	-	-	kg	2.09E-4		1	1.63	(2,3,2,3,3,5);	Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
emission water, ground-	Phosphate	-	-	kg	7.63E-5		1	1.63	(2,3,2,3,3,5);	Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004)	95.0	5.0
emission water, ground-	Nitrate	-	-	kg	1.24E-2		1	1.64	(3,2,3,3,3,5);	Annahme 10% geht ins Grundwasser (keine spezifischen Angaben zu Zeitpunkt, Niederschläge, ...)	95.0	5.0
emission water, river	Phosphorus	-	-	kg	2.41E-4		1	1.63	(2,3,2,3,3,5);	Formula as in ecoinvent report No. 15, chapter 4.4 (Nemecek et al. 2004), assumption land cover similar than beans	95.0	5.0
	COD, Chemical Oxygen Demand	-	-	kg	1.35E-1		1	1.65	(3,3,3,3,3,5);	Assumption about 5% of total harvested kenaf	95.0	5.0
	BOD5, Biological Oxygen Demand	-	-	kg	1.19E-1		1	1.65	(3,3,3,3,3,5);	Assumption about 4% of total harvested kenaf	95.0	5.0
emission soil, agricultural	Cadmium	-	-	kg	1.25E-9		1	1.64	(2,3,3,3,3,5);	Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Chromium	-	-	kg	2.01E-5		1	1.64	(2,3,3,3,3,5);	Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Copper	-	-	kg	-6.22E-6		1	1.64	(2,3,3,3,3,5);	Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Mercury	-	-	kg	-5.34E-8		1	1.64	(2,3,3,3,3,5);	Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Nickel	-	-	kg	-1.18E-8		1	1.64	(2,3,3,3,3,5);	Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Lead	-	-	kg	-4.56E-6		1	1.64	(2,3,3,3,3,5);	Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Zinc	-	-	kg	-2.92E-5		1	1.64	(2,3,3,3,3,5);	Estimation for calculation input - output balance using data about metal content of different plants and fertilizers summarized in Nemecek et al. (2004)	37.0	63.0
	Oils, unspecified	-	-	kg	8.57E-4		2	1.69	(4,3,3,3,3,5);	Assumption input pesticides = input in soil, 90% oil	37.0	63.0
resource, in air	Carbon dioxide, in air	-	-	kg	4.86E+0		1	1.24	(3,1,1,3,1,5);	Estimation for CO2, 1.8 kg CO2 per kg kenaf	37.0	63.0
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	4.86E+1		1	1.24	(3,1,1,3,1,5);	Estimation for energy 18 MJ / kg kenaf	37.0	63.0
resource, land	Occupation, arable, non-irrigated	-	-	m2a	1.79E+0		1	1.26	(2,2,3,2,1,5);	Cultivation of one ha for the time of 4-5 months (nicht bewässert)	95.0	5.0
	Transformation, from arable, non-irrigated	-	-	m2	4.76E+0		1	1.32	(2,2,3,3,1,5);	Assumption transformation from arable land to arable land, (not irrigated)	95.0	5.0
	Transformation, to arable, non-irrigated	-	-	m2	4.76E+0		1	1.32	(2,2,3,3,1,5);	Assumption transformation from arable land to arable land, (not irrigated)	95.0	5.0

3.4 Data Quality Considerations

There is a wide range concerning the use of fertilizers and pesticides as well as the yields depending on the region and the type of agriculture. Especially concerning the use of pesticides there are only very few data available. Large uncertainties exist for emissions on the field. The inventory is based on published data, statistics and expert interviews. Nevertheless the data are acceptable for an average production of fibres. To quantify the uncertainties the simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Abbreviations

BOD Biological oxygen demand

COD Chemical oxygen demand

DM Dry Matter

EcoSpold Meta Information

Tab. A. 4 EcoSpold Meta Information of jute production

ReferenceFunction	401	Name	jute production, irrigated system
Geography	662	Location	IN
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	2.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	69
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	The inventories include the mainly manual cultivation of Jute from conventional production standards. Included steps are soil cultivation, pestizides fertilisation (mineral fertilizer), harvest, loading for transport and extraction of the fibres after retting process.
	404	Amount	1
	490	LocalName	Juteproduktion, Anbau bewässert
	491	Synonyms	
	492	GeneralComment	Production of Jute delivers the co-products ' jute fibres' and 'jute stalks'. An allocation was carried out based on economical and mass criteria
	494	InfrastructureIncluded	1
	495	Category	agricultural production
	496	SubCategory	plant production
	497	LocalCategory	Landwirtschaftliche Produktion
	498	LocalSubCategory	Pflanzenbau
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	1995
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	
Geography	663	Text	Data from expert gremiums Indian Jute Insitute and experts assumptions. India is the most important producers of Jute fibres
Technology	692	Text	Cultivation of Jute from conventional production standards
Representativeness	722	Percent	55
	724	ProductionVolume	The world production 2006 was 2'466'620 t Jute (1'415'700 t from India)
	725	SamplingProcedure	Expert interviews, literature data and assumptions.
	726	Extrapolations	
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	69
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	40
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
ProofReading	762	PageNumbers	biofibres
	5616	Validator	8
	5615	Details	Passed
	5619	OtherDetails	none

Tab. A. 5 EcoSpold Meta Information of kenaf production

ReferenceFunction	401	Name	kenaf production
Geography	662	Location	IN
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	2.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	69
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	The inventories include the mainly manual cultivation of Kenaf from conventional production standards. Included steps are soil cultivation, pesticides fertilisation (mineral fertilizer), harvest, loading for transport and extraction of the fibres after retting process.
	404	Amount	1
	490	LocalName	Kenaf Produktion
	491	Synonyms	
	492	GeneralComment	Production of Kenaf delivers the co-products 'kenaf fibres, at farm' and 'kenaf stalks, from fibre production, at farm'. An allocation was carried out based on economical and mass criteria
	494	InfrastructureIncluded	1
	495	Category	agricultural production
	496	SubCategory	plant production
	497	LocalCategory	Landwirtschaftliche Produktion
	498	LocalSubCategory	Pflanzenbau
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	1995
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	
Geography	663	Text	Data from expert panel Indian Jute Institute and experts assumptions. India is the most important producers of Kenaf fibres
Technology	692	Text	Cultivation of kenaf from conventional production standards
Representativeness	722	Percent	55
	724	ProductionVolume	World production 2000: 0.51 Mio. t (China: 0.244 Mio t / India: 0.2 Mio t)
	725	SamplingProcedure	Expert interviews, literature data and assumptions.
	726	Extrapolations	
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	69
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	48
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	bio fibres
ProofReading	5616	Validator	8
	5615	Details	Passed
	5619	OtherDetails	none

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4 Bast fibres, yarn and textile production

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Review: Hans-Jörg Althaus, Empa

Last changes:

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Abstract

The inventory of the processing steps for bast fibres ‘yarn production, bast fibres’, ‘weaving, bast fibres’ refer to production from India.

The inventory of the bast fibre processing include data about the transport to the factories, infrastructure of plant and process specific energy and material need apart from the processed bast fibres themselves. The inventory for the resulting products for different bast fibres ‘yarn, at plant’ and ‘textile, at plant’ include additionally the production of the required bast fibres themselves considering material loss of processing.

The reference function of the processing is 1 kg of related output good respectively bast fibre type. The reference function of the resulting products is 1 kg yarn or 1 kg woven textile respectively.

4.1 Bast fibres products and processing

Jute and Kenaf is mainly used in the domestic markets of the producing countries. The following commercially important products are manufactured from jute fibres:

- Plied Yarn
- Twine yarn
- Woven Sacking fabric
- Woven Hessian fabric
- Woven Canvas fabric
- Woven Tarpaulin fabric
- Woven Carpet
- Woven Carpet Backing fabric
- Decorative fabrics
- Woven and Nonwoven Geotextiles
- Composites

In these work datasets for the production of yarns have been worked out. Yarns are the base for all woven application as sacks, geotextiles or woven carpets. The data for the production have been collected in jute mills producing sacks and hessian. The process has been split up in the processes shown below to give the possibility to adopt it to other products if necessary.

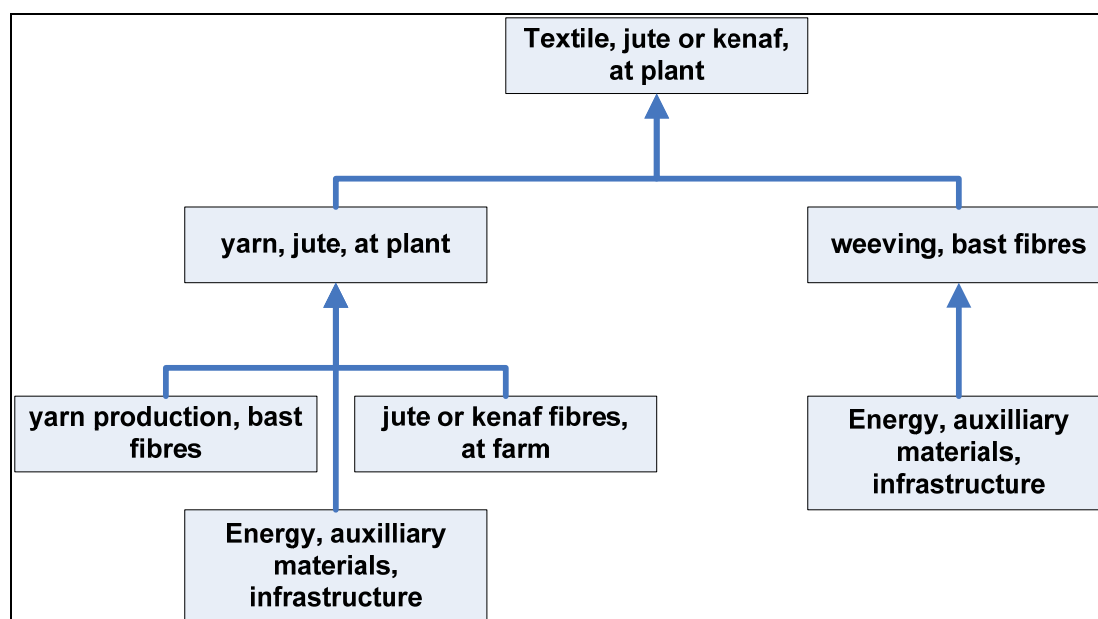


Fig. 4.1 Structure of the processes for the production of textiles from bast fibres

The inventories of processing include data of processing steps apart from the fibres production itself. The inventories of output products for different bast fibres ‘yarn, at plant’ and ‘textile, at plant’ include the required bast fibres input good inclusive material loss of processing. Data about different process steps are listed in Tab. 3.1. The data is based on expert interviews with persons from Jute mill in Kolkata India (S.K. Dokania, General Manager, The Hooghly Mills Co. Ltd., 2003) and experts from “Central Research Institute for Jut & Allied Fibres”, Dr. H.S. Sen.. India is one of the main producer of bast fibre products (Jute and Kenaf mainly).

The total amount of energy and auxiliary products applied were split on different process stages according to expert judgements. Data for infrastructure was added for consistency with ecoinvent processes. The infrastructure of the mechanical steps spinning and weaving was considered to be similar as in the packaging industry. Generally infrastructure was assumed to contribute less than 5% of impacts.

Average transport distances were assumed to be about 250 km from the local trader to the bast fibres mills. Yarn production and if required weaving were assumed to be carried out on the same site mainly, as in the example of Hessian production in Jute mill ‘The Hooghly Mills Co. Ltd.’ In Kolkata, India

No data was available for required water and waste water treatment of processing, and therefore not included.

Tab. 4.1 Energy and material need split for processing steps in jute mill (Total energy and material need from expert interview Jute Mill, S.K. Dokania, General Manager, The Hooghly Mills Co. Ltd., 2003)

Process	Fibre loss	Energy need	Material need	Assumptions
Hessian for textile production: - Opening/Batching - Carding - Combing/flyer - Spinning - Weaving	18 %	Ø 1250 MJ/t (electricity mainly) 20% 40% 10% 10% 20%	9.4 kg/t batching oil (petroleum based or vegetable oil as for example rice bran oil)	Transport 250 km local trader to bast fibres mill (lorry) Infrastructure similar size and equipment than packaging plant

4.1.1 Input from techno sphere

Existing ecoinvent data inventories were applied for the energy and material consumption (water, salt, auxiliary products), transports, assumption of infrastructure and waste water treatment in the processing of bast fibres.

4.1.2 Emissions, CO₂-binding and Solar Energy in Biomass of bast fibres

Emissions, CO₂-binding and solar energy are included in the used data inventories. No direct emissions were considered in the processing.

4.1.3 Land occupation

Land occupation is included in the used data inventories of fibres cultivation and infrastructure of processing (infrastructure of mechanical processing of bast fibres assumed similar than packaging industry 'packaging box production unit').

4.2 Statistics of yarn and textile production

Statistics of world production of Jute, Kenaf and allied bast fibres are published in the internet by the FAO³². The world production 2006 was 2'466'620 t Jute (1'415'700 t from India) and 359'260 t Kenaf and allied fibres (159'300 t from India).

The processed Jute from India is exported 35% as yarn, 30% as hessian and 35% other products as carpet, shopping bags, floor coverings, sacking, geotextile etc (P.K. Banerjee, 2003).

4.3 Characterisation and Use of bast yarn and textiles³³

Bast fibres, mainly Jute and Kenaf, are cultivated almost exclusively in developing countries of East Asia and in some parts of Latin America. Bangladesh, India and Thailand account for over 90 percent of world production. The fibre is processed mainly in the producing countries themselves and is used for the manufacturing of traditional products such as hessian cloth, food grade bags, carpet backing and other floor covering. Diversified jute products, such as geo-textiles and composites are also manufactured in relatively small quantities. Jute constitutes a low proportion of the value of world trade, but its cultivation and processing is labour-intensive and therefore provides a livelihood and an important source of food security for many farmers and their families in Asia.

4.4 Life Cycle Inventories of bast fibre products and processing

The inventories of processing refer to the spinning of yarn and weaving of textiles. The inventories of processing steps include data about the transport to the factories, the energy and material need apart from fibres as well as estimations about the related infrastructure. The reference function refers to the processing of 1 kg input good. The inventories of resulting products yarn and textile include the production of fibres itself. The reference function is 1 kg yarn or textile respectively.

³² FAO statistics about world production and price from <http://ftp.fao.org/docrep/fao/009/j8117m/j8117m00.pdf> (August 2007)

³³ Text about Jute, kenaf and allied fibres from <http://www.fao.org/docrep/006/y5143e/y5143e1g.htm> (August 2007)

Tab. 4.2 Unit process of yarn production from jute

	Name	Location	Unit	yarn, jute, at plant	Uncertainty Type	Standard Deviation 95%	General Comment
	Location InfrastructureProcess Unit			IN 0 kg			
product	yarn, jute, at plant	IN	kg	1			
technosphere	jute fibres, irrigated system, at farm	IN	kg	6.96E-1	1	1.16	(2,2,3,1,1,4); Fibres loss Jute mill 18% (expert interview general manager of Jute mill, 2003). Assumption Carbotech AG, loss mainly in yarn production and only 2% in further processing (weaving). 60% fibres from rainfed and 40% from irrigated systems (FAO statistics)
technosphere	jute fibres, rainfed system, at farm	IN	kg	4.64E-1	1	1.16	(2,2,3,1,1,4); Fibres loss 18% (expert interview general manager of Jute mill, 2003). Assumption Carbotech AG, loss mainly in yarn production and only 2% in further processing (weaving). 60% fibres from rainfed and 40% from irrigated systems (FAO statistics)
technosphere	yarn production, bast fibres	IN	kg	1.16E+0	1	1.17	(2,2,3,3,1,4); Fibres loss jute mill about 18% (expert interview general manager of Jute mill, 2003). Assumption Carbotech AG, loss mainly in yarn production and only 2% in further processing (weaving).
technosphere	disposal, paper, 11.2% water, to sanitary landfill	CH	kg	1.60E-1	1	1.17	(2,2,3,3,1,4); 0

Tab. 4.3 Unit process of yarn production from kenaf

	Name	Location	Unit	yarn, kenaf, at plant	Uncertainty Type	Standard Deviation 95%	General Comment
	Location InfrastructureProcess Unit			IN 0 kg			
product	yarn, kenaf, at plant	IN	kg	1			
technosphere	kenaf fibres, at farm	IN	kg	1.16E+0	1	1.16	(2,2,3,1,1,4); Fibres loss Jute mill 18% (expert interview general manager of Jute mill, 2003). Assumption Carbotech AG, loss mainly in yarn production and only 2% in further processing (weaving). 60% fibres from rainfed and 40% from irrigated systems (FAO statistics)
technosphere	yarn production, bast fibres	IN	kg	1.16E+0	1	1.17	(2,2,3,3,1,4); Fibres loss jute mill about 18% (expert interview general manager of Jute mill, 2003). Assumption Carbotech AG, loss mainly in yarn production and only 2% in further processing (weaving).
technosphere	disposal, paper, 11.2% water, to sanitary landfill	CH	kg	1.60E-1	1	1.17	(2,2,3,3,1,4); 0

Tab. 4.4 Unit process of weaving of bast fibres

	Name	Location	Unit	weaving, bast fibres	Uncertainty Type	Standard Deviation 95%	GeneralComment
	Location InfrastructureProcess Unit			IN 0 kg			
product	weaving, bast fibres	IN	kg	1			
technosphere	electricity, production mix UCTE	UCTE	kWh	7.44E-1	1	1.16	(2,2,3,1,1,4); Expert Interview, S.K. Dokania, General Manager of Jute Mill 'The Hooghly Mills co. LTD' (2003). Normally the electricity is produced by diesel engines.
emission air, unspecified	Heat, waste	-	MJ	2.68E+0	1	1.16	(2,2,3,1,1,4); Heat waste derived from electricity consumption
technosphere	transport, lorry >16t, fleet average	RER	tkm	5.00E-2	1	2.03	(2,2,3,3,1,4); Assumption Carbotech AG, weaving and yarn production generally at same site 80%, 10% inland transport of yarn to further processing and 10% inland transport to port and overseas transport to further processing
	transport, transoceanic freight ship	OCE	tkm	1.20E+0	1	2.03	(2,2,3,3,1,4); Assumption Carbotech AG, weaving and yarn production generally at same site 80%, 10% inland transport of yarn to further processing and 10% inland transport to port and overseas transport to further processing
	packaging box production unit	RER	unit	5.00E-10	1	3.50	(5,1,1,3,4,5); Estimation infrastructure of Jute mill comparable to packaging plant (building, equipment, electronics). 50% accounted for spinning and 50% for weaving. Assumption life time 50 years with an output of 20'000 t of textile per year. Totally max. 5% of impacts.

Tab. 4.5 Unit process of textile production from jute yarn

	Name	Location	Unit	textile, jute, at plant	Uncertainty Type	Standard Deviation 95%	GeneralComment
	Location InfrastructureProcess Unit			IN 0 kg			
product	textile, jute, at plant	IN	kg	1			
technosphere	yarn, jute, at plant	IN	kg	1.02E+0	1	1.30	(4,2,2,1,1,5); Fibres loss Jute mill 18% (expert interview general manager of Jute mill, 2003). Assumption Carbotech AG, loss mainly in yarn production and only 2% in further processing (weaving).
technosphere	weaving, bast fibres	IN	kg	1.02E+0	1	1.30	(4,2,2,1,1,5); Fibres loss Jute mill 18% (expert interview general manager of Jute mill, 2003). Assumption Carbotech AG, loss mainly in yarn production and only 2% in further processing (weaving).
technosphere	disposal, paper, 11.2% water, to sanitary landfill	CH	kg	2.00E-2	1	1.30	(4,2,2,1,1,5); The fibre losses are disposed or given for free to the workers for different uses.

4.5 Data Quality Considerations

There is a wide range concerning the use of fertilizers and pesticides as well as the yields depending on the region and the type of agriculture. Especially concerning the use of pesticides there are only very few data available. Large uncertainties exist for emissions on the field. The inventory is based on published data, statistics and expert interviews. The data of jute fibre processing are also connected with high uncertainties, because there is a large amount of jute mills and no statistic data are available e.g. on energy consumption. Nevertheless the data are acceptable for an average production. To quantify the uncertainties the simplified approach with a pedigree matrix has been used for calculating the standard deviation.

EcoSpold Meta Information

Tab. A. 6 EcoSpold Meta Information textile production from jute

ReferenceFunction	401	Name	textile, jute, at plant
Geography	662	Location	IN
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	2.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	69
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	The inventorie includes the cultiavtion of bast fibres and its further spinning to yarn and weaving to textile.
	404	Amount	1
	490	LocalName	Gewebe, Jute, ab Werk
	491	Synonyms	
	492	GeneralComment	Inventory refers to 1 kg woven textile
	494	InfrastructureIncluded	1
	495	Category	textiles
	496	SubCategory	production
	497	LocalCategory	Textile
	498	LocalSubCategory	Gewinnung
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	
Geography	663	Text	Most important producer of bast fibres procduts is in India
Technology	692	Text	Weaving was considered for the textile production in the present study.
Representativeness	722	Percent	
	724	ProductionVolume	
	725	SamplingProcedure	Data were compiled from literature.
	726	Extrapolations	
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	69
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	48
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	
ProofReading	5616	Validator	8
	5615	Details	Passed
	5619	OtherDetails	none

Tab. A. 7 EcoSpold Meta Information of yarn production

ReferenceFunction	401	Name	yarn production, bast fibres
Geography	662	Location	IN
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	2.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	69
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	The inventories include energy consumption, transport and infrastructure related to the processing of bast fibres to yarn (opening, batching, cardening, spinning).
	404	Amount	1
	490	LocalName	Garnherstellung, Bastfasern, Spinnerei
	491	Synonyms	
	492	GeneralComment	Inventory refers to the processing of 1 kg fibres only, without the production of the fibres itself.
	494	InfrastructureIncluded	1
	495	Category	textiles
	496	SubCategory	production
	497	LocalCategory	Textile
	498	LocalSubCategory	Gewinnung
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	
Geography	663	Text	Most bast fibres are processed in Asia . India was considered for the processing of bast fibres in this study.
Technology	692	Text	Opening, batching, cardening, combing and spinning are the main processes of the yarn production. Mechanical cleaning and no chemical cleaning was assumed for this study.
Representativeness	722	Percent	
	724	ProductionVolume	
	725	SamplingProcedure	Data are from expert interviews and literature.
	726	Extrapolations	Infrastructure of mechanical processings of fibres was assumed similar to packaging plant
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	69
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	48
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	
ProofReading	5616	Validator	8
	5615	Details	Passed
	5619	OtherDetails	none

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5 Polylactide, granulate

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Review:

Last changes:

10 August 2007

Abstract

For this polymer two inventories have been developed. One inventory refers to the PLA production from the world largest PLA plant in Blair (Nebraska). This production site belongs to the company NatureWorks®. From this data another process for a general production plant has been derived.

The eco-profile of the NatureWorks® polylactide (PLA) production was published 2007 in the journal of Industrial Biotechnology (Vol. 3 No.1, spring 2007). The publication does not give the input amounts but only the resulting eco-profile for the total of evaluated processes. The system boundaries include the corn production, harvest, drying and transport as well as the further processing of corn to dextrose, lactic acid, lactide and finally polylactide. The related data inventories of those processes include from the cradle-to-polymer-factory-gate the consumption of resources and energy as well as emissions and depositions of waste.

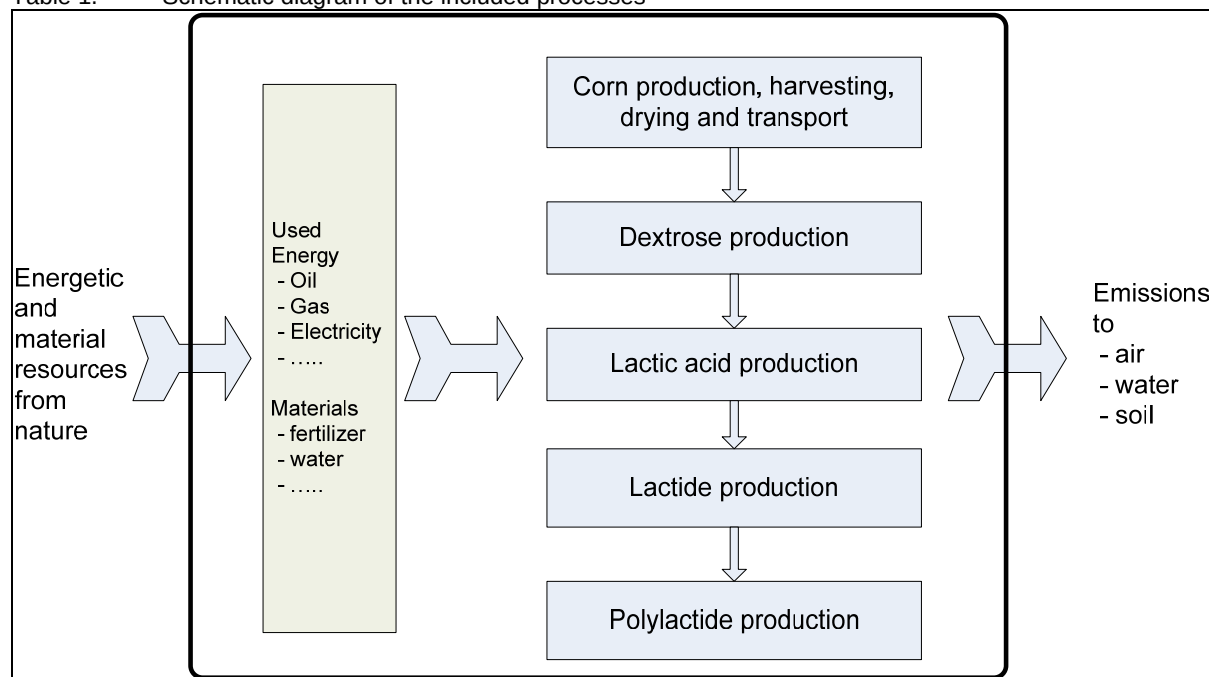
The eco-profile of the NaturWorks® production was optimised in the evaluated year 2006 by purchasing renewable energy certificates (REC) for the equivalent of the overall required electricity of the PLA production. The REC were bought in form of wind power electricity replacing the regional electricity mix from fossil fuels (mainly coal, oil, gas and nuclear). Fuel consumption and emissions were adapted respectively, reducing energy demand, CO₂, CO, NO_x, SO_x and HC emissions significantly.

The inventory of PLA production is 1 kg PLA granulate from the NatureWorks® plant in Blair, Nebraska. Especially because of the REC's this inventory is only valid for this plant, being the largest, but not the only and it is planned to realize other plants in the future. This is the reason a general PLA inventory was elaborated using a general electricity mix instead of the wind-mix used for the PLA of NatureWorks®.

5.1 Characteristic of PLA production

The main processes of the PLA production are illustrated in Table 1. For further details of the PLA production see the two articles from E. Vink: in "Journal of Industrial Biotechnology" (Vol. 3 No.1, spring 2007) and in "Polymer degradation and stability", 2002.

Table 1. Schematic diagram of the included processes



The publication does not give data for the input amounts but only the resulting eco-profile. According to the methodology used from Boustead for the European plastics Industry the figures are given for used resources and emissions separately for the following categories:

- fuels production,
- fuels use,
- transports and
- processes (feedstock for the energetic resources).

Evaluated processes within the systems boundaries are the corn production, harvest, drying and transport as well as the further processing of corn to dextrose, lactic acid, lactides and finally polylactide. The related data inventories of those processes include from the cradle-to-polymer-factory-gate the consumption of resources and energy as well as emissions and depositions of waste.

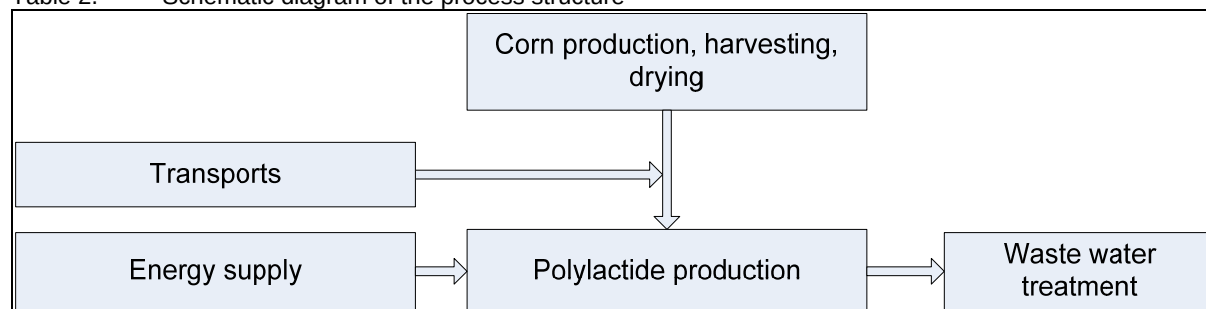
The eco-profile of the NaturWorks production was optimised in the evaluated year 2006 by purchasing renewable energy certificates (REC) for the equivalent of the overall required electricity of the PLA production. The REC were bought in form of wind power electricity replacing the regional electricity mix from fossil fuels (mainly coal, oil, gas and nuclear). Fuel consumption and emissions were adapted respectively, reducing energy demand, CO₂, CO, NO_x, SO_x and HC emissions significantly.

For the following reasons the published inventory has not been taken as it is but split up in different processes as far as possible:

- To make it consistent with the ecoinvent processes
- To make it more transparent
- To introduce the infrastructure
- To give the possibility to create a general PLA process

To elaborate a more transparent process the overall process has been split up in the processes shown in Table 2. It was not possible to do it more detailed because no detailed data were available.

Table 2. Schematic diagram of the process structure



The procedure how the data for the different sub processes was obtained from the aggregated data is described in chapter 5.5.

5.2 Allocation to co-products

The data in the publication are allocated to the different by products so no further allocation was necessary.

5.3 Statistics of PLA production

NatureWorks® is the main producer for PLA. The total production amount is about 140'000t per year.

5.4 Characterisation and Use of PLA

Polylactideacid is used as a polymer for applications in the areas of fibres and non wovens, films, extruded and thermoformed containers, and extrusion and emulsion coatings. The PLA polymer fits for many applications currently using polyester, polyolefin's, polystyrene and cellulosics. The polylactid acid of the main producer NatureWorks® is derived from corn.

5.5 Life Cycle Inventories of PLA from NatureWorks®

The inventory refers to the production of 1 kg PLA granulate. It includes all the processes from the agriculture to the plant gate. The following procedure has been carried out to divide the overall data in the different processes shown in Table 2.

- **Transport**
The resources used for transport are given in the publication. Assuming that the given amount of oil for transport was used for lorry transport the distances has been evaluated taking into account the precombustion of the diesel.
- **Corn production**
The amount of used corn could be calculated from the data given in the publication and so the process of corn growing in the USA from ecoinvent could be used. Because of the fact that the given data considers the allocation of the by products no further allocation was necessary.
- **Waste water treatment**
The amount of waste water was taken from a written communication from NatureWorks. Main emissions like COD and BOD from the process waste water treatment have been compared to the values given in the publication showing a good correspondence.
- **Energy supply**
To avoid double counting the energetic resources for corn growing and waste water treatment

could be evaluated and subtracted from the values given in the publication for the production process. The resulting energetic resources are used for the process according to the publicised data. For these energetic resources the corresponding processes to produce heat were chosen from ecoinvent.

- Electricity

The amount of electricity could be evaluated from the given carbon dioxide emissions for electricity and the local electricity mix (MAPP) used to calculate these emissions, see K. Suong (2005) and the given wind energy used. For the electricity wind power has been used according to the REC purchased by NatureWorks®. To be in accordance with ecoinvent the production and the use infrastructure for the electricity network have been taken into account.

- Disposal

For all the processes described above the disposal has been evaluated and subtracted from the values given in the publication. The remaining amounts have been used.

The data on energetic resources and on CO₂ equivalents given detailed in the publication has been used as the main quality check for the generated process. This comparison showed a very good accordance with the emissions contributing to climate change. The non renewable resources are slightly higher, around 10%, because of higher amount of resources used for precombustion in ecoinvent compared to the resources given in the publication and because of the included infrastructure. These results show that the generated process seemed to be a good description of the PLA production. Further the ozone formation of the created process has been compared to the impacts given in the publication. The ozone formation given in the publication was higher than the one from the ecoinvent process because of NMVOC emissions in the production processes. The difference has been added to the process to get the published value. Differences result for some agricultural emissions because of the fact that the maize production in ecoinvent shows higher emissions than these used in the publication.

The purchase of CERs is very positive for the climate and other environmental impact, but on the other hand it is not a intrinsic property of PLA and is possible for all production and services. To generate a general data inventory of PLA the process for PLA, NaturWork was changed by using the European electricity mix instead of the wind power.

Tab. 5.1 Unit process of PLA production, NatureWorks Nebraska (inclusive compensation of purchased REC)

	Name	Location	Unit	poly lactide, granulate, NatureWorks Minnesota	Uncertainty Type	Standard Deviation 95%
	Location InfrastructureProcess Unit			US 0 kg		
product	polylactide, granulate, NatureWorks Minnesota	US	kg	1		
technosphere	treatment, maize starch production effluent, to wastewater treatment, class 2	CH	m3	3.20E-3	1	1.13
	electricity, at wind power plant	RER	kWh	1.99E+0	1	1.22
	transmission network, long-distance	UCTE	km	3.17E-10	1	3.05
	transmission network, electricity, medium voltage	CH	km	3.24E-8	1	3.05
	transmission network, electricity, high voltage	CH	km	8.44E-9	1	3.05
	distribution network, electricity, low voltage	CH	km	0.000'000'2	1	3.05
	sulphur hexafluoride, liquid, at plant	RER	kg	1.00E-7	1	1.22
emission air, low population density	Heat, waste	-	MJ	7.17E+0	1	1.22
technosphere	natural gas, at long-distance pipeline	RER	Nm3	3.60E-3	1	1.22
	naphtha, at refinery	RER	kg	7.00E-3	1	1.22
	transport, lorry >16t, fleet average	RER	tkm	2.00E-1	1	2.05
	natural gas, burned in industrial furnace >100kW	RER	MJ	1.85E+1	1	1.22
	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	MJ	1.59E-1	1	1.22
technosphere	chemical plant, organics	RER	unit	4.00E-10	1	3.27
	corn, at farm	US	kg	1.51E+0	1	1.21
	disposal, plastics, mixture, 15.3% water, to sanitary	CH	kg	1.00E-3	1	1.09
	disposal, hazardous waste, 25% water, to hazardous	CH	kg	6.40E-3	1	1.09
emission air, low population density	NM VOC, non-methane volatile organic compounds, unspecified origin	-	kg	2.52E-3	1	1.62
emission air, unspecified	Sulfur hexafluoride	-	kg	1.00E-7	1	1.62
	Ozone	-	kg	4.60E-6	1	1.62
	Heat, waste	-	MJ	1.05E-1	1	1.30

Tab. 5.2 Unit process of PLA production

	Name	Location	Unit	poly(lactide, granulate, at plant)	Uncertainty Type	Standard Deviation 95%
	Location InfrastructureProcess Unit			US 0 kg		
product	polylactide, granulate, at plant	US	kg	1		
technosphere	treatment, maize starch production effluent, to wastewater treatment, class 2	CH	m3	3.20E-3	1	1.24
technosphere	electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.83E+0	1	1.24
emission air, low population density	Heat, waste	-	MJ	6.58E+0	1	1.24
technosphere	natural gas, at long-distance pipeline	RER	Nm3	3.60E-3	1	1.24
	naphtha, at refinery	RER	kg	7.00E-3	1	1.24
	transport, lorry >16t, fleet average	RER	tkm	2.00E-1	1	2.06
	natural gas, burned in industrial furnace >100kW	RER	MJ	1.85E+1	1	1.24
	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	MJ	1.59E-1	1	1.24
technosphere	chemical plant, organics	RER	unit	4.00E-10	1	3.28
	corn, at farm	US	kg	1.51E+0	1	1.24
	disposal, plastics, mixture, 15.3% water, to sanitary landfill	CH	kg	1.00E-3	1	1.09
	disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	6.40E-3	1	1.09
emission air, low population density	NMVOC, non-methane volatile organic compounds, unspecified origin	-	kg	2.52E-3	1	1.58

5.6 Data Quality Considerations

Tab. 5.1 and Tab. 5.2 show data quality indicators for the inventory of PLA production systems. The simplified approach with a pedigree matrix has been used for calculating the standard deviation. The inventory is based on published data, so the data quality is quite reliable.

EcoSpold Meta Information

Tab. A. 8 EcoSpold Meta Information of cotton production and related co-products

Name	polylactide, granulate, NatureWorks Minnesota	polylactide, granulate, at plant
Location	US	US
Infrastructure	0	0
Unit	kg	kg
Type	1	1
Version	1.0	1.0
energyValues	0	0
LanguageCode	en	en
LocalLanguageCode	de	de
Person	69	69
QualityNetwork	1	1
DataSetRelatesToProduct	1	1
IncludedProcesses	The publication contains only cumulative results. To get more transparency and to be in accordance with other eco invent processes and the guideline the data has been splitted up in maize production, energy use, transport and waste water treatment. The infrastructure has been added. So the emissions from energy consumption, waste water treatment and raw material inputs have been subtracted resulting in a difference in NMVOC emissions shown in this process.	
Amount	1	1
LocalName	Poly lactide, Granulat, NatureWorks Minnesota	Poly lactide, Granulat, ab Werk
Synonyms		
GeneralComment	Inventory refers to the production of 1 kg PLA in the world largest PLA plant. The inventories include the LCI data from the report of the producer NatureWorks. Inventory refers to the production of 1 kg PLA. It is based on data from the world largest PLA plant. The inventories include the LCI data from the report of the producer NatureWorks. In this publication only aggregated data are reported. To get more transparency and to be in accordance with other eco invent processes and the guideline the data has been splitted up in maize production, energy use, transport and waste water treatment. The infrastructure has been added. In the published data from the plant in Nebraska the carbon dioxide emissions are offset by wind power certificates. This is not an intrinsic property of the PLA production, but specific to this plant. It is why in this process for the electricity production the UCTE Mix has been chosen. If the polymer is produced by NatureWorks in Nebraska the specific process should be used.	
InfrastructureIncluded	1	1
Category	plastics	plastics
SubCategory	polymers	polymers
LocalCategory	Kunststoffe	Kunststoffe
LocalSubCategory	Polymere (Granulate)	Polymere (Granulate)
Formula		
StatisticalClassification		
CASNumber		
StartDate	2006	2006
EndDate	2007	2007
DataValidForEntirePeriod	1	1
OtherPeriodText		
Text	Refers to a production of 1 kg PLA in Nebraska by NatureWorks. This is the largest PLA plant in the world. (taking into account the plant specific bonus from special energy certificates, CO2 emissions and other emissions recompensation by wind power investments) Based on the data of the production of 1 kg PLA in Nebraska by NatureWorks. The electricity mix was changed to UCTE Mix to get a more general process being a better approach for other geographical locations, than this specific site.	
Text	Evaluation of plant site in Minnetonka	Evaluation of plant site in Nebraska
Percent		
ProductionVolume		
SamplingProcedure	Data were compiled from LCI report of producer	Data were compiled from LCI report of producer
Extrapolations		
UncertaintyAdjustments	none	none
Person	69	69
DataPublishedIn	2	2
ReferenceToPublishedSource	48	48
Copyright	1	1
AccessRestrictedTo	0	0
CompanyCode		
CountryCode		
PageNumbers		
Validator	8	8
Details	automatic validation in Excel	automatic validation in Excel
OtherDetails	none	none

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6 Viscose fibres, at plant

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 Review: Hans-Jörg Althaus, Empa

Last changes:

12 December 2007

Abstract

The inventory refers to the viscose fibres production from sulphat pulp. For the evaluation data from Lenzing AG in Austria published by the Federal Environment Agency of Austria was considered³⁴.

The inventory includes the production of required sulphate pulp, resources and energy need for the transformation of the pulp to spin able viscose as well as estimations about material transports, infrastructure, waste and waste water treatment. Site specific characteristics of the plant in Lenzing were adapted to a more general standard (i.e. use of purchased pulp completing production on site³⁵). This inventory is valid for a average viscose production.

The reference function of the inventory is 1 kg viscose fibres, at plant.

6.1 Characteristics and data of viscose production

The main process of the viscose production is the transformation of sulphate pulp to spin able viscose using CS₂. High water amount and energy in form of heat and electricity are typical for the transformation process. Required resources and energy in Tab. 3.1 are from the Lenzing production of viscose fibres 2001 report in 'Medienübergreifende Umweltkontrolle in ausgewählten Gebieten' (Federal Environment Agency of Austria, 2004).

Tab. 6.1 Characteristics of viscose fibre production, Lenzing AG 2001, Austria

Multitoutput process	Total production 2001 in t	Input materials	Energy need	Assumptions
Viscose production: - viscose fibres - sulphuric acid - sodium sulfat	160'855 126'306 138'844	Sulphate pulp: about 163'910 t Water river: 32'849'000 m ³ Sodium chloride (33%): 6'572 t Sodium hypochloride: 4'520 t Caustic soda s. (50%): 117'703 t Liquid sulfur: 33'215 t Oxygen : 2'994 t Nitrogen: 7'543 t Zinc sulfate: 2'377 t Auxiliaries: about 2'500 t	Electricity: 164'421 MWh Heat: 894'450 MWh Assumption: Electricity mix Austria. Heat production as Lenzing AG from gas, oil, waste wood, coal.	50% of pulp produced on site, 50% purchased over sea. Waste water: 50% of used water to waste water plant

³⁴ The data about Lenzing production was collected for the year 2001 and published 2004 in 'Medienübergreifende Umweltkontrolle in ausgewählten Gebieten'. Report downloaded from homepage in August 2007 http://www.umweltbundesamt.at/fileadmin/site/umweltthemen/industrie/pdfs/MUK/07_Lenzing.pdf

³⁵ The production of Viscose by Lenzing AG has been completed since 2001 with new production sites in Asia (China, India, Indonesia). The production of Indonesia is based for example on pulp bought from USA, Brasil and South Africa (see http://www.lenzing.com/images/lenzing_kk/gesamt_end.pdf). High production costs in Europe and increasing wood price are relevant factors for the transfer of viscose production to Asia.

Existing ecoinvent data inventories were considered for sulphate pulp, input materials, energy supply, infrastructure, transport and waste treatment. Fuel mix for heat production was chosen as reported from Lenzing AG (coal, waste wood, gas, oil, sludge and others).

Whereas Lenzing AG in Austria produced 2001 the sulphate pulp on site, purchasing of sulphate pulp is reported for production sites in Asia (USA, Brazil and South Africa). No data was available for a general share of pulp production on site compared to use of purchased pulp. Purchasing of sulphate pulp with overseas transports was considered for 50% of processed sulphate pulp. An average transport distance of 15'000 km was added considering for overseas transport (as for example from USA, Brazil and South Africa to Asia). Further assumptions for transports were based on ecoinvent guidelines (600 km train and 100 km lorry 32 t for used chemicals and twice transport distance of about 200 km for sulphate pulp from overseas).

Data of on site waste water treatment from Lenzing AG Austria does not indicate specific values for the viscose fibres production but only for all activities in the production site (waste water treatment for about 35% of total water consumption on site). For the reported water consumption of fibre production was assumed 50% to be treated in communal waste water treatment plant.

Infrastructure of fibre production was assumed to be comparable to pulp production with a similar output amount per year. Amount and process were chosen as in the inventory of sulphate pulp.

6.2 Allocation to co-products

The co-product 'sulphuric acid' and 'sodium sulphate' were considered by an allocation upon economical criteria. The allocation was based on the reported production amount of Lenzing AG 2001 and related market prices from internet sources.

Viscose fibres were reported about 0.43 US \$ or 0.6 Euro/kg in Chinese import evaluation of May 2007³⁶. Sulphuric acid is reported about 55 US \$ or 40 Euro/t and sodium sulphate about 113 US \$ or 85 Euro/t³⁷.

From the calculated results the economic allocation of 85% to viscose fibres, 10% to sodium sulphate and 5% to sulphuric acid. The allocation was not considered for the sulphate pulp, considering the organic material as basic material for fibre production mainly.

6.3 Statistics of viscose production

The production amount of Lenzing AG Austria in 2001 was about 160'855 t of viscose fibres (Federal Environment Agency of Austria, 2004). Since 2001 Lenzing AG has been growing steadily in the global market investing in production sites in Asia (China and Indonesia).

The total world production of viscose staple fibres was estimated in a press release by Lenzing AG about 1.8 millions tons for the year 2003³⁸. No further statistics were found for the development of the world production of viscose fibres of the past years. China's recently fast developing fibre industry is actually one of the main producers of viscose yarn and textiles.

³⁶ China Chemical fibres and Textiles www.ccfgroup.com (July 2007)

³⁷ Data from <http://www.the-innovation-group.com/ChemProfiles> (July 2007)

³⁸ <http://www.lenzing.com/press/en/842.jsp> data from article 'The Lenzing Group Confirms 2003' (August 2007)

6.4 Characterisation and Use of Viscose

Viscose is used as polymer for application in the areas of fibres and non woven products. The Viscose fibres (rayon) are derived from sulphate pulp and are highly absorbent, soft and comfortable, easy to dye and drapes well.

Major fibre uses are the apparel, home furnishings, industrial uses as for example medical surgical products and non woven products as well as other uses as in feminine hygiene products.

6.5 Life Cycle Inventory of Viscose fibres

The inventory given in Tab. 6.2 refers to the production of 1 kg viscose fibres. It includes the production and transformation of sulphate pulp to spin able viscose, the spinning process as well as related transport and waste and waste water treatment.

6.6 Data Quality Considerations

Tab. 6.2 shows data quality indicators for the inventory of viscose production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation. The inventory is based on published data from Lenzing AG in Austria, statistics and extrapolations. Uncertainties exist regarding main origin of pulp³⁹ and for the processing due to market development with increasing transfer of production to Asian production sites. So there can be relevant differences to this average inventory.

³⁹ Influence of development of wood price in Europe (increasing use for heat production relevant for European market)

Tab. 6.2 Unit process of Viscose production

	3702		3703	3508	3706	3707	3708	3709	3792		2404	2404	2404
	Name	Location	Infrastructure	Processes	Unit	viscose	Uncertainty	Standard deviation	GeneralComment	viscose	sodium sulphat	sulphuric acid	
						production				fibres,	from viscose	from viscose	
						AT				at plant	production, at	production, at	
0	0	0	0										
kg	kg	kg	kg										
allocated products	viscose fibres, at plant	AT	0	kg	1					100	0	0	
	sodium sulphat from viscose production, at plant	AT	0	kg	8.63E-1					0	100	-	
	sulphuric acid from viscose production, at plant	AT	0	kg	7.85E-1					0	-	100	
technosphere	sulphate pulp, TCF bleached, at plant	RER	0	kg	1.02E+0	1	1.23	(2.3.2.3.1.5): Pulp amount and type from fibres production Lenzing AG 2001 (Federal Environment Agency- Austria, 2004). Use of existing data inventory 'sulphate pulp, TCF bleached, at plant'		100.0	-	-	
	pulp plant	RER	1	unit	5.00E-11	1	3.05	(2.3.2.3.1.5): Assumption Carbotech AG, infrastructure similar than pulp plant with comparable output fibres respectively pulp		100.0	-	-	
	transport, lorry 16-32t, EURO4	RER	0	tkm	3.53E-1	1	2.06	(2.3.2.3.1.5): Transport distance of chemicals 100 km Lkw 32t as in guidelines eco-invent, sulfat pulp about 200 km considered twice as in guidelines ecoinvent for oversea purchased goods. Assumption Carbotech AG 50% of fibres from oversea.		85.0	10.0	5.0	
	transport, freight, rail	RER	0	tkm	8.85E-1	1	2.06	(2.3.2.3.1.5): Assumption Carbotech AG 50% of fibres purchased from oversea, 50% produced on site. Estimations oversea transport distance about 15000 to 20000 km for transport from USA, Brasil and South Africa to Asia.		85.0	10.0	5.0	
	transport, transoceanic freight ship	OCE	0	tkm	4.46E+0	1	2.06	(2.3.2.3.1.5): Assumption Carbotech AG 50% of fibres purchased from oversea, 50% produced on site. Estimations oversea transport distance about 15000 to 20000 km for transport from USA, Brasil and South Africa to Asia.		85.0	10.0	5.0	
	electricity, low voltage, production RER, at grid	RER	0	kWh	1.02E+0	1	1.25	(2.3.3.3.1.5): Electricity amount for fibre production from Lenzing AG 2001 (Federal Environment Agency- Austria, 2004), general electricity mix UCTE		85.0	10.0	5.0	
emission air, unspecified	Heat, waste	-	-	MJ	3.68E+0	1	1.25	(2.3.3.3.1.5): Heat waste derived from electricity consumption		85.0	10.0	5.0	
technosphere	heat, light fuel oil, at industrial furnace 1MW	RER	0	MJ	1.74E-1	1	1.25	(2.3.3.3.1.5): Oil consumption for heat production from Lenzing AG 2001 (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	heat, natural gas, at industrial furnace low-NOx >100kW	RER	0	MJ	3.98E+0	1	1.25	(2.3.3.3.1.5): Gas consumption for heat production from Lenzing AG 2001 (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	wood chips, from industry, hardwood, burned in furnace 1000kW	CH	0	MJ	7.01E+0	1	1.25	(2.3.3.3.1.5): Wood consumption for heat production from Lenzing AG 2001 (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	heat, at hard coal industrial furnace 1-10MW	RER	0	MJ	6.64E+0	1	1.25	(2.3.3.3.1.5): Coal consumption for heat production from Lenzing AG 2001 (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	heat, digester sludge, at incineration plant, future, allocation price	CH	0	MJ	9.33E-1	1	1.25	(2.3.3.3.1.5): Sludge and others for heat production from Lenzing AG 2001 (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	heat, unspecific, in chemical plant	RER	0	MJ	7.72E-1	1	1.25	(2.3.3.3.1.5): Heat unspecific assumed for not specified energy in fibres production Lenzing AG 2001 (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
resource, in water	Water, river	-	-	m3	2.04E-1	1	1.25	(2.3.3.3.1.5): Water consumption for fibre production Lenzing AG 2001 (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
technosphere	sodium chloride, powder, at plant	RER	0	kg	1.24E-1	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	sodium hypochlorite, 15% in H2O, at plant	RER	0	kg	1.56E-1	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	7.32E-1	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	sulphur dioxide, liquid, at plant	RER	0	kg	2.06E-1	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	carbon disulfide, at plant	GLO	0	kg	9.11E-2	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	sulphuric acid, liquid, at plant	RER	0	kg	7.00E-2	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	oxygen, liquid, at plant	RER	0	kg	1.86E-2	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	nitrogen, liquid, at plant	RER	0	kg	4.69E-2	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	zinc monosulphate, ZnSO4.H2O, at plant	RER	0	kg	1.48E-2	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004)		85.0	10.0	5.0	
	chemicals inorganic, at plant	GLO	0	kg	1.55E-2	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004), no specific data about used auxiliaries, assumption Carbotech AG 50% organic and 50% anorganic		85.0	10.0	5.0	
	chemicals organic, at plant	GLO	0	kg	1.55E-2	1	1.25	(2.3.3.3.1.5): Data 2001 for fibre production Lenzing AG (Federal Environment Agency- Austria, 2004), no specific data about used auxiliaries, assumption Carbotech AG 50% organic and 50% anorganic		85.0	10.0	5.0	
	treatment, sewage, to wastewater treatment, class 3	CH	0	m3	1.02E-4	1	1.25	(2.3.3.3.1.5): Assumption Carbotech AG, 50% of total water consumed to waste water treatment ARA (in Lenzing total waste water treatment about 35% of total water consumption of company)		85.0	10.0	5.0	
	disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	0	kg	1.78E-5	1	1.25	(2.3.3.3.1.5): Mainly batteries and accumaltors, assumption Carbotech AG 1/3 of Lenzing waste materials from fibres production, 50% in recycling and 50% as hazardous waste Lenzing AG icineration of hazardous waste		85.0	10.0	5.0	
	disposal, municipal solid waste, 22.9% water, to sanitary landfill	CH	0	kg	7.43E-3	1	1.25	(2.3.3.3.1.5): Assumption Carbotech AG, 1/3 of Lenzing waste materials from fibres production, delivery non-hazardous waste Lenzing AG in deposition site		85.0	10.0	5.0	

EcoSpold Meta Information

Tab. A. 9 EcoSpold Meta Information of viscose production and related co-products

ReferenceFunction	401	viscose production
Geography	662 Location	AT
ReferenceFunction	493 InfrastructureProcess	0
ReferenceFunction	403 Unit	kg
DataSetInformation	201 Type	5
	202 Version	1.0
	203 energyValues	0
	205 LanguageCode	en
	206 LocalLanguageCode	de
DataEntryBy	302 Person	69
	304 QualityNetwork	1
ReferenceFunction	400 DataSetRelatesToProduct	1
	402 IncludedProcesses	This data includes the transformation of sulphate pulp to spinnable viscose using CS2
	404 Amount	1
	490 LocalName	Viskoseproduktion
	491 Synonyms	
	492 GeneralComment	The multioutput-process "viscose fibres" delivers the co-products "viscose fibres, at plant", "sodium sulphate from viscose production, at plant" and "sulphuric acid from viscose production, at plant". The allocation is based on economic criteria and was applied except for the used sulphate pulp.
	494 InfrastructureIncluded	1
	495 Category	textiles
	496 SubCategory	production
	497 LocalCategory	Textile
	498 LocalSubCategory	Gewinnung
	499 Formula	
	501 StatisticalClassification	
	502 CASNumber	
TimePeriod	601 StartDate	1997
	602 EndDate	2007
	603 DataValidForEntirePeriod	1
	611 OtherPeriodText	Time of publications.
Geography	663 Text	Data are from an Austrian company havin production sites in different countries, so the data can be taken as an global average..
Technology	692 Text	Transformation of sulfat pulp by CS2 to spinnable viscose
Representative	722 Percent	10%
	724 ProductionVolume	World production about 1.8 million tons
	725 SamplingProcedure	Literature data.
	726 Extrapolations	Some data are derived from other technologies.
	727 UncertaintyAdjustments	none
DataGeneratorAn	751 Person	69
	756 DataPublishedIn	2
	757 ReferenceToPublishedSource	48
	758 Copyright	1
	759 AccessRestrictedTo	0
	760 CompanyCode	
	761 CountryCode	
	762 PageNumbers	
	Validator	8
Länge 724	Details	automatic validation in Excel
	OtherDetails	none

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Federal Environment Agency Austria (2004). Data from Lenzing AG from year 2001 published in 'Medienübergreifende Umweltkontrolle in ausgewählten Gebieten'. Download of report as pdf from www.umweltbundesamt.at/fileadmin/site/umweltthemen/industrie/pdfs/MUK/07_Lenzing.pdf

7 Modified starch, at plant

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Review: Hans-Jörg Althaus, Empa

Last changes:

12 Dezember 2007

Abstract

The inventory refers to the production of modified starch from corn.

An eco-profile of modified starch was published 2004 by Novamont in the environmental product declaration (EPD) of MATER-BI granulate NF07U from production in Terni Italy. Calculations and assumptions in the present study are based mainly on the highly aggregated data about resource and electricity consumption of this EPD because no detailed data was available.

The inventory includes input materials corn starch, plasticizers and complexing agents, transports of input materials, energy consumption in the processing and packaging and regarding waste treatment.

The reference function is 1 kg granulate of modified starch, at plant.

7.1 Characteristic and data of modified starch production

The inventory 'modified starch, at plant' includes data about input materials corn starch, plasticizers and complexing agents, transports of input materials, energy consumption of processing and packaging as well as waste treatment. Data based on the highly aggregated data about resource and electricity consumption from the EPD of MATER-BI granulate NFO7U (Novamont, 2004)⁴⁰.

For the following reasons the published inventory has not been taken as it is but split up in different processes as far as possible:

- To make it consistent with the ecoinvent processes
- To make it more transparent
- To introduce the infrastructure

The assumptions and calculations summarized in Tab. 7.1 were applied for the use of the aggregated data from the EPD respectively to split up inventory. It was not possible to do it more detailed because no detailed data were available.

Existing data inventories for corn starch, fossil components of plasticizers and additives (assumption naphtha and natural gas), electricity and heat supply as well as transports and waste treatment were applied. Infrastructure was added assuming a similar but smaller site as in chemical industry. Additional to the EPD, land use is included within applied inventories.

The data on energetic resources and on CO₂ equivalents given detailed in the publication has been used as the main quality check for the generated process. This comparison with the EPD showed a very good accordance with the emissions contributing to climate change. Significant differences result for potential emissions compared to the aggregated data in the EPD of MATER-BI granulate NFO7U. Higher eutrophication and acidification potential result from field emissions included in the applied inventory 'corn starch, at plant'. The ozonformation given in the publication was higher than the one

⁴⁰ This EPD and relative Life Cycle assessment study have been approved by accredited Certification Body RINA S.p.A. in accordance with MSR 1999:2 standard provided by Swedish Environmental Management Council. Registration number: S-P-00102 Validity period: October, 2008 Controlled and verified by RINA S.p.A. (www.rina.it), Verification Body accredited by SWEDAC (Registration n. 1812)

from the ecoinvent process because of NMVOC emissions in the production processes. The difference of NMVOC has been added to the process. (0.7 g NMVOC/kg granulate). Further interpretations lack background information about calculation models and system boundaries in the EPD.

Tab. 7.1 Information from EPD mater-bi granulate⁴¹ and carried out calculations and assumptions

	Corn starch	Plasticizers and additives	Electricity	Heat	Transport	Waste
Data EPD	Total resource consumption as aggregated data (with energy content: oil, gas, coal, uranium)					
	0.59 kg CO ₂ fixation per kg granulate	Crude oil mentioned as plasticizers for Mater-bi bags ⁴² .	Total direct electricity consumption 0.56 kWh	-	Oil consumption given for transports	Amount hazardous and others.
Assumption	CO ₂ fixation from use of corn starch	Plasticizers and additives based on fossil components (gas, oil)	Electricity supply mix of Italy	Remaining oil, coal, gas for heat production	lorry 16 t	Incineration hazardous, deposition other.
Calculation	¹⁾ Fixation of 0.59 kg CO ₂ corresponds to 340 g modified starch. (CO ₂ fixation per kg corn starch 1.74 kg CO ₂)	²⁾ About 660 g plasticizers and additives (340 g corn starch per kg granulate)	³⁾ 0.56 kWh supply mix Italy	Total oil, gas and coal consumption minus ¹⁾⁻⁴⁾ Oil (3.4 MJ) Coal (5 MJ)	⁴⁾ transport distance related to oil consumption 0.04 km LkW 16t	9 g hazardous and 2.9 g other waste from granulate production/packaging

7.2 Allocation to co-products

No co-products are reported for the production of modified starch.

7.3 Statistics of production

Novamont is the main producer of modified starch (MATER-BI). The capacity of their factory in Terni, Italy, is about 20'000 ton/year.

7.4 Characterisation and Use of modified starch

Modified starch represents a family of biodegradable thermoplastic materials obtained from renewable raw materials, covering a variety of applications in the fields of waste collection, hygiene, agriculture, catering and packaging. The evaluated MATER-BI granulate NF07U is mainly used for the production of films.

7.5 Life Cycle Inventories of modified starch

Inventory refers to the production of modified starch from corn starch using plasticizers and additives. Inventories include data about input materials corn starch and fossil components (plasticizers and

⁴¹ Data as in publication on http://bio4eu.jrc.es/documents/e_epd102.pdf (August 2007)

⁴² www.novamont.com, link to LCA and EPD, figure under LCA for Mater-bi bags (August 2007)

complexing agents), transports of input materials, energy consumption of processing and packaging as well as waste treatment.

Tab. 7.2 Unit process of modified starch, at plant

	Name	Location	Unit	modified stark, at plant	UncertaintyType	Standard Deviation 95%	GeneralComment
	Location InfrastructureProcess Unit			RER 0 kg			
product	modified stark, at plant	RER	kg	1			
technosphere	maize starch, at plant	DE	kg	3.36E-1	1	1.57	(2,2,2,3,4,5); Calculation portion starch from reported CO ₂ fixation per kg granulate (0.59 kg CO ₂) compared to CO ₂ fixation in corn starch (1.74 kg CO ₂). 0.59 kg CO ₂ per kg Materbi reported in EDP Materbi Granulate NF07U (Novamont, 2004)
	electricity, low voltage, production IT, at grid	IT	kWh	5.55E-1	1	1.57	(2,2,2,3,4,5); Value for direct electricity consumption from EDP Materbi Granulate NF07U (Novamont, 2004)
	naphtha, APME mix, at refinery	RER	kg	3.09E-1	1	1.57	(2,2,2,3,4,5); Total oil consumption from EDP Materbi Granulate NF07U (Novamont, 2004). Assumption Carbotech AG share between fuel and material use (=plasticizer for Materbi)
	natural gas, high pressure, at consumer	IT	MJ	1.25E+1	1	1.57	(2,2,2,3,4,5); Total gas consumption from EDP Materbi Granulate NF07U (Novamont, 2004). Assumption Carbotech AG material use mainly (=plasticizers for Mater-bi)
	heat, at hard coal industrial furnace 1-10MW	RER	MJ	3.40E+0	1	1.57	(2,2,2,3,4,5); Total coal consumption from EDP Materbi Granulate NF07U (Novamont, 2004). Assumption Carbotech AG use as fuel mainly.
	heat, light fuel oil, at industrial furnace 1MW	RER	MJ	5.02E+0	1	1.57	(2,2,2,3,4,5); Total oil consumption from EDP Materbi Granulate NF07U (Novamont, 2004). Assumption Carbotech AG for share between use as fuel and material use as plasticizer for Materbi
	disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	9.00E-3	1	1.57	(2,2,2,3,4,5); Data for granulate processing and packaging from EDP Materbi Granulate NF07U (Novamont, 2004), assumption incineration
	disposal, municipal solid waste, 22.9% water, to sanitary landfill	CH	kg	2.90E-3	1	1.57	(2,2,2,3,4,5); Data for granulate processing and packaging from EDP Materbi Granulate NF07U (Novamont, 2004), assumption landfill
	transport, lorry >16t, fleet average	RER	tkm	4.19E-2	1	2.28	(2,2,2,3,4,5); Data for oil consumption in Transport from EPDM Materbi Granulated Novamont (Novamont, 2004)
	chemical plant, organics	RER	unit	5.00E-10	1	3.51	(5,4,1,1,4,5); Assumption infrastructure similar than chemical plant, but only half size. 50 years life time and output 20'000 t as Novamont in Italy
emission air, low population density	Heat, waste	-	MJ	2.00E+0	1	1.57	(2,2,2,3,4,5); Total oil consumption from EDP Materbi Granulate NF07U (Novamont, 2004). Assumption Carbotech AG share between fuel and material use (=plasticizer for Materbi)
	NMVOC, non-methane volatile organic compounds, unspecified origin	-	kg	7.00E-4	1	2.08	(5,4,1,1,4,5); Correction for lacking potential emissions photochemical oxidants compared to aggregated data in EDP Materbi Granulate NF07U (Novamont, 2004). Assumption Carbotech AG VOC emissions as typical in plastic industry

7.6 Data Quality Considerations

Tab. 7.2 shows data quality indicators for the inventory of modified starch, which are based on calculations on highly aggregated data of MATER-BI production of Novamont in Italy. The simplified approach with a pedigree matrix has been used for calculating the standard deviation. Uncertainties are not mentioned in the data from the EPDM and background information about system boundaries lack transparency. Nevertheless data quality was assumed to be comparable to other aggregated data for polymer granulates from the plastic industry.

Abbreviations

DM Dry Matter

EPD Environmental Product Declaration

EcoSpold Meta Information

Tab. A. 10 EcoSpold Meta Information of modified starch

ReferenceFunction	401	Name	modified stark, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	69
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Only high aggregated data were available. So the emissions from energy consumption, waste water treatment and raw material inputs have been subtracted resulting in a difference in NMVOC emissions.shown in this process.
	404	Amount	1
	490	LocalName	Modifizierte Stärke, ab Werk
	491	Synonyms	
	492	GeneralComment	Inventory refers to the production of 1 kg granulate modified starch. The inventories is based on calculations and extrapolations using highly aggregated background data from the environmental product declaration of Materbi (Novamont, 2004, Italy). Included processes in the inventory of modified starch granulate are the production of input materials corn starch and fossil components (plasticizers and complexing agents), transports of input materials, energy consumption in the processing and packaging at plant aswell as waste treatment.
	494	InfrastructureIncluded	1
	495	Category	plastics
	496	SubCategory	polymers
	497	LocalCategory	Kunststoffe
	498	LocalSubCategory	Polymere (Granulate)
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	
Geography	663	Text	Modified starch "mater-bi" is produced mainly by Novamont in Terni, Italy
Technology	692	Text	Production of modified starch out of corn starch, plasticizers and complexing agent.
Representativeness	722	Percent	
	724	ProductionVolume	20'000 t/year produced by Novamont in Italy.
	725	SamplingProcedure	Data were compiled from environmental product declaration of producer Novamont in Italy
	726	Extrapolations	
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	68
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	48
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	
ProofReading	5616	Validator	8
	5615	Details	automatic validation in Excel
	5619	OtherDetails	none

References

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